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**Investigations of partonic energy loss and initial state effects in small systems via dijet measurements with the ATLAS detector**

by

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Dissertation

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## Abstract

This dissertation presents two experimental studies of QCD using data recorded by the ATLAS experiment at the Large Hadron Collider. The first study investigates proton-size fluctuations in proton–lead collisions. The internal momentum configuration of a proton can modify its effective transverse size, particularly when a single parton carries an unusually large fraction of the proton's momentum. Dijet production provides a means of probing this configuration because the dijet kinematics constrain the momentum fraction extracted from the proton. The relationship between this momentum fraction and both bulk particle production and forward neutron production is measured. These observables provide information about proton-size fluctuations and enable comparisons with theoretical models of the proton's internal structure. They also provide important constraints for interpreting proton–lead collisions as a control system for heavy-ion measurements. The second study examines jet quenching in light-ion collisions using the dijet balance observable, defined as the ratio of the transverse momentum of the lower-momentum jet to that of the higher-momentum jet. Interactions between jets and the QGP modify this distribution by reducing the balance between the two jets. Measurements in oxygen–oxygen and neon–neon collisions provide access to a collision system smaller than previously studied heavy-ion systems while retaining a substantial transverse overlap between the colliding nuclei. A rigorous statistical analysis establishes that the dijet balance is significantly modified in light-ion collisions relative to proton–proton collisions. Together, these studies use dijets to investigate both the structure of cold QCD matter and the energy loss of hard-scattered partons traversing the QGP.

## Acknowledgements

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*A dedication is optional.*

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## List of Abbreviations

| Abbreviation | Definition |
|---|---|
| ATLAS | A Toroidal LHC ApparatuS |
| LHC | Large Hadron Collider |
| QCD | Quantum Chromodynamics |
| QGP | Quark-Gluon Plasma |
| ZDC | Zero Degree Calorimeter |

## List of Symbols

| Symbol | Definition |
|---|---|
| $\sqrt{s_{\mathrm{NN}}}$ | Nucleon-nucleon center-of-mass energy |
| $p_{\mathrm{T}}$ | Transverse momentum |
| $\eta$ | Pseudorapidity |
| $\phi$ | Azimuthal angle |

# Chapter 1 Introduction

The purpose of this dissertation is to describe a set of measurements that experimentally study the strong force in two very different temperature regimes using data recorded by the ATLAS experiment at the Large Hadron Collider (LHC). Quantum Chromodynamics (QCD) is the theory of the strong force, which acts between the fundamental constituents of matter, quarks and gluons, collectively called partons. QCD is characterized by weak interactions at high energy scales and strong interactions at low energy scales. This behavior gives rise to the interesting feature of confinement, in which quarks and gluons in a vacuum are always found in a bound, color-neutral state. However, understanding QCD in a vacuum is not enough. When QCD is coupled to an external system, thermodynamic quantities such as energy density and temperature become relevant. At sufficiently high temperature and energy density, QCD predicts a transition from confined nuclear matter to a deconfined phase near $T = 160$ MeV.

The curious nature of this phase change is that, above this temperature, the confinement principle described above no longer holds, and quarks and gluons move about freely in a state known as the quark-gluon plasma (QGP). This phase change defines two separate regimes of QCD, both worthy of study. **Cold QCD** is the state of nuclear matter in which the universe has existed for the vast majority of its history. It is important to study because it is the state we most often interact with as humans. **Hot QCD**, also known as the QGP, is the state of nuclear matter that existed in the very early universe. Since quarks and gluons are not confined to bound states, the QGP provides a laboratory for studying their interactions. Over the past 50 years, experimental heavy-ion physicists have built and used massive machines capable of probing both Cold and, more recently, Hot QCD through high-energy particle collisions.

Jets are highly energetic sprays of collimated particles that originate from high-energy scatterings in particle collisions. Due to momentum conservation, they are most commonly produced in pairs through a highly energetic $2 \to 2$ scattering process between

two fundamental particles. These jet pairs form a dijet. This dissertation presents two measurements that use dijets as experimental probes to study each phase independently.

Protons are composite particles made up of three valence quarks and a number of 'sea' quarks and gluons. When protons are given a certain amount of energy, such as in a particle accelerator, conservation of momentum ensures that the total momentum of each proton is shared among its constituent partons, but it does not ensure equal sharing between them. Constituent partons share the momentum of the parent proton through quantum-mechanical processes. The momentum configuration of the partons can affect the overall shape of the proton. When, by chance, one parton monopolizes more than its share of momentum, the color fields of the other partons are shadowed, giving the appearance of a smaller-than-average proton.Citation needed Dijets have the unique experimental ability to measure the fraction of the momentum extracted from a projectile proton in a high-energy nuclear collision. In addition, by colliding protons with a very large target ion (Pb) the size of the projectile can be studied by measuring quantities proportional to the size of the hole left in the target. Two such quantities, bulk particle production and forward neutron production, are studied as a function of the momentum fraction extracted from the proton. The novel measurement presented in this dissertation allows detailed comparisons with theoretical models that attempt to capture this effect and provides crucial input for the heavy-ion community, which uses proton–Pb collisions as a control case for Pb–Pb collisions.

Because dijets originate from $2 \to 2$ scattering processes, conservation of momentum requires each jet in the pair to have a similar energy. The observable $x_{\mathrm{J}}$ represents the ratio of the lower-momentum jet to the higher-momentum jet and is called the dijet balance. The distribution of $x_{\mathrm{J}}$ is known to be modified by the presence of final–state effects, i.e. the presence of a QGP medium surrounding the dijet. citation needed The stochastic manner in which jet-medium interactions occur produces dijets that are less balanced on average. New light-ion data (O+O and Ne+Ne) provide a laboratory smaller

than previous heavy-ion systems (Pb+Pb or Au+Au). Additionally, unlike asymmetric systems ($p$+Pb or $d$+Au), light ions have a sizeable transverse overlap. Results presented in this dissertation take a well established observable and apply it to a new collision system. For the first time, a rigorous statistical analysis is presented establishing that the dijet balance is significantly modified in light-ion collisions with respect to $pp$. They also help inform models that constrain how jets lose energy to the QGP.

The dissertation is organized as follows. First, the relevant theoretical background of QCD, nuclear structure, jets, and heavy-ion physics is established in Chapter 2. Chapter 3 discusses the LHC accelerator complex and the ATLAS experiment as well as details some technical work done by the author of this dissertation. In Chapter 4, the analysis of Cold QCD proton-size fluctuations in $p$+Pb collisions is presented and the results are discussed. Then, the observation of jet energy loss in light-ion collisions via a dijet-balance measurement is reported in Chapter 5. Finally, the key results of this dissertation and their implications for the field are summarized in Chapter 6.

# Chapter 2 Background

## 2.1 Quantum Chromodynamics

Quantum Chromodynamics (QCD) has been tested experimentally in many different collision systems and at many different energies. It is a field theory describing the interaction between quarks, the fermionic building blocks of matter, and gluons, the bosonic force carriers. This section introduces the basic features of QCD that are relevant background for the investigations presented in this dissertation.

Gell-Mann successfully described the observed spectrum of hadrons using different flavors of *quarks* with different fractional charges [1]. This quark model was highly successful, but it predicted fermions that appeared to violate the spin-statistics theorem, such as the $\Delta^{++}$ baryon, which contains three $u$ quarks with parallel spins [2]. This problem was resolved by introducing an additional quantum number called color charge, consisting of three color degrees of freedom, conventionally denoted $r$, $g$, and $b$. QCD describes the observed hadron spectrum in terms of colorless bound states. Mesons are colorless because they consist of a color–anticolor pair, while baryons are colorless because they contain all three colors or all three anticolors.

### 2.1.1 QCD Lagrangian

QCD is a gauge theory based on a local $SU(3)$ color charge symmetry. Just as the $U(1)$ symmetry of Quantum Electrodynamics (QED) implies the existence of a gauge field, the photon, which couples to electrically charged fermions, the $SU(3)$ symmetry of QCD implies the existence of gauge fields, the gluons, which couple to colored fermions (quarks) [3]. The elements of $SU(3)$ do not necessarily commute with one another; therefore, the $SU(3)$ symmetry is non-Abelian. The non-Abelian nature of QCD allows gluons to self-interact, unlike photons in QED.

The QCD Lagrangian is given in Equation 1 [4].

$$\mathcal{L}_{\mathrm{QCD}} = -\frac{1}{4} F^{a\,\mu\nu} F^{a}_{\mu\nu} + \sum_{f} \overline{\Psi}_{f} \left( i\gamma^{\mu} D_{\mu} - m_{f} \right) \Psi_{f}, \tag{1}$$

$D_{\mu}$ is the gauge-covariant derivative, defined by

$$D_{\mu} = \partial_{\mu} - i g_{s} T^{a} G^{a}_{\mu}. \tag{2}$$

$\Psi_{f}$ is the quark field of flavor $f$, $m_{f}$ is its mass, $T^{a}$ is a generator of the $SU(3)$ color group, $G^{a}_{\mu}$ is the gluon field, and $g_{s}$ is the QCD coupling constant. The gluon field-strength tensor is

$$F^{a}_{\mu\nu} = \partial_{\mu} G^{a}_{\nu} - \partial_{\nu} G^{a}_{\mu} + g_{s} f^{abc} G^{b}_{\mu} G^{c}_{\nu}, \tag{3}$$

where $f^{abc}$ are the structure constants of $SU(3)$. The first term in Equation 1 describes the gluon kinetic term and gluon self-interactions, while the second term describes quark propagation, quark–gluon interactions, and the quark mass term. The index $f$ runs over the active quark flavors, and the index $a$ runs over the eight generators of $SU(3)$. The eight gluon fields transform according to the adjoint, or octet, representation of the color group.

### 2.1.2 $\alpha_{s}$, asymptotic freedom, and perturbative QCD

The QCD coupling constant $g_{s}$ from Equation 1 can be used to define $\alpha_{s} = g_{s}^{2}/4\pi$, which is the QCD analogue of the fine-structure constant. However, the effective coupling of QCD depends not only on quark–gluon and gluon–gluon interactions, but also on higher-order diagrams involving interaction loops, which must be accounted for through renormalization. The renormalization scale, $\mu_{R}$, sets the scale at which divergences are subtracted. As a result, $\alpha_{s}$ depends on the momentum-transfer scale of the QCD

interaction, $Q$, according to Equation 4 [2]. At one-loop order, the running coupling is

$$\alpha_s(Q) = \frac{4\pi}{\beta_0 \ln\left(Q^2/\Lambda^2_{\mathrm{QCD}}\right)}, \qquad \beta_0 = 11 - \frac{2}{3}n_f. \tag{4}$$

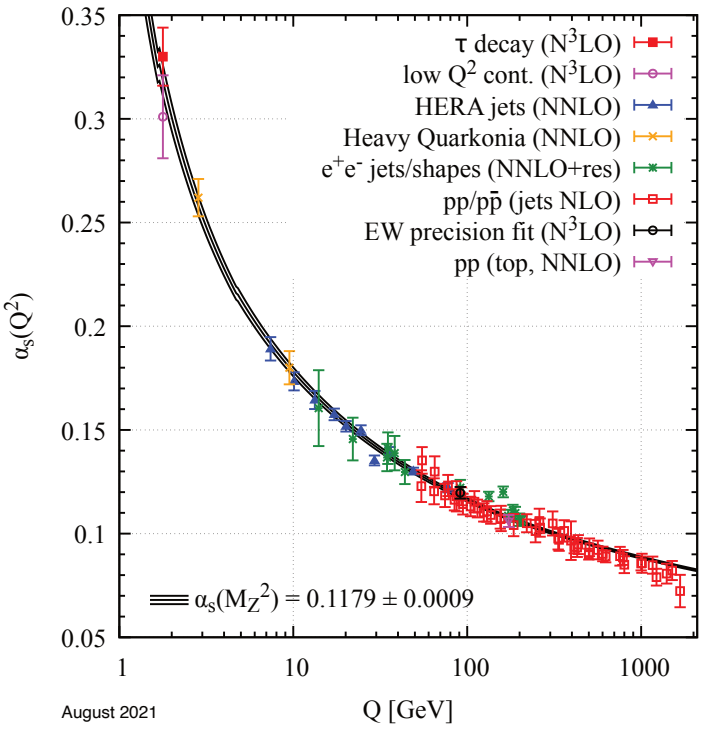



Figure 1: The running of the strong coupling $\alpha_s$ as a function of the momentum-transfer scale, illustrating asymptotic freedom. Taken from Reference [5]

Here, $n_f$ is the number of quark flavors active at the scale $Q$, and $\Lambda_{\mathrm{QCD}}$ is the characteristic energy scale of QCD, approximately 200 MeV, depending on the renormalization scale ($\mu_R$). For momentum transfers well above $\Lambda_{\mathrm{QCD}}$, $\alpha_s$ is small and the QCD coupling becomes weak. This phenomenon is known as *asymptotic freedom* and is shown in Figure 1. In the asymptotically free regime, QCD calculations become directly accessible through perturbation theory, where observables can be estimated as a series in $\alpha_s$. Perturbative QCD (pQCD) allows for highly precise predictions of many observables measured at high-energy particle colliders. Conversely, for collisions outside the perturbative regime, corresponding to low momentum transfer, calculating

observables becomes much more difficult. Sophisticated theoretical approaches such as lattice QCD can perform these calculations, but they are computationally expensive.

### 2.1.3 Confinement and Hadronization

The observed hadron spectrum consists of colorless (color-singlet) QCD bound states. In fact, no colored bound state has ever been observed, a principle known as *confinement*. The principle of confinement can be illustrated by trying to pull a quark–antiquark pair apart. As the $q\bar{q}$ pair is separated, gluon field lines appear between the two particles, similar to the field lines between an electric dipole. Self-interactions between the gluon field lines keep them close together, creating a tube of color field between the two quarks. Figure 2 illustrates this process by comparing the gluon self-interactions in a quark–antiquark system with the field lines of an electric dipole and the compressed field lines of a color dipole.

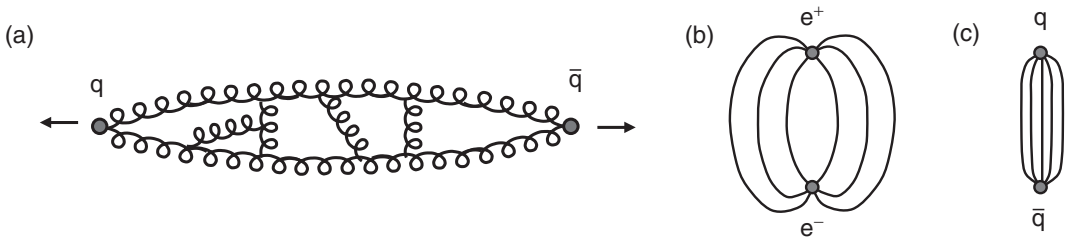



Figure 2: (left) Illustration of gluon self-interactions in a $q\bar{q}$ system. (middle) Field lines for an electric dipole. (right) Compressed field lines for a color dipole. Taken from Reference [3].

As the separation increases, this tube becomes longer; therefore, the energy required to maintain this configuration is linearly proportional to the distance. The energy density of this tube is experimentally found to be approximately 1 GeV/fm, which explains why it is energetically favorable for colored objects to form colorless states [3].

Now that the principle of confinement has been established, consider the situation in which each (anti-)quark in a $q\bar{q}$ pair is given a large momentum that pulls the pair apart, as in sufficiently hard $e^+e^- \to q\bar{q}$ scattering. As described above, the $q\bar{q}$ pair forms a tube

of gluon flux with highly concentrated energy. At some point, the energy of the gluon flux becomes large enough to produce new $q\bar{q}$ pairs, which in turn break the original flux tube into two smaller flux tubes, or 'strings.' The original highly energetic quarks lose some energy in this breaking process because energy is required to produce the new strings. This string lengthening and binary breaking process will repeat itself until all the $q\bar{q}$ pairs are formed into colorless hadrons. The result is two streams of highly collimated hadrons, one in the direction of the original high-energy quark and the other in the direction of the high-energy antiquark. The formation of these hadron streams, which are what experimentalists detect, is called *hadronization* and is illustrated in Figure 3 [3].

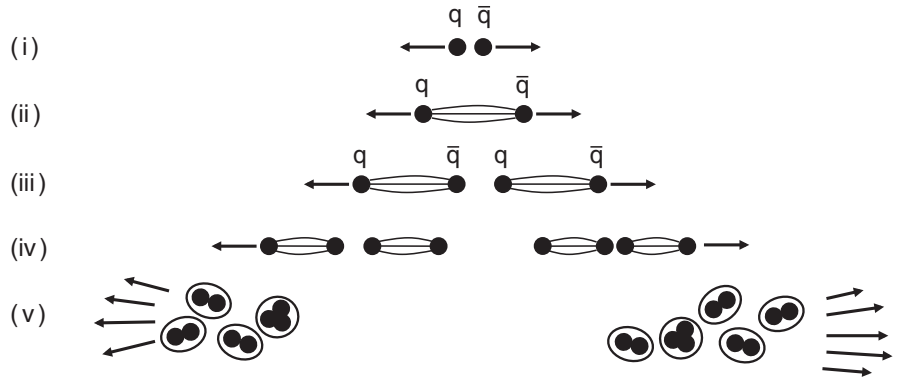



Figure 3: Schematic illustration of hadronization through the breaking of a color flux tube into quark–antiquark pairs and collimated streams of hadrons. Taken from Reference [3].

### 2.1.4 The Parton Model and Deep-Inelastic Scattering

The parton model provides a description of the internal structure of hadrons by imagining the hadron as a box of free point-like quarks and gluons, collectively called partons [6]. Since the hadron has momentum $p$, each parton has momentum $\xi_i p$, where $\sum_{\text{partons}} \xi_i = 1$. The partons themselves are imagined to be practically free of interactions, which requires that the hadron have a large momentum. The fast-moving nature of the hadron ensures that, in the laboratory frame, the internal dynamics between partons occur over a relatively long timescale. When a probe collides with the

hadron, the internal structure is effectively frozen over the timescale of the collision.

The parton model was introduced by Feynman [6], but did not gain acceptance until later experimental developments. High-energy lepton-nucleon scattering played a fundamental role in determining the internal configuration of nucleons. Taking $e^- + p \to e^{-\prime} + X$ scattering as an example, it is illustrative to define some kinematics using the diagram shown in Figure 4.

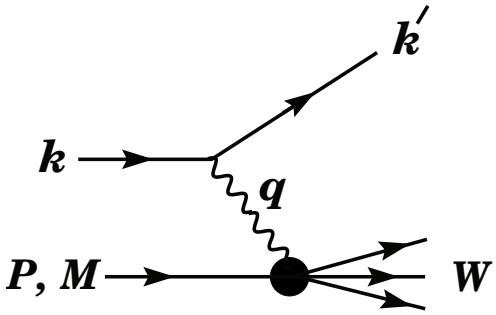



Figure 4: Schematic diagram of deep-inelastic $e^- + p$ scattering, where the electron has initial four-momentum $k$ and the proton has mass $M$ and four-momentum $P$. Taken from Reference [5].

$$Q^2 = -q^2, \qquad x = \frac{Q^2}{2P \cdot q}, \qquad y = \frac{q \cdot P}{k \cdot P}, \qquad W^2 = (P + q)^2 \tag{5}$$

Here, $q$ is the four-momentum transferred by the exchanged virtual photon, $Q^2$ is the squared momentum-transfer scale, $x$ is the Bjorken scaling variable, $y$ is the fractional energy loss of the electron, and $W^2$ is the invariant mass squared of the hadronic final state. The Bjorken scaling variable, typically called Bjorken-$x$, has the physical interpretation of the partonic momentum fraction ($\xi_i$) carried by the struck parton in the proton. The above process is called deep ($Q^2 \gg M^2$) inelastic ($W^2 \gg M^2$) scattering (DIS) [5].

Experimentally, one can measure the scattering angle ($\theta$) and the initial ($E_e$) and final ($E'_e$) energies of the electron, which can be mapped onto the above kinematic

variables using the following expressions.

$$Q^2 = 2E_e E'_e(1+\cos\theta), \qquad x = \frac{Q^2}{2\mathbf{M}\,(E_e - E'_e)}, \qquad y = 1 - \frac{E'_e}{E_e} \tag{6}$$

For unpolarized electromagnetic DIS in the one-photon-exchange approximation, the cross section can be written differentially in $x$ and $Q^2$ as

$$\frac{\mathrm{d}^2\sigma}{\mathrm{d}x\,\mathrm{d}Q^2} = \frac{4\pi\alpha^2}{Q^4}\left[(1 - y - \frac{\mathbf{M}^2 y^2}{Q^2})F_2(x,Q^2) + y^2 F_1(x,Q^2)\right] \tag{7}$$

where $\alpha$ is the electromagnetic fine-structure constant. $F_1$ and $F_2$ are *structure functions* that encode information about the internal charge configuration of the hadron. Bjorken predicted that, in the high-energy limit, the structure functions depend only on Bjorken-$x$ and not on $Q^2$ (i.e., $F(x,Q^2) \to F(x)$) [7]. This is consistent with the parton model: if the internal structure of the proton is described by point-like partons, scattering at different (sufficiently large) momentum scales does not resolve any additional structure. Further confirmation that DIS involves lepton scattering from quarks inside the hadron was provided by the observation of the Callan–Gross relation [8], $F_2(x) = 2xF_1(x)$, at the Stanford Linear Accelerator Center (SLAC) [9].[1] This relation is a prediction for electron scattering from spin-1/2 constituents, whereas $F_1(x) = 0$ would indicate scattering from spin-0 constituents.

[1] Gluons cannot directly interact with the virtual photon in DIS, but they can through fluctuations into a $q\bar{q}$ dipole which we call a 'resolved' photon.

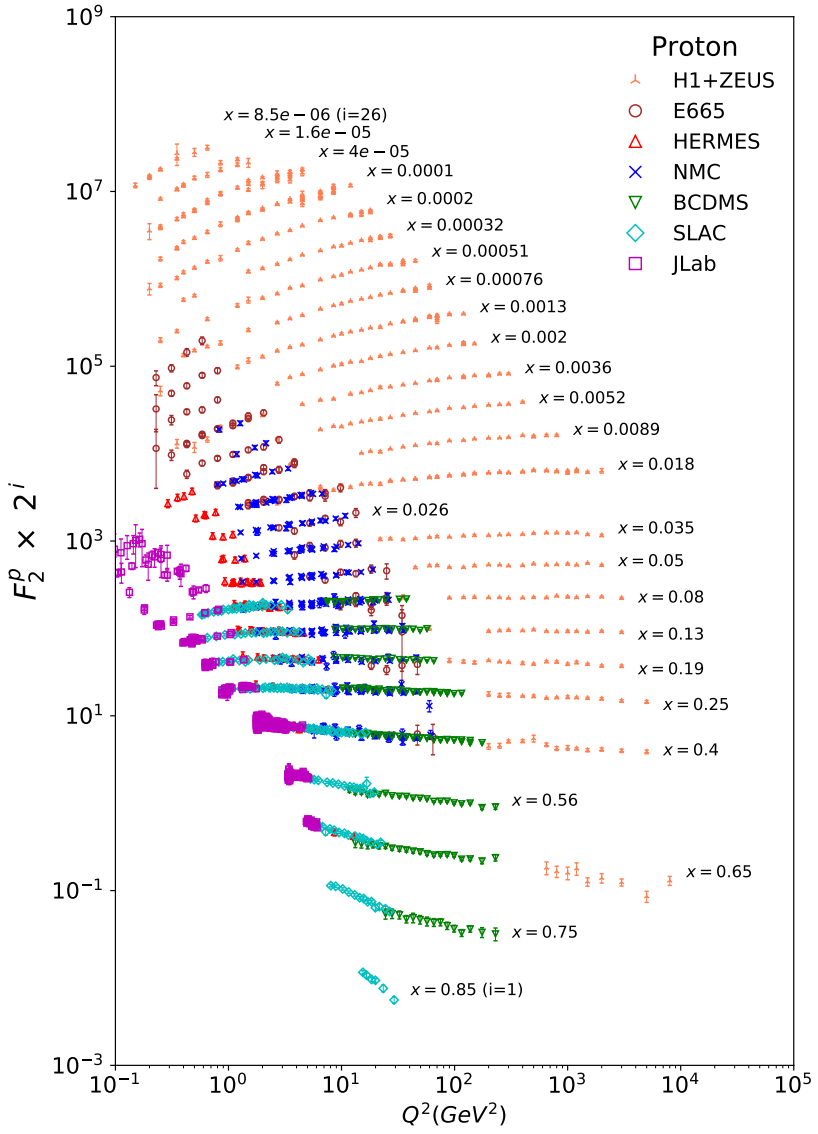



Figure 5: The proton structure function $F_2^p$ measured in electromagnetic scattering of electrons, positrons, and muons on protons. The data are plotted as a function of $Q^2$ in bins of fixed $x$, with $F_2^p$ multiplied by $2^{i_x}$ for plotting, where $i_x$ indexes the $x$ bins. Only data with $W^2 > 3.5\ GeV^2$ are included. Some points are offset slightly in $Q^2$ for clarity. Taken from the Review of Particle Physics [5].

As the range of $x$ and $Q^2$ accessible to experiments increased, the Bjorken-scaling argument held over a wide range of $x$ values ($0.03 \leq x \leq 0.4$), as shown by DIS measurements of $F_2(x)$ in Figure 5 taken from Hadron Electron Ring Accelerator [10] (HERA) data and other fixed target experiments. However, the independence of $F_2(x)$ from $Q^2$ breaks down outside this range of $x$, necessitating a description of the structure

function as $F_2(x, Q^2)$. This is because the simple parton model underlying the Bjorken-scaling argument does not account for interactions among partons within the hadron. In DIS, the photon can probe constituents with a spatial resolution of $O(1/Q)$. At relatively low $Q^2$, this resolution reveals only the rough structure of a hadron, such as the three so-called *valence* quarks ($uud$) in a proton, shown in the left panel of Figure 6. However, when the proton is probed at higher $Q^2$, a more complicated substructure is revealed, including so-called *sea quarks* and gluons that fluctuate in and out of existence within the proton, as shown in the right panel of Figure 6. These sea partons were always present, but at the lower energies of DIS, the probe was not powerful enough to resolve them separately from the valence quarks.

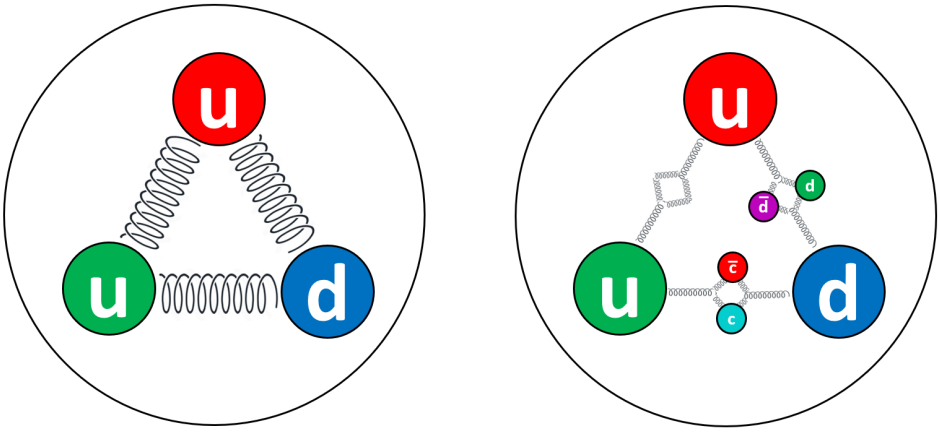



Figure 6: Schematic illustration of proton structure. The simple model on the left contains valence quarks held together by gluons, while the more complete model on the right also includes sea quarks and gluons. Illustration provided by A. Tate and inspired by Y. Kulinich.

This increased resolution has a feed-down effect on the structure function of the proton. At high $Q^2$, a very high-$x$ parton at low $Q^2$ can be resolved as a medium-high-$x$ parton and an additional soft parton, decreasing $F_2$ at high $x$ and increasing $F_2$ at low $x$, exactly the behavior observed by the DIS experiments in Figure 5. The fact that Bjorken scaling holds at moderate $x$ comes from a detailed balance of these two competing effects. DIS provides direct evidence that the substructure of the proton, and of all other hadrons, consists of interacting point-like partons.

### 2.1.5 Factorization

While the matrix element for a given QCD process may be calculable to some order in $\alpha_s$, an observable involving hadrons is also affected by long-distance interactions before it can be observed experimentally. Additionally, for collision systems involving hadrons, the internal partonic configuration of the reactants must be considered. Therefore, any measurement involving hadrons in the initial or final state is sensitive to low-energy (nonperturbative) QCD interactions. The running coupling of $\alpha_s$ in QCD can be exploited to derive predictions for lepton-hadron and hadron-hadron cross sections at large momentum transfer. This is done using *factorization* theorems, which systematically separate (factorize) long-distance from short-distance behavior [11]. The factorization theorem requires $Q^2 \gg M^2$, and corrections are of $O\left(M^2/Q^2\right)$. Additionally, a factorization scale, $\mu_F$, denotes the point at which collinear radiation is considered part of the perturbative hard scattering ($Q > \mu_F$) rather than absorbed into the nonperturbative long-range interactions ($Q < \mu_F$). Typically, the factorization scale ($\mu_F$) is roughly the scale of the probe ($Q$). Schematically, this separation is represented in Figure 7, which depicts a $pp$ collision containing a hard scattering.

Mathematically, the factorized cross section can be written schematically for inclusive hadron production as

$$\begin{aligned} \mathrm{d}\sigma_{pp\to h+X} \approx \sum_{ijkl} \int & \mathrm{d}x_i \mathrm{d}x_j \mathrm{d}z_k \, f_i\left(x_i, \mu_F^2\right) f_j\left(x_j, \mu_F^2\right) \\ & \otimes \mathrm{d}\hat{\sigma}_{ij\to kl}\left(x_i, x_j, \alpha_s\left(\mu_R\right), Q^2\right) \\ & \otimes D_{k\to h}\left(z_k, \mu_F^2\right) \end{aligned} \tag{8}$$

The left-hand side of Figure 7 and the top line of Equation 8 represent individual partons being extracted from a proton to participate in the scattering. The partons are drawn from the proton according to a number-density function known as a *parton distribution function* (PDF), $f_i\left(x_1, \mu_F^2\right)$, which describes the average number of partons

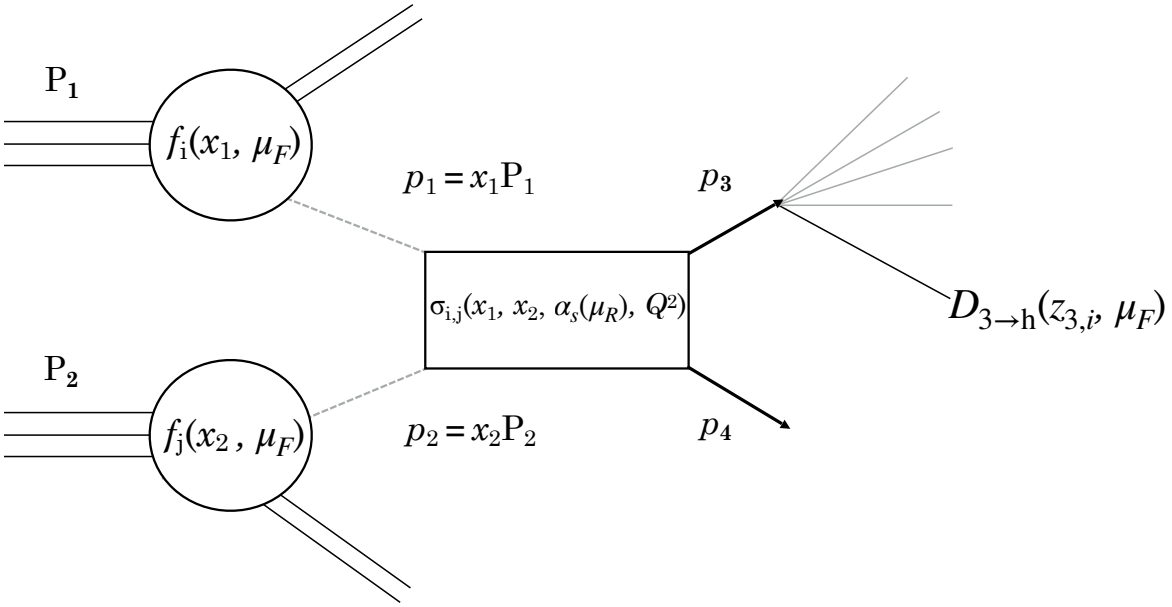



Figure 7: Schematic representation of QCD factorization.

(quarks or gluons) that can be found at a given momentum fraction ($x$). The box in the middle of Figure 7 and the hard-scattering term in Equation 8 express the hard scattering itself, which is directly calculable to some order in $\alpha_s\,(\mu_R)$ using pQCD. The right-hand side of Figure 7 illustrates how the products of the hard scattering hadronize into a final-state hadron measured in a detector. This step is parameterized by so-called *fragmentation functions*, which are number-density functions representing the expected number of final-state hadrons carrying a fraction of the outgoing parton's momentum ($z$). Equation 8 integrates from $0 \rightarrow 1$ over the momentum fractions $x$ extracted from the colliding hadrons and over the momentum fraction $z$ of the final observed hadron. Both the PDFs and fragmentation functions capture nonperturbative QCD physics and are obtained from fits to experimental data. The factorization theorem requires a separation of scales between the hard and soft parts of the collision and is essential for making precise QCD predictions.

## 2.2 Descriptions of Nucleon Structure

### 2.2.1 Parton Distribution Functions

The PDF of a given hadron parameterizes the number density of a given type of parton at a given Bjorken-$x$, as formally introduced in Section 2.1.5. All the possible partons are the 6 flavors of quarks and the one gluon, so for a given hadron there are 13 different parton distributions (counting anti-quarks). For heavy-flavor quarks ($c$, $b$, $t$), whose masses are well above $\Lambda_{\text{QCD}}$, their PDFs can be computed directly using pQCD. For the light flavor quarks and gluon, the PDFs themselves must be determined from experimental data.

From DIS data, it is possible to obtain structure functions that, at leading order (LO), are sensitive to the quark content of the hadron across a wide range of $x$ and $Q^2$. At LO, the structure functions are related to the quark PDFs according to Equation 9 [12].

$$F_2(x, Q^2) = \frac{1}{2} \sum_q e_q^2 \left[ x f_q(x, Q^2) + x f_{\bar{q}}(x, Q^2) \right] \tag{9}$$

where $e_q$ is the fractional charge of a given quark flavor. The PDFs also have several sum rules that offer additional constraints to any fit [11]. For illustrative purposes, all the following sum rules will apply to protons.

$$\sum_i \int_0^1 x f_i(x, Q^2) dx = 1, \qquad \sum_i \int_0^1 e_q f_i(x, Q^2) dx = 1 \tag{10}$$

Equation 10 displays the sum rule conserving the total proton momentum (left) and the sum rule conserving the total proton charge (+1) (right).

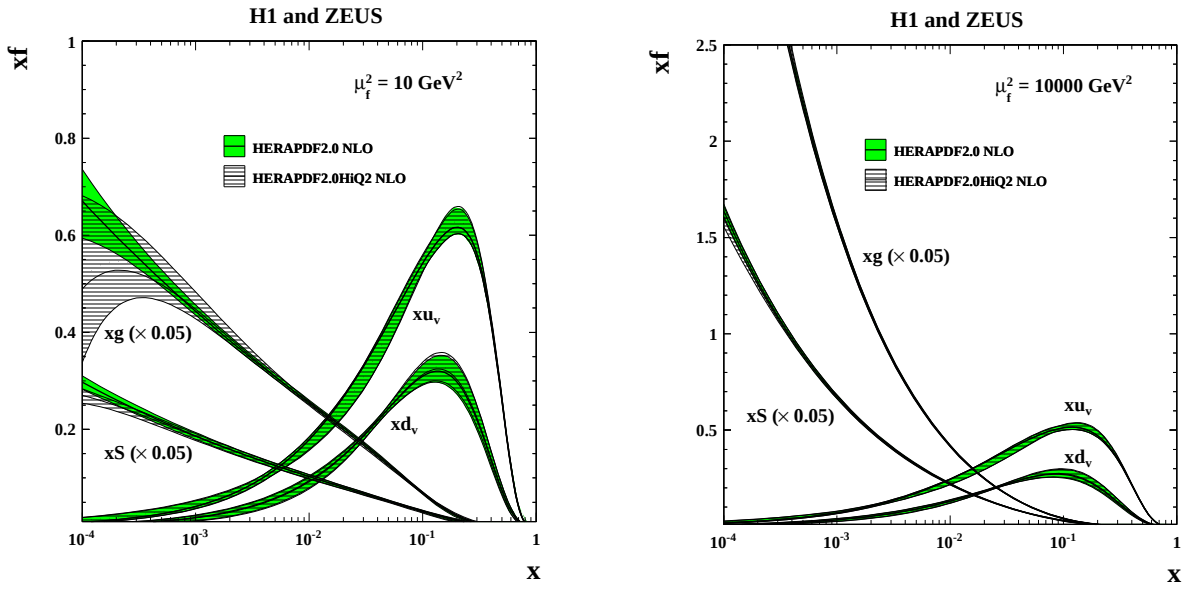
$$\int_0^1 (f_u - f_{\bar{u}}) dx = 2, \ \int_0^1 (f_d - f_{\bar{d}}) dx = 1, \ \int_0^1 (f_{\geq s} - f_{\geq \bar{s}}) dx = 0, \tag{11}$$

Equation 11 shows the flavor-conservation sum rules for a proton with two $u$ valence

quarks, one $d$ valence quark, and no strange or heavier valence quarks. Under all of these constraints, dedicated PDF fitting groups will use fits of the form given in Equation 12 [5].

$$f_i = x^a \left(\cdots\right) \left(1 - x\right)^b \tag{12}$$

In total the PDF fitting groups typically have approximately 14-32 free parameters in their fits. The fits are optimized to minimize $\chi^2$ with respect to the input data, and the errors are determined using the Hessian method [5]. For the light flavor quarks and gluon, while pQCD *cannot* directly compute PDFs it *can* describe how they should evolve with $Q^2$ through the Dokshitzer–Gribov–Lipatov–Altarelli–Parisi (DGLAP) evolution equations [13]. The DGLAP equations evolve a prediction for a PDF at a scale $\mu_0$ to a different scale $\mu$. PDF fitting groups generally start their PDFs at a scale of $\mu_0^2 \approx 1$–$4\ GeV^2$ and evolve them from there. The evolution of the parton distribution functions with the factorization scale is illustrated in Figure 8 for two representative values of $Q^2$ [10]. The PDF fits were extracted using data from Figure 5, among other sources.




(a) $Q^2 = 10\ GeV^2$ (b) $Q^2 = 10\ TeV^2$

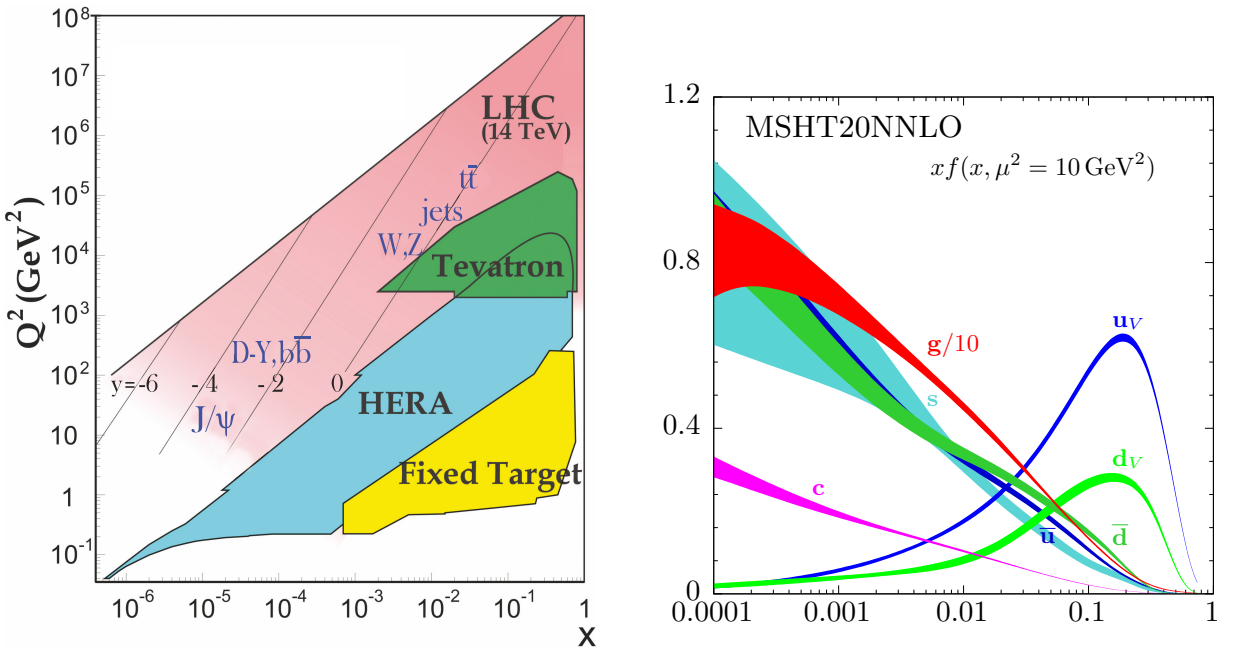
Figure 8: Proton PDFs at two factorization scales, showing the scale evolution predicted by the HERAPDF analysis. Figure from H1 and ZEUS Collaborations, arXiv:1506.06042v3 [10].

Similarly to DGLAP evolution, the Balitsky-Fadin-Kuraev-Lipatov (BFKL) equations describe the evolution of parton densities at small $x$ [14, 15].

In general, many different processes and experiments contribute to global PDF analyses, and the neutral-current DIS example outlined above is merely illustrative. The broad kinematic reach of the data used in global PDF analyses is shown in the left panel of Figure 9 [5], which includes data from fixed-target experiments, $e^- p$ collisions at HERA, and $pp$ ($p\bar{p}$) collisions at the LHC and Tevatron.

The right panel of Figure 9 displays another example of a PDF fit. Note that the valence and sea quarks are separate in the fit. The valence quarks tend to carry a significant fraction of the parton's momentum and are peaked at $x = 0.2$, while the contribution from sea quarks and gluons rises at low $x$. In particular, this low-$x$ region is very interesting because of the behavior of the gluon PDF, which rises immensely due to soft-gluon radiation. This rise at low $x$ is more prominent at high $Q^2$, where the higher-resolution probe can distinguish increasingly soft gluons. By unitarity constraints,

this gluon splitting cannot continue indefinitely as $x$ decreases. At sufficiently small $x$, the gluon number density must decrease due to the recombination of these soft gluons in a phenomenon called *gluon saturation* [16].




(a) Kinematic reach of fixed-target and collider experiments.

(b) NNLO MSHT20 PDFs at $Q^2 = 10^2\ GeV^2$.

Figure 9: Examples of the experimental kinematic coverage and global PDF results. The right panel shows the parton distribution functions obtained from the NNLO MSHT20 global analysis at $Q^2 = 10^2\ GeV^2$. Figure from the Reference [5].

**Nuclear PDFs** The partonic content of protons and neutrons (nucleons) is modified when they are bound into nuclei. This modification is typically quantified through a nuclear modification factor, which compares the parton distribution function in a nucleus with that in a free nucleon. The nuclear modification factor is defined as

$$R_i^A(x, Q^2) = \frac{f_i^A(x, Q^2)}{f_i(x, Q^2)}, \tag{13}$$

where $f_i^A$ is the PDF per nucleon in a nucleus with mass number $A$, and $f_i$ is the corresponding PDF in a free nucleon. Values of $R_i^A$ below or above unity indicate

suppression or enhancement, respectively, relative to the free-nucleon reference [17]. Figure 10 shows an illustration of the general shape of a nuclear modification factor, which can be decomposed into four main regions.

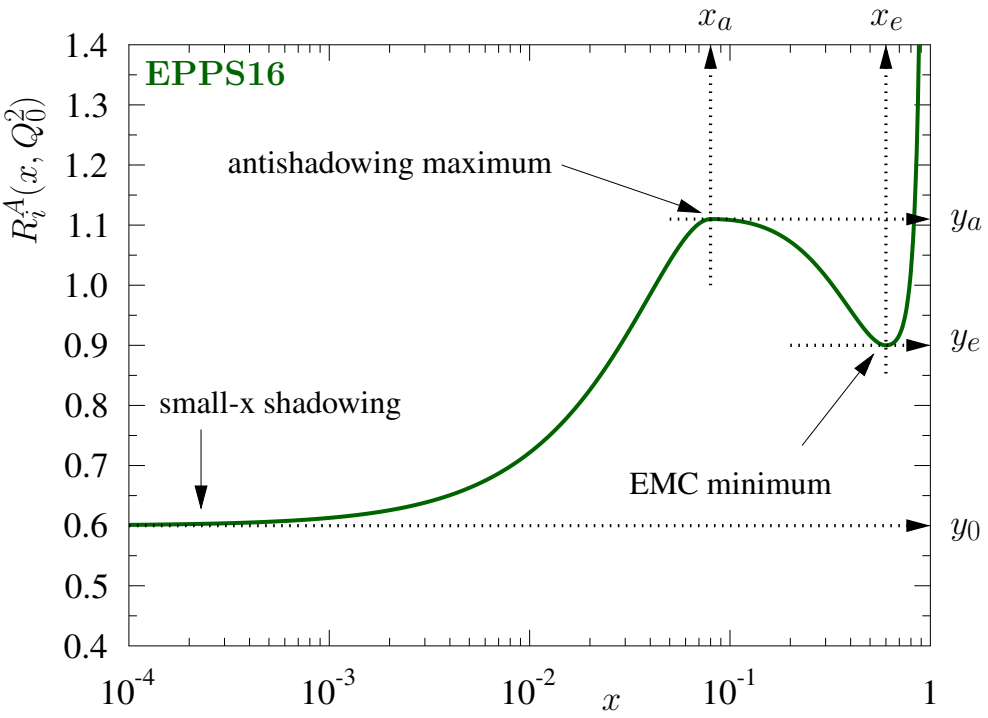



Figure 10: Illustration of the EPPS16 $R_i^A(x, Q^2)$ fit function. Figure taken from Reference [17].

1. **Shadowing:** At low $x$, the density of very soft partons decreases in the nuclear environment because of a crowding effect from surrounding nucleons [18].

2. **Antishadowing:** At more moderate $x$, the parton density is enhanced as partons redistributed from the shadowing region migrate to higher $x$ [18].

3. **EMC effect:** Measured by the European Muon Collaboration [19], this modification is not yet fully understood but is believed to be caused by short-range correlations between nucleon pairs [20].

4. **Fermi motion:** The quantum-mechanical motion of nucleons bound in the nucleus

produces an enhancement at large $x$ [21].

In general, nPDFs are less well determined than their free-proton counterparts, especially for nuclear species for which experimental data are sparse.

**Fragmentation Functions** Fragmentation functions are the final-state analogues of PDFs and are written as $D_i^h(z, \mu_F^2)$. They were introduced formally in Section 2.1.5 as number-density functions representing the expected number of final-state hadrons carrying a momentum fraction $z$ of an outgoing parton. Like PDFs, they are nonperturbative objects that must be determined from experiment. A clean probe of fragmentation functions comes from the $e^+e^- \to h + X$ process, since there is no hadronic initial state and hence no PDF dependence. The total cross section for this process can be used to determine the fragmentation functions in a manner similar to that used to determine structure functions in DIS. The fragmentation functions can then be extracted from these measurements [5].

### 2.2.2 Generalized Parton Distributions

We have discussed already how PDFs encode the partonic content of hadrons as a function of the total *longitudinal* momentum fraction of the hadron. Further properties, such as the spatial distribution of partons, can also be of interest.

Electromagnetic form factors (FFs) describe the effective charge ($F_1(Q^2)$) and magnetic moment ($F_2(Q^2)$)[2] of a hadron as a function of the momentum transfer $Q^2$. For a proton, these factors are equivalent to the Mott-scattering limit as $Q^2 \to 0$, where $F_1(0) = 1$ and $F_2(0) = \kappa_p$[3] [22]. As $t$ increases, the FFs decrease as the scattering resolves further details of the charge and magnetic moments within the proton.

Next, we consider elastic scattering of a virtual photon off a proton. Working in the

[2]Despite the similar notation, these electromagnetic FFs differ from the structure functions introduced earlier.

[3]$\kappa_p$ is the anomalous magnetic moment of the proton.

light-front formalism, the momentum transfer ($Q^2$) lies in the transverse plane of the collision, so $Q^2 = -q_\perp^2$. In this setting, a charge density can be extracted for the proton using the $F_1(Q^2)$ form factor through a Fourier transform from momentum to position space, as shown in Equation 14 [23].

$$\rho^p(\mathbf{b}_\perp) = \int \frac{d^2\mathbf{q}_\perp}{(2\pi)^2} \exp\left(-i\mathbf{b}_\perp \cdot \mathbf{q}_\perp\right) F(q^2) \tag{14}$$

Deeply virtual Compton scattering (DVCS) is a type of deep ($Q^2 \gg M^2$) exclusive scattering (DES) in which a photon or meson is electroproduced by scattering from a nucleon.

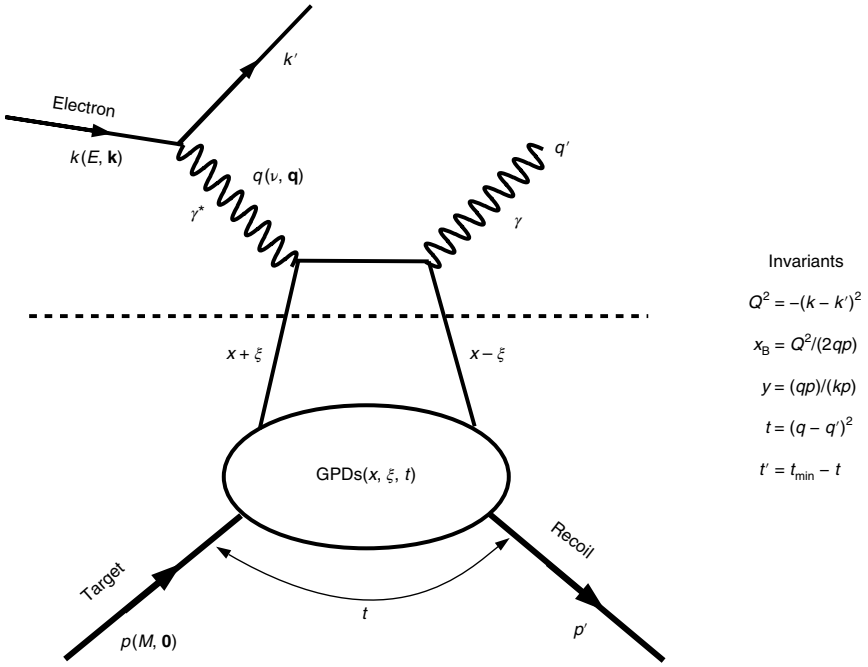



Figure 11: Handbag representation of deeply virtual Compton scattering, in which a virtual photon scatters from a parton in a nucleon and a real photon is produced. The process is described by generalized parton distributions [24].

As shown in Figure 11, this process has a factorized structure described by Generalized Parton Distributions (GPDs). GPDs are powerful tools because they simultaneously access both the longitudinal momentum and transverse spatial

distributions of partons within a hadron, unifying FFs and PDFs in one theoretical framework. There are four GPDs for each quark and gluon, corresponding to the four independent helicity-spin combinations of the quark–nucleon system. This discussion will focus on the GPD corresponding to an average over quark helicities and an unflipped nucleon spin, written as $H(x, \xi, t)$ [25]. The three kinematic variables that parameterize GPDs are $x$, the average longitudinal momentum fraction of the struck quark (i.e., Bjorken $x$)[4]; $x + \xi$ ($x - \xi$), the longitudinal momentum fraction of the struck quark before (after) the scattering; and $t$, the squared four-momentum transfer given to the nucleon [24]. The quark PDFs and FFs can both be derived from the GPDs through the relations given in Equation 15 [23, 25]

$$H^q(x,0,0) = \begin{cases} f_q(x) & x > 0 \\ -f_q(-x) & x < 0 \end{cases} \qquad \int_{-1}^{1} dx H^q(x,0,t) = F_1^q(t) \tag{15}$$

Experimentally, GPDs can be determined by measuring DVCS rates, such as those measured at Jefferson Lab [24]. GPDs provide a richer description of nucleon structure than PDFs or FFs alone and can be used to understand parton dynamics in hadron collisions more closely.

### 2.2.3 Color Transparency

*Color transparency* (CT) is the tendency for a hadron in a configuration with a very high-$x$ parton to have a spatially compact transverse density [23]. The remnants of the struck hadron can fluctuate into a reduced-size, color-neutral configuration [26]. CT can be considered a QCD analogue of the charge-transparency phenomenon in QED [27], but this comparison is complicated by the non-Abelian nature of QCD. The dynamics underlying CT are closely related to GPDs, because GPDs encode both the transverse

[4]In the language of GPDs, $x$ can be negative. A negative $x$ corresponds to an antiquark with momentum fraction $|x|$.

spatial density and the longitudinal momentum configurations of partons within hadrons.

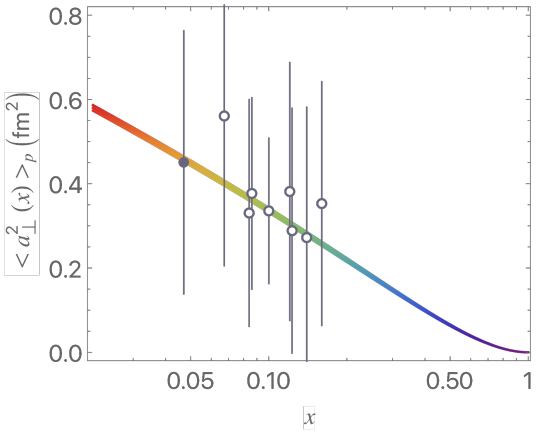
Consider a scattering with one struck parton in a proton and $n-1$ spectator partons. The transverse coordinate $\mathbf{a}_\perp = \sum_{j=1}^{n-1} x_j \mathbf{b}_{\perp,j}$ represents the $x$-weighted transverse position of all the spectator partons. Similarly to Equation 14, a Fourier transform of the GPD gives direct access to a *transverse-distance-dependent PDF*, $f_q(x, \mathbf{a}_\perp)$, shown in Equation 16.

$$f_q(x, \mathbf{a}_\perp) = \int \frac{d^2\mathbf{q}_\perp}{(2\pi)^2} \exp\left(-i\mathbf{a}_\perp \cdot \mathbf{q}_\perp\right) H^q(x, 0, \mathbf{q}_\perp) \tag{16}$$

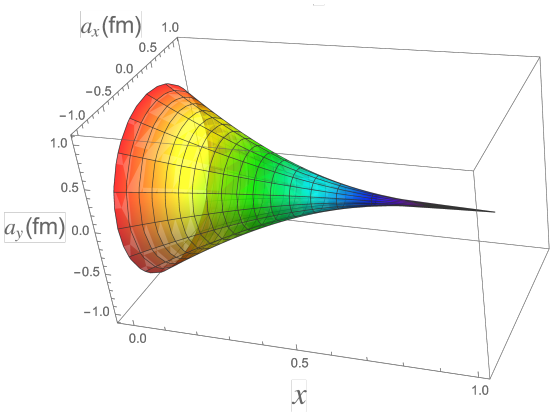
Using this distribution, it is possible to derive the squared transverse spatial distance of the spectator partons as a function of the $x$ of the struck parton using Equation 17 [23].

$$\langle \mathbf{a}_\perp^2(x) \rangle_q = \frac{\int d^2\mathbf{a}_\perp \mathbf{a}_\perp^2 f_q(x, \mathbf{a}_\perp)}{\int d^2\mathbf{a}_\perp f_q(x, \mathbf{a}_\perp)} \tag{17}$$

CT effects become relevant as the $x$ of the struck quark approaches unity. According to the PDFs, the high-$x$ struck quark is very likely to be a valence quark. Therefore, considering only the valence quarks, the transverse radius of the proton can be written as a charge-weighted sum, $\langle \mathbf{a}_\perp^2(x) \rangle_p = 2e_u \langle \mathbf{a}_\perp^2(x) \rangle_u + e_d \langle \mathbf{a}_\perp^2(x) \rangle_d$. Figure 12 shows a prediction for $\langle \mathbf{a}_\perp^2(x) \rangle_p$ that assumes a phenomenological functional form for how the GPD encodes the spatial distribution of the spectator quarks as a function of $x$. The prediction is compared with data from the CLAS and HERMES experiments and shows good agreement within the experimental uncertainties.




(a) Model prediction compared with CLAS and HERMES data.




(b) Color-transparency behavior at large light-front momentum fraction.

Figure 12: The $x$ dependence of the proton transverse size, $\langle \mathbf{a}_\perp^2(x) \rangle_p$, determined from the hadron profile function. The left panel compares the model with CLAS and HERMES data. The right panel shows an illistraion of the CT phenomenon. At large $x$, the transverse size approaches that of a point-like color-singlet configuration. Figure taken from Reference [23].

The CT phenomenon has also been theoretically framed in terms of color fluctuations (CF) [28–30], which parameterize the reduced interaction strength of the struck-hadron remnant after the high-$x$ parton has been removed. Whichever term is preferred, this phenomenon is particularly relevant for proton–nucleus collisions. When the projectile proton undergoes a hard scattering and enters a spatially compact state, the probability of interacting with other nucleons in the nucleus decreases. This is schematically shown in Figure 13.

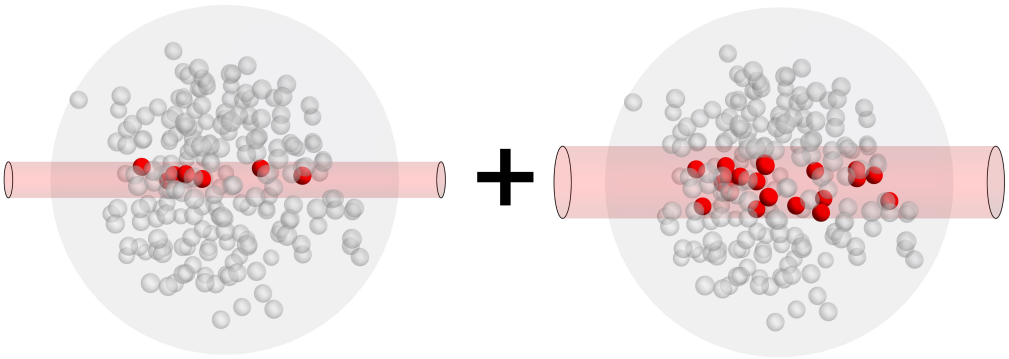


Figure 13: The same target geometry is shown for a compact proton configuration with a smaller interaction probability and for a larger configuration that interacts with more nucleons. The red tube indicates the projectile's transverse extent through the nucleus, while red nucleons mark interactions. Taken from Reference [30].

### 2.3 Jets

Precise tests of QCD are challenging because the hard-scattered partons hadronize and fragment into many different hadrons. To accurately test pQCD predictions, each final-state hadron would have to be associated with a parent parton from the hard scattering, which is challenging experimentally. Fortunately, in sufficiently hard processes, the hard-scattered partons are sufficiently boosted in the laboratory frame that the characteristic emissions and splittings during hadronization occur at relatively small angles. The consequence of this boost is that the final-state hadrons tend to form highly collimated sprays of particles known as *jets*. At leading order, jet production in $l^+l^-$ annihilation and in hard parton–parton scattering produces two outgoing partons. In the transverse plane, momentum conservation makes these partons approximately back-to-back in azimuth. The resulting jets are observed in back-to-back pairs called *dijets*. Jets were first observed at SLAC in $e^+e^-$ collisions [31]. Since jets are complex multi-object observables, they must be defined precisely, both theoretically and experimentally [32].

**Requirements for Jets** In order to be calculable to next-to-leading order (NLO) and beyond, jets must be both infrared and collinear (IRC) safe [5]. The infrared condition requires that a parton in the shower emitting a very soft ($z \to 0$) parton should not change the observable. The collinear condition requires that the splitting of one parton into two collinear partons ($\theta \to 0$) should not change the observable. IRC safety ensures that divergences from the infrared and collinear contributions can cancel and give finite predictions. Beyond being IRC safe, the physics community also defined some proposed standard requirements for jets at the Snowmass conference in 1990 [33]. These are as follows:

1. Simple to implement in experimental analyses
2. Simple to compute theoretically
3. Well defined at any order of perturbation theory
4. Yield finite cross sections at all orders of perturbation theory
5. Yield a cross section mostly insensitive to hadronization effects

As the field of jet physics has matured, the community has converged on a set of standard practices for defining and measuring jets.

**Jet Kinematics** Like any physics object, the kinematics of a jet can be described by a four-vector. The most commonly used set of variables to describe a jet is $(m, p_{\mathrm{T}}, y, \phi)$, where $m$ is the jet mass, $p_{\mathrm{T}}$ is the transverse momentum, $y$ is the rapidity, and $\phi$ is the azimuthal angle around the beam axis. The jet variables can be obtained from the energy-momentum four-vector $\left(E, p_x, p_y, p_z\right)$ through Equation 18[5]

$$m = \sqrt{E^2 - p^2}, \quad p_{\mathrm{T}} = \sqrt{p_x^2 + p_y^2}, \quad y = \frac{1}{2} \ln \left( \frac{E + p_z}{E - p_z} \right), \quad \phi = \operatorname{atan2}\left(p_y, p_x\right) \tag{18}$$

[5]Here, we assume that the beam is oriented along the $z$ axis.

For relativistic ($E \approx p$) jets, the rapidity can be approximated by the pseudorapidity ($\eta = -\ln\tan(\theta/2)$). Here, $\theta$ represents the polar angle with respect to the beamline. Working with $\phi$ and $\eta$ is convenient because they map straightforwardly onto a detector apparatus that measures angles relative to an interaction point. Values of $|\eta|$ close to zero are typically referred to as "mid-rapidity", while large values of $|\eta|$ are referred to as "forward". Figure 14 shows an example of a dijet event recorded by the ATLAS detector, with the kinematics labeled.

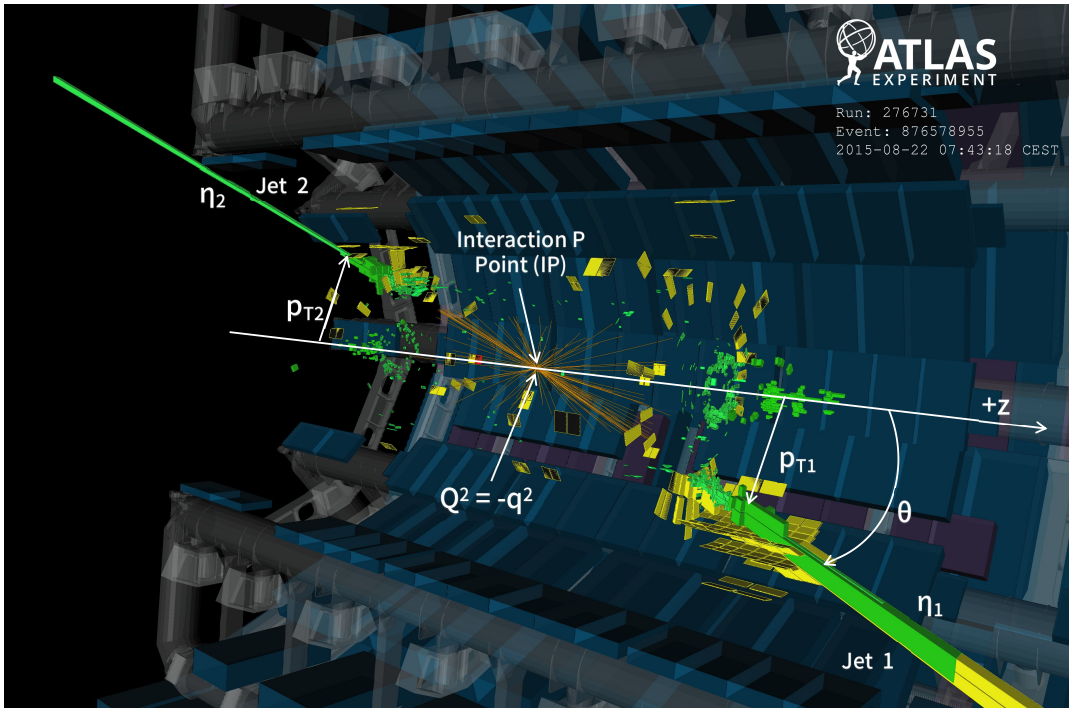



Figure 14: Example of a $pp$ dijet event recorded by the ATLAS detector. The two leading jets are visible as prominent, approximately back-to-back deposits of energy in the detector. Event display provided by the ATLAS experiment [34]. The labeling on the diagram is courtesy of Y. Kulinich.

### 2.3.1 Jet Clustering Algorithms

A jet is a complex final-state object, and the question of exactly how these sprays of hadrons are grouped into a single jet requires a precise answer. First, let us define the spray of hadrons more generally as *jet constituents*, since an experiment may not always

be able to measure each hadron directly and may instead rely on experimental proxies, which are discussed later. Jet clustering algorithms determine how constituents are grouped into jets and can generally be divided into two main classes: cone "top-down" algorithms and sequential recombination "bottom-up" algorithms [35]. These algorithms use information about the angular positions $(\eta - \phi)$ and momenta $(p_{\mathrm{T}})$ of constituents to form jets. The distance between constituents, expressed in units of angular distance, $\Delta R_{ij}^2 = \left(\eta_i - \eta_j\right)^2 + \left(\phi_i - \phi_j\right)^2$, is also highly relevant. Algorithms also rely on *recombination schemes*, which dictate how constituent kinematics are combined to form the jet kinematics. The most common and simplest recombination scheme is the "$E$-scheme", which sums the four-vectors of the constituents.

Cone algorithms are typically iterative: a jet captures all constituents within a radius $R$ of a seed constituent. The constituent momenta are combined to determine a net direction, toward which the seed is moved. This process is repeated until the jet direction stabilizes. One drawback of cone algorithms is that they tended to have problems with IRC safety. The "midpoint algorithm" was developed as a patch for cone algorithms and saw widespread use during Run II of the Tevatron [35]. A solution to the IRC-safety problem is a *seedless cone* algorithm, which avoids seeds and iterations and instead finds all stable cones. The `SISCone` procedure provides a computationally efficient implementation of this approach [36].

Sequential recombination jet algorithms do away with the top-down approach of assigning particles to a cone of radius $R$ and instead follow simple pairwise rules for combining constituents. These algorithms define the following quantities:

$$d_{ij} = \min\left(p_{\mathrm{T}i}^{2t}, p_{\mathrm{T}j}^{2t}\right)\frac{\Delta R_{ij}^2}{R^2} \qquad d_{iB} = p_{\mathrm{T}i}^{2t} \tag{19}$$

Here, $t$ is a parameter that specifies the algorithm variant, and $R$ is a radius parameter. The algorithms are defined as follows:

1. Find all $d_{ij}$ and $d_{iB}$ values for the constituents in the event.

2. Order the $d_{ij}$ and $d_{iB}$ values and find the smallest.

3. If the minimum is a $d_{ij}$, recombine $i$ and $j$ into a single new constituent and return to step one.

4. If the minimum is a $d_{iB}$, declare constituent $i$ to be a jet, remove it from the list, and return to step one.

5. Stop when no constituents remain in the list.

The procedure continues until all constituents have been assigned to final-state jets. Because a very soft, isolated constituent can be classified as a jet in this framework, a minimum transverse momentum is typically specified for jets of interest. The $k_t$ algorithm corresponds to $t = 1$ and clusters soft constituents first [37]. The `Cambridge/Aachen` algorithm corresponds to $t = 0$ and clusters constituents in an energy-independent way, using only their angular proximity [38]. Both the $k_t$ and `Cambridge/Aachen` algorithms cluster constituents in a way that can lead to non-conical jet shapes in $\eta - \phi$ space. Finally, the anti-$k_t$ algorithm corresponds to $t = -1$ and clusters the hardest constituents first [39]. This makes the jets grow outward from hard "seeds", reminiscent of cone algorithms but without the IRC-safety concerns. The anti-$k_t$ algorithm with $R = 0.4$ and $E$-scheme recombination is the standard for jet physics at the LHC, including the two measurments presented in this dissertation[6].

Figure 15 displays an example of different jet clustering algorithms each applied to the same event. The `FastJet` package includes a publicly avalible, highly optimized version of jet clustering algoritms [40].

[6]The choice of anti-$k_t$ with $E$-scheme recombination is heavily favored in the field, however, many interesting results have come from varying the $R = 0.4$ choice.

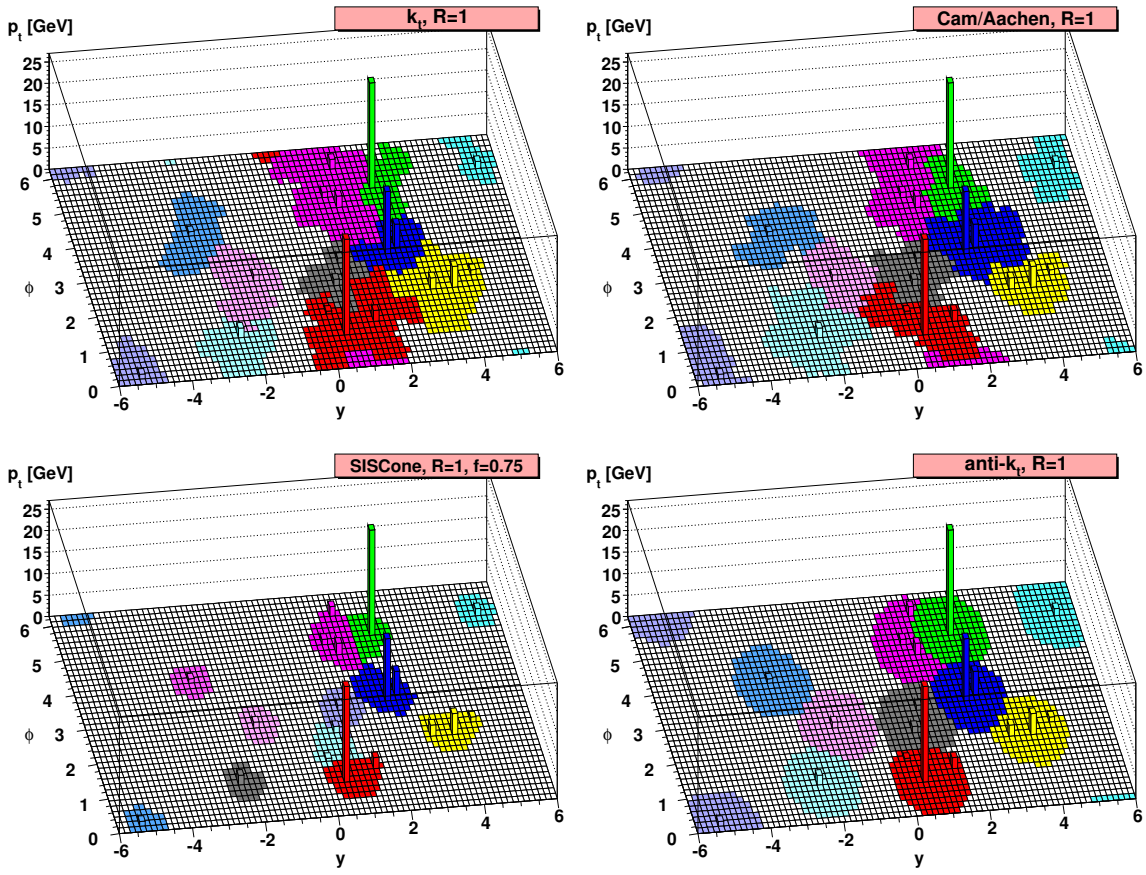



Figure 15: [35].

### 2.3.2 Jet constituents

As previously mentioned, jets are reconstructed using underlying objects called jet constituents. From a theortical perpective jets are simply clusters of hadrons produced in the collision, each with an exact four-momentum that can be calculated and input into the `FastJet` package. From an experimental perspective, the picture is much more complicated. Below we will discuss several types of jets measured by experiments and their positives and negatives.

**Calorimeter Jets** Calorimeter jets are reconstructed from energy deposits measured in the calorimeter, commonly grouped into topological clusters or towers and represented by four-vectors. They provide sensitivity to both charged and neutral particles, although the measured jet energy must be calibrated to account for detector response and energy losses [41].

**Track Jets** Track jets are reconstructed by clustering charged-particle tracks measured in the inner tracking detector. Their directions and momenta are measured with high precision, but they do not include neutral particles and are limited by the tracking detector's geometric acceptance and track-reconstruction thresholds.

**Particle Flow Jets** Particle-flow jets are built from particle-flow objects that combine tracking and calorimeter information to estimate the momenta of individual particles. Using tracks for charged particles and calorimeter clusters for neutral particles helps avoid double-counting charged-particle energy deposits while retaining sensitivity to both charged and neutral components.

### 2.3.3 Factorization of Jet Observables

Similar to Equation 8 jets observables can also be factorized into nonperturbative and perterbative parts [11]. Here the PDF and hard scattering element stay the same. In the

ideal case where the jet, captures all of the fragmented hadrons from the parton leaving the hard scattering, the factorization is complete. However, real jets do not always capture the product of the scattering as it is possible for a realivly hard radiaion to push some jet energy outside of the perscribed cone. To account for these real-world effects, perturbativly calculable semi-inclusive jet functions ($J_i\left(z, R, \mu_R\right)$) take the place of fragmentation functions in Equation 8 to account for effects on the jets [42]. It parameterizes the likleihood for a jet with a certain clustering algoirthm with radius $R$ to be produced in the final-state carrying a fraction of the energetic partons momentum $z$.

$$\begin{aligned} \mathrm{d}\sigma_{pp\to jj} \approx \sum_{ijkl} \int \mathrm{d}x_i \mathrm{d}x_j \mathrm{d}z_k \, \mathrm{d}z_l \, f_i\left(x_i, \mu_F^2\right) f_j\left(x_j, \mu_F^2\right) \\ \otimes \, \mathrm{d}\hat{\sigma}_{ij\to kl}\left(x_i, x_j, \alpha_s\left(\mu_R\right), Q^2\right) \\ \otimes \, J_k\left(z_k, R, \mu_R^2\right) \otimes J_l\left(z_l, R, \mu_R^2\right) \end{aligned} \tag{20}$$

### 2.4 Relativistic Ion Collisions

Relativistic ion collisions provide experimentalists with a new laboratory for studying QCD. There are two main classes of collisions that can occur and they are differentiated by the impact parameter ($b$) between the colliding nucleons. Ultra-peripheral collisions (UPCs) occur when $b > 2R_A$. Even though the interacting nuclei do not overlap, a large flux of nearly real photons is present between them. These photons can interact with each other in light-by-light ($\gamma\gamma$) scattering or with the other nucleus in photonuclear ($\gamma A$) processes. Hadronic collisions, on the other hand, occur when $b < 2R_A$. These collisions are characterized by high multiplicities in the final state. Hadronic collisions can have sufficient energy and density to briefly form a state of deconfined partonic matter called the Quark-Gluon Plasma (QGP); see Section 2.5. Collisions across these regimes can produce final states ranging from very sparse to high multiplicity. Experimentally, measuring these collisions is challenging because the environment can be

extremely sparse in UPCs or extremely complex in hadronic collisions; both regimes push the limits of detectors. Since this dissertation documents measurements of observables in hadronic nuclear collisions, this section focuses on their properties. UPCs enter as background to the presented measurements, and steps are taken to remove them from the analysis samples.

### 2.4.1 Geometry of Ion Collisions and the Glauber Model

In a hadronic relativistic ion collision, the two colliding nuclei overlap with some impact parameter $b$. For simplicity, consider modeling the ions as spheres that are Lorentz-contracted along the beam axis. In this picture, the ions resemble two flat discs about to collide head-on; see the left panel of Figure 16.

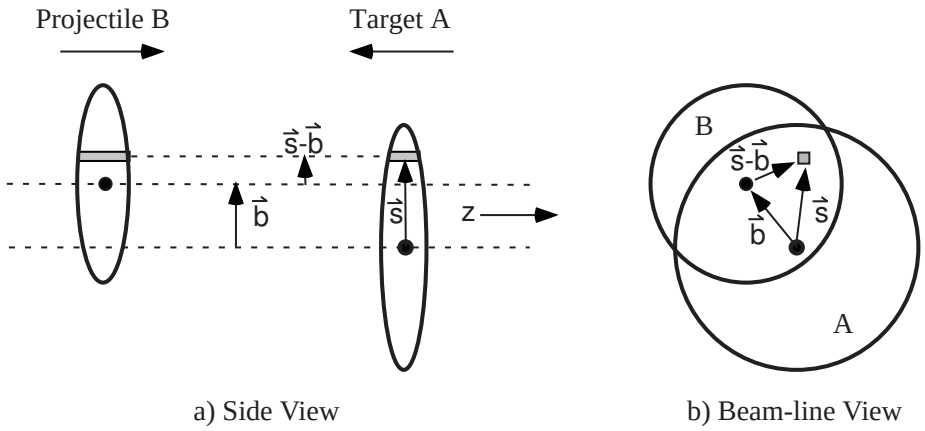



Figure 16: Schematic illustration of the ion collision geometry, reproduced from Ref. [43].

In this picture, a lower $b$ leads to greater overlap between the two ions. The greater overlap leads to more nucleon–nucleon interactions and, on average, more final-state particles in the detector. Figure 17 compares an O+O collision at $b = 2$ fm with a Pb+Pb collision at $b = 8$ fm.

This increase in event activity with the initial-state parton flux can be parameterized by the *Glauber model* [43]. In its simplest form, the model treats a collision of ions as a superposition of many individual nucleon–nucleon collisions. Physicists are typically

interested in how observables differ from this naive assumption, which provides information about the physics of ion collisions. The nuclear charge density is typically modeled by a Woods-Saxon distribution:

$$\rho(r) = \rho_0 \frac{1}{1 + \exp\left(\frac{r-R}{a}\right)} \tag{21}$$

where $\rho_0$ is the central density, $R$ is the nuclear radius, and $a$ is the skin depth. For the $^{208}$Pb ion used in this dissertation, the values are $R = 6.62 \pm 0.06$ fm and $a = 0.546 \pm 0.01$ fm [44]. For the $^{16}$O and $^{20}$Ne species used in this dissertation, more sophisticated density profiles derived from dedicated quantum Monte Carlo simulations account for correlations between nucleons in small and deformed nuclei [45]. From the density profiles, one can construct a probability distribution for finding a nucleon at a transverse radius $s$ using Equation 22.

$$\hat{T}_A(s) = \int \hat{\rho}_A(s, z_A)\, dz_A \tag{22}$$

Here, $\hat{\rho}$ is the normalized version of the density in Equation 21. Using these probability distributions, one can construct a joint probability for two nuclear species, A and B. Integrating over this joint probability gives the "thickness function" shown in Equation 23.

$$\hat{T}_{A,B}(b) = \int \hat{T}_A(s)\hat{T}_B(s-b)d^2s \tag{23}$$

$\hat{T}_{A,B}(b)$ has units of inverse area. Thus, the probability for a nucleon in ion A to interact with a nucleon in ion B is $\hat{T}_{A,B}(b)\sigma_{\text{inel}}^{\text{NN}}$, where $\sigma_{\text{inel}}^{\text{NN}}$ is the inelastic nucleon–nucleon cross section. Given A nucleons in ion A and B nucleons in ion B, the average number of nucleon–nucleon collisions is given by Equation 24.

$$N_{\text{coll}}(b) = AB\hat{T}_{A,B}(b)\sigma_{\text{inel}}^{\text{NN}} \tag{24}$$

The total number of nucleons participating in the collision is given by Equation 25.

$$N_{\text{part}}(b) = A \int \hat{T}_A(s) \left\{ 1 - \left[ 1 - \hat{T}_B(s-b)\sigma_{\text{inel}}^{\text{NN}} \right]^B \right\} d^2s + B \int \hat{T}_B(s-b) \left\{ 1 - \left[ 1 - \hat{T}_A(s)\sigma_{\text{inel}}^{\text{NN}} \right]^B \right\} d^2s \tag{25}$$

Everything discussed so far has implicitly used the "optical form" of Glauber theory, which models the nucleus as a sphere with a continuous nucleon density. This approach allows analytical formulas for $N_{\text{coll}}$ and $N_{\text{part}}$, but it does not describe all features of ion collisions. The modern approach uses Glauber Monte Carlo. In this approach, nucleon positions are sampled according to the nuclear density distributions. If the positions of two nucleons are unphysically close (i.e. almost on top of one another) the position for one of them is redrawn. Next, a random impact parameter is drawn, and the colliding nuclei are placed accordingly. The nucleons at their sampled positions in A and B travel along straight-line trajectories. If two nucleons from the colliding ions come closer than $\sqrt{\sigma_{\text{inel}}^{\text{NN}}/\pi}$ along these paths, they are considered to have undergone a nucleon–nucleon collision, and $N_{\text{coll}}$ is incremented by one. The Monte Carlo approach can be thought of as a summation of these individual collisions, where any nucleon that enters one contributes to $N_{\text{part}}$. From many Monte Carlo events, one can build distributions of $N_{\text{coll}}$ and $N_{\text{part}}$, from which the averages $\langle N_{\text{coll}} \rangle$ and $\langle N_{\text{part}} \rangle$ are extracted. The optical and Monte Carlo approaches tend to differ most in situations where fluctuations in nucleon positions are important, such as in simulations of light ions or large-$b$ heavy-ion collisions. Single event Glauber Monte Carlo profiles for O+O and Pb+Pb collisions are shown in Figure 17.

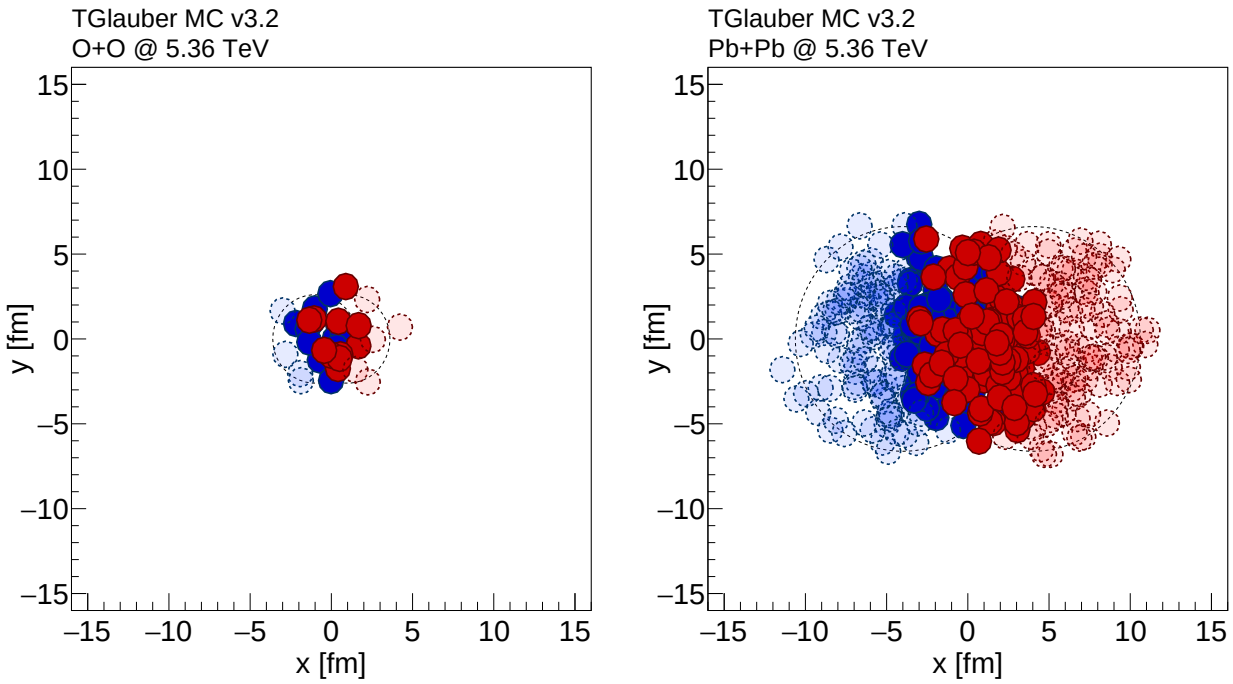



Figure 17: Distributions of nucleons generated with Glauber Monte Carlo in the $x - -y$ transvrse plane. The two nuclei are shown in different colors, with the particiapents shown as darker circules. Results for O+O collisions (left) and Pb+Pb collisions (right) are shown. Figures provided by R. Longo.

Since the impact parameter between colliding nuclei is not experimentally accessible, one needs some type of modeling to connect the impact parameter to something measurable. This is typically done through so-called "global observables" like forward transverse energy or charged particle multiplicity, which are thought to be monotonically related to $b$. The mapping between impact parameter in a Glauber simulation and the global observable is given by a *centrality* calibration; this is why these global observables are also referred to as *centrality estimators* (CE). Additionally, the data used as input to these centrality calibrations consist of minimum-bias inelastic collisions.

The two-component model is used for the measurements presented in this dissertation. This model postulates that the CE in ion collisions scales with the effective number of *production elements*.

$$N_{\text{elem}} = xN_{\text{coll}} + (1-x)N_{\text{part}}/2 \tag{26}$$

where $x \in (0,1)$. First, a distribution of the CE is fit to *pp* data, which represents $N_{\text{elem}} = 1$. Then, the CE distribution in ion collisions can be modeled as an $N_{\text{elem}}$-fold convolution of the *pp* fit, with the distribution of $N_{\text{elem}}$ obtained from the Glauber Monte Carlo. The templated ion Monte Carlo CE distribution is then fit to the corresponding distribution in ion data, which is histogrammed above a low CE threshold. This threshold represents the lowest CE value for which a contribution from UPC events can be neglected, so that the centrality calibration corresponds only to hadronic collisions. The free parameters in this fit are the $x$ in the $N_{\text{elem}}$ extraction, and the specific fit parameters for the single element CE extraction which were initialized on *pp* data. Once the ion Monte Carlo CE fit has converged, typically to a value of $x \approx 0.1$, the fraction of the total hadronic cross section sampled by the experiment, $f_{\text{samp}}$, can be extracted. This is done by looking at the fraction of the ion Monte Carlo CE distribution above the low CE threshold; typical values are $f_{\text{samp}} \approx 0.8\%$. The centrality percentiles can then be extracted from the data by dividing it into $f_{\text{samp}} \times 100$ equal-sized percentile bins. Centrality percentiles are defined such that low percentiles correspond to collisions with large parton flux (low $b$), typically called *central collisions*, while high percentiles correspond to collisions with low parton flux (high $b$), typically called *peripheral collisions*. The most central percentile bin is $[0\%, 1\%]$, corresponding to the highest CE values. The most peripheral percentile bin is $\left[f_{\text{samp}} \times 100 - 1\ \%, f_{\text{samp}} \times 100\ \%\right]$, corresponding to the lowest CE values measured above the low threshold. Once these bins have been defined, the $\langle N_{\text{coll}} \rangle_{\text{bin}}$, $\langle N_{\text{part}} \rangle_{\text{bin}}$, and $\langle T_{A,B} \rangle_{\text{bin}}$ can be extracted per-bin. In this way, subsets of the experimental data can be associated with geometric quantities from Glauber modeling, and observables can be studied as a function of centrality.

### 2.4.2 Control Systems: *pp* and *p*+A collisions

Reletvistic ion collisions are primarily oriented towards the goal of studying the QGP. To isolate the effects of the QGP, it is often the case than an obervable is measured in A+A

collisions and also in a control system.

It is an open question if any (or how much) QGP is produced in control systems. This topic will be discussed more later in Sction 2.6. For this broad introduction, it is sufficient to note that any QGP formed in $pp$ or $p$+A collisions is expected to be small enough that its effects on many final-state observables are limited. These observables can often be described well by theoretical models that do not include QGP effects. The above statement is often not true for A+A collisions, which is why we can say that $pp$ and $p$+A systems can be used as control systems. Additinally, systems colliding a very large and very small ion, like $d$+Au collisions at the Reletivistic Heavy Ion Collider (RHIC), also behave as control systems for the same reason.

$p$+A collisions can be very useful to study nPDF effects coming from the A nucleus. Cross sectional measurments perfomred in $p$+A collisions can directly constrain the nPDFs. These constraints improve predictions for A+A measurements and help disentangle QGP effects from potentially substantial nPDF effects.

### 2.4.3 Centrality biases from color transparency in $p$+A collisions

Centrality can be defined in $p$+A collisions in a similar way to A+A collisions. As the projectile (proton) impacts the target (ion) at different impact parameters it will on avergae encounter fewer target nucleons at high-$b$ than it will at low-$b$. This creates the normal scaling we expect, where the CE scales invserly with the impact paramter. A key feature of $p$+A collisions is the asymmetry of the incoming system: a single proton collides with a nucleus. As the proton traverses the nucleus, successive nucleon–nucleon interactions can reduce its available energy. Meanwhile, target nucleons that have not yet interacted retain their beam energy, contributing to a net energy-flow imbalance toward the target direction. This means that measurments of CE are typically restricted to the nucleus-going direction, as the proton-going energy flow is less corralated with $N_{\text{elem}}$.

The Glauber Monte Carlo can be run for $p$+A in much the same way it was for A+A.

Here the relationship between $N_{\text{coll}}$ and $N_{\text{part}}$ is trivial $N_{\text{coll}} = N_{\text{part}} - 1$.

However, impact parameter is not the only variable that impacts the $N_{\text{coll}}$. As discussed in Section 2.2.3, color transparency effects impact the interaction strength of the proton, leading to fewer wounded nucleons in the same target nucleus, all else being equal, as shown in Figure 13. The impact on $N_{\text{coll}}$ is exactly the same as if the event was more periphiphral than it really is. This leads to a systematic bias, when events are partitioned in both centrality and some quantity that can be mapped onto the Bjorken-$x$ of the proton[7].

Introduce Aric measurment here or before paper?

### 2.4.4 Luminosity

Placeholder text for Luminosity

## 2.5 The Quark-Gluon Plasma

The prior discussion of QCD involved hadrons in a vacuum at zero temperature; however, studying strong dynamics at nonzero (high) temperatures or baryon densities provides a novel environment for testing QCD. This QCD environment is very rare in nature and is found in only two places [46]. The first is in the interior of neutron stars, where the baryon density greatly exceeds normal nucleon densities. The second is around $10^{-5}$ s after the Big Bang, when the temperature of the entire universe was very large, comparable to the nucleon rest mass. Additionally, this high-temperature regime can be synthesized in a laboratory with collisions of relativistic heavy ions.

Models have long shown that nuclear matter is influenced by its thermodynamic environment. Hagedorn's Bootstrap model [47], proposed before the acceptance of the parton model, posits that the density of available hadronic states increases exponentially with the temperature of a system. The result is that, at a given temperature $T_H$, the

[7] For example jet $p_{\text{T}}$ and $\eta$.

partition function for the hadronic states diverges, and any additional energy input into the system will create additional hadrons but will not increase the temperature. The Bootstrap model estimated this temperature to be roughly $T_H = 160$ MeV.

With the discovery of asymptotic freedom, it was found that the hadronic couplings become weak at high temperatures, leading to a state of matter in which the degrees of freedom change from hadrons to quarks and gluons. This transition is related to the restoration of chiral symmetry in QCD [48]; in cold hadronic matter, the pions are the Goldstone bosons of a spontaneously broken chiral symmetry. This state of nuclear matter, originally described in Reference [49] as "quark soup", has become known as the Quark-Gluon Plasma (QGP). This led to the reinterpretation of $T_H$ as a phase transition point, i.e., the critical temperature above which quarks are liberated from hadrons [50].

### 2.5.1 Phase Diagram of Nuclear Matter

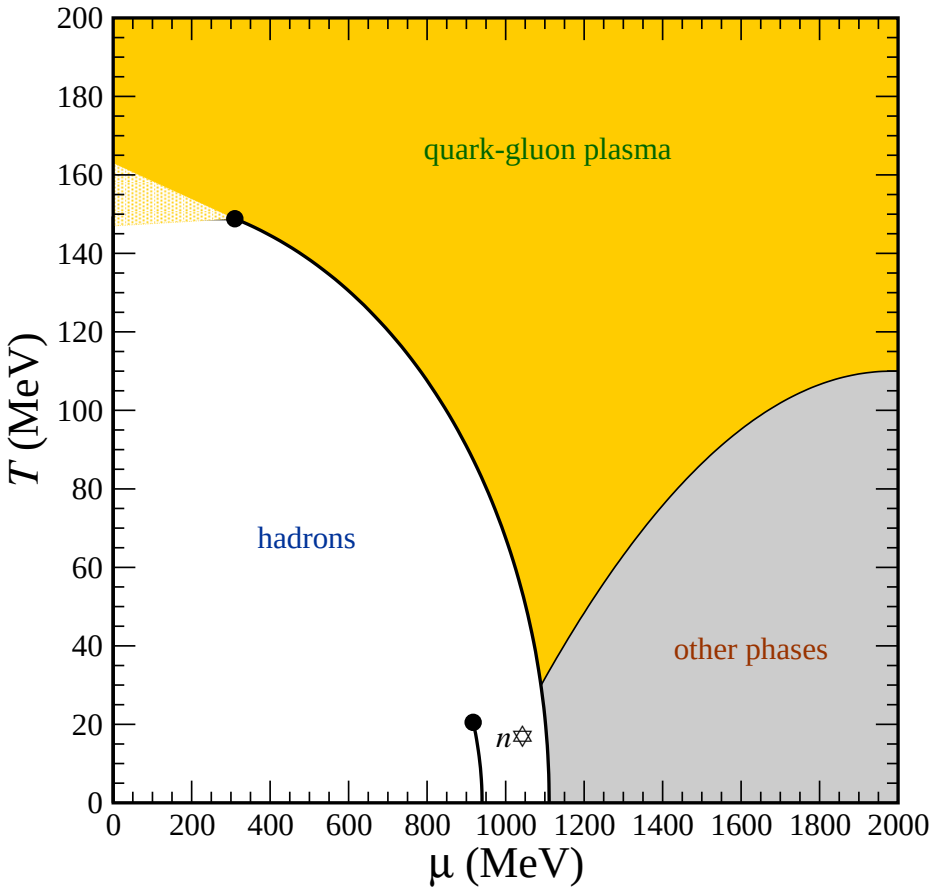



Figure 18: Phase diagram of nuclear matter in the baryon chemical potential ($\mu$) and temperature ($T$) plane. Taken from Ref. [51].

The phase diagram for QCD matter is shown in Figure 18. The baryon chemical potential is related to the net baryon density. At low baryon density, the crossover between the hadronic phase and the QGP is a smooth transition. The nature of this transition is related to the nonzero quark masses. As the phase diagram moves towards higher $\mu$, lattice calculations indicate that the transition may become a first-order phase transition [51]. At very high baryon density, exotic states such as a color superconductor are postulated to exist, but this portion of the phase diagram is less well studied [52]. Heavy-ion collisions at increasingly higher energies probe the low-$\mu$, high-$T$ region of the

QGP diagram because the ratio of baryon number to entropy scales inversely with collision energy.

### 2.5.2 Timeline of an ion collision

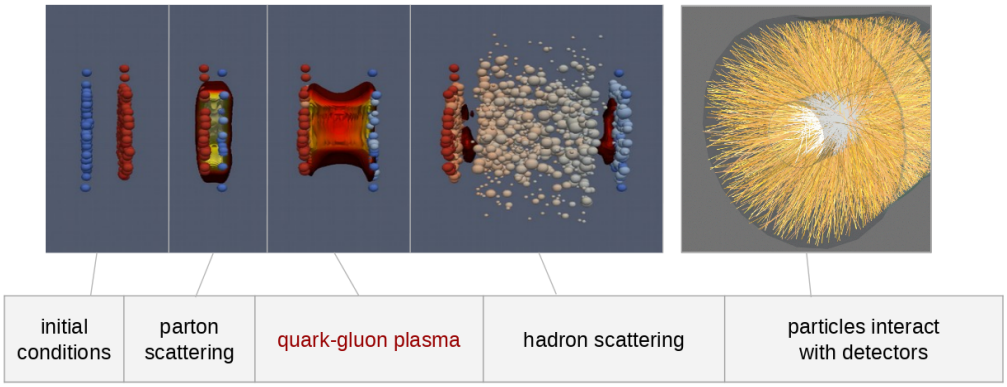



Figure 19: Cartoon illustrating the stages of a heavy-ion collision. Image from Reference [53] originally by H. Petersen and J. Bernhard, MADAI Collaboration.

The QGP exists for only a short time during a heavy-ion collision. To study its properties, it is important to understand how the collision evolves into and out of the QGP state. The different stages of an ion collision are described in Figure 19.

**Initial Conditions** The collision consists of two ions overlapping with some impact parameter $b$, which can be related to the centrality of the collision (as discussed in Section 2.4.1). Centrality affects the transverse volume and shape of the later QGP medium: a central collision has a roughly circular overlap region, while a peripheral collision has a more lenticular overlap region. The partonic content of the interacting nucleons is also modified by nPDFs, as discussed in Section 2.2.1.

**Pre-equilibrium/parton scattering** As the nuclei cross each other at a time of 0 fm/$c$, most nucleon–nucleon collisions are of the low-$Q^2$ variety. Nucleon–nucleon collisions also happen at a much higher $Q^2$, but they are a small fraction of the total

$N_{\text{coll}}$. At $t > 0$ fm/$c$, the two nuclei begin to separate at the speed of light, carrying most of the baryon density away from the collision point. As the two receding nucleons are pulled apart, the *glasma* state forms. It can be described as a "color capacitor plate" in which longitudinal color field lines form between the two ions [54]. The Glasma is not yet an equilibrated QGP, but it is an early state dominated by strong longitudinal color fields that evolve and produce particles.

**QGP formation** Over roughly the first 1 fm/$c$ after the initial collision, the Glasma evolves into the hot, dense matter that forms the QGP. The system hydrodynamizes, meaning that viscous hydrodynamics becomes a useful macroscopic description. The QGP behaves like a nearly perfect fluid, with a small shear-viscosity-to-entropy-density ratio, $\eta/s$ [55]. As time moves forward, the QGP expands hydrodynamically according to the pressure gradients in the system.

**Hadronization and freezeout** The QGP cools as it expands, with the temperature eventually falling below the crossover from QGP to the hadronic phase of matter. During this phase, the unbound, interacting quarks and gluons start to hadronize. At a temperature called the *chemical freezeout temperature*, the partons have mostly become confined into hadrons, and no further inelastic interactions occur between them. At this stage, the number of specific hadron types in the final state is fixed, although they may still elastically interact with each other. At a later stage, kinetic freezeout occurs at the *kinetic freezeout temperature*. At this point, the spacing between the final-state hadrons becomes so large that interactions are negligible. At this stage, the momentum spectrum of the hadrons becomes fixed, and the final-state particles free-stream towards the detector, where they can be observed experimentally.

### 2.5.3 Soft probes of the QGP

Observables related to the low-momentum bulk particles produced in a heavy-ion collision can provide sensitive probes of the nature, size, and evolution of the QGP. Specifically, the azimuthally differential yields of final-state particles, typically referred to as *flow*, are related to the transverse extent of the medium. These yields can be written as a Fourier expansion of the $\frac{\mathrm{d}N}{\mathrm{d}\phi}$ distribution:

$$\frac{\mathrm{d}N}{\mathrm{d}\phi} \propto 1 + 2\sum_{n=1}^{\infty} v_n \cos\left(n(\phi - \Psi_n)\right) \tag{27}$$

where $N$ is the number of final-state hadrons, $\phi$ is the azimuthal angle, $\Psi_n$ is the event-plane angle, and $v_n$ are the $n$th-order flow coefficients. The event-plane angle is an estimate of the direction of maximal particle emission for a specific harmonic $n$. For $n = 2$, the event-plane angle roughly tracks the azimuthal angle of the impact-parameter vector $\vec{b}$. The sum in Equation 27 is over all particles in a given event, though typically only charged particles are available experimentally. The flow coefficients observed experimentally are averages of the coefficients extracted event by event.

For heavy-ion events with non-negligible $b$, the resulting fireball is less circular (left side of Figure 20) and more lenticular (right side of Figure 20). The pressure of the QGP fluid is at a maximum in the center of the medium and is zero in the vacuum outside it. Therefore, pressure gradients will form across the medium, which will be counteracted by the shear viscosity of the medium. This gradient is larger across the shorter axis of the medium than across the longer axis, inducing more rapid hydrodynamic expansion along the shorter axis. After hadronization and freezeout, the result of this anisotropic expansion, called elliptic flow, is encoded in the $v_2$ flow coefficient of Equation 27. Measurements of elliptic flow provide important input into calculations of the shear viscosity of the medium [55]. In fact, early RHIC data seem to favor a very low shear viscosity of the QGP fluid ($\eta/s \approx 0.1$) [56], earning the QGP the name "the most perfect

fluid." Figure 21 displays a measurement of $v_2$ (among other coefficients) as a function of both charged-particle $p_{\mathrm{T}}$ and event centrality. For very central events (small $b$), the medium formed is roughly circular, and therefore there is little azimuthal anisotropy in the pressure gradients, leading to smaller elliptic flow. In more peripheral selections (higher $b$), the fireball takes on a more lenticular shape, which leads to a higher $v_2$.

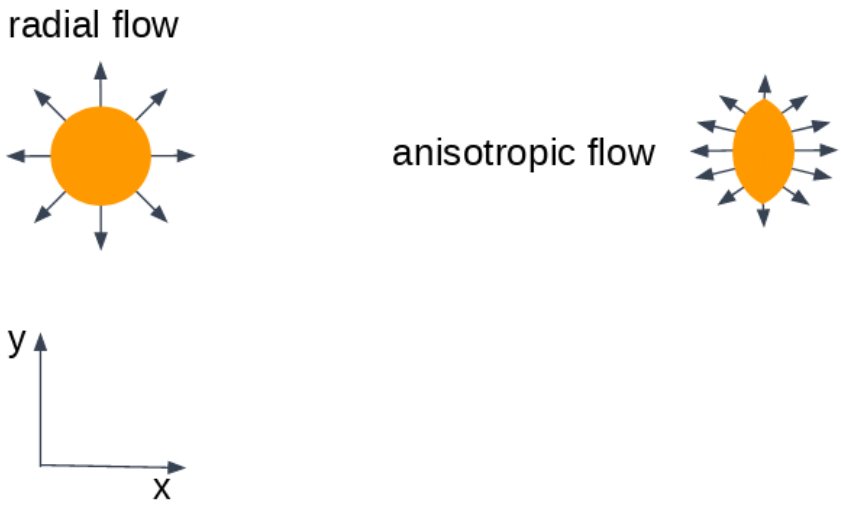



Figure 20: Schematic illustration of particle flow in a heavy-ion collision. Taken from Ref. [53].

The zeroth-order flow coefficient, called radial flow, is not included in the Fourier decomposition of Equation 27 because it represents the collective isotropic expansion and does not vary with $\phi$. The bulk viscosity of the QGP counteracts this radial acceleration; therefore, measurements of $v_0$ can help constrain this quantity [55]. The left side of Figure 20 shows the isotropic expansion of the (orange) medium leading to radial flow. The ATLAS experiment recently measured radial flow directly [57] in collisions with very small $b$, i.e., very central collisions. Anisotropy in the $\frac{\mathrm{d}N}{\mathrm{d}\phi}$ distribution can result from either the geometry or fluctuations in the initial state.

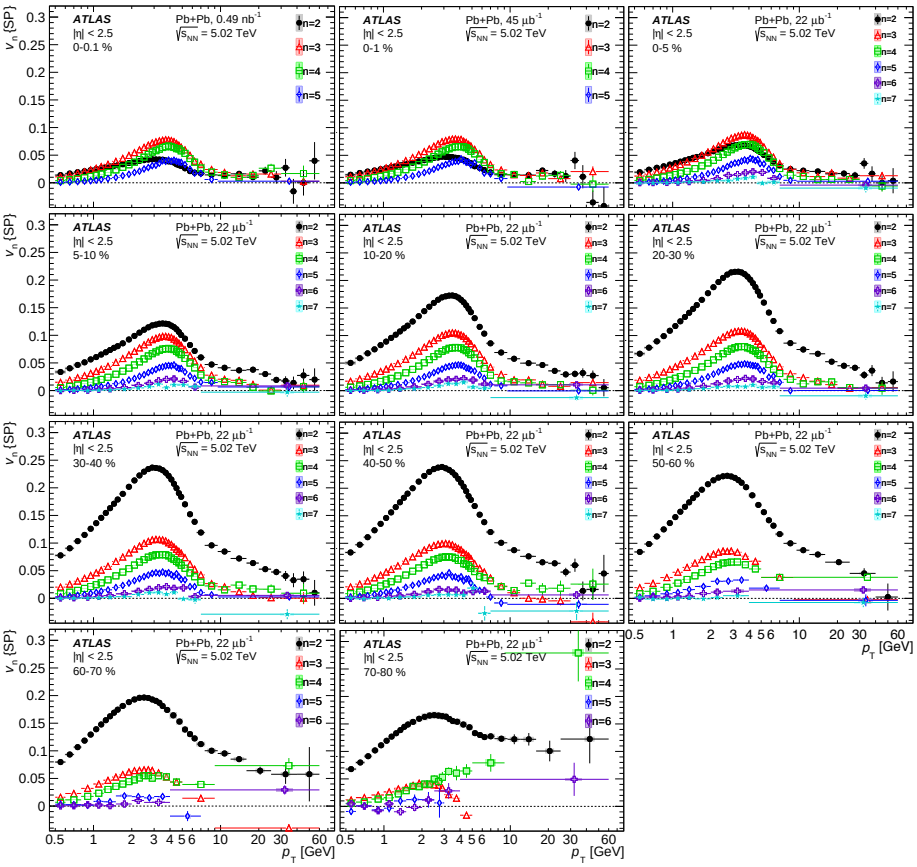



Figure 21: The $v_n$ obtained as a function of transverse momentum $p_T$ in 11 centrality intervals, from the most central at the top left panel to the most peripheral at the bottom right panel. Taken from Ref. [58].

Another source of transverse anisotropy in the QGP comes from fluctuations in the initial state. If the optical Glauber approximation of smooth nuclear densities were correct, then all odd $v_n$ coefficients should vanish. However, due to event-by-event fluctuations in the nucleon positions, these odd harmonics end up being non-zero. The $v_3$ is termed triangular flow, and it reflects triangular structures inside the QGP fireball. Figure 21 shows that, in central collisions, azimuthal particle production is dominated by fluctuations in the initial state because $v_{3,4} > v_2$ for many bins of $p_{\mathrm{T}}$. Experimentally, measuring flow signals can be complicated by the presence of high-$Q^2$ processes. For example, a dijet will produce some azimuthal anisotropy that does not come from the QGP. Methods for subtracting such "non-flow" contributions are used to circumvent this

problem.

Other bulk observables, such as the charged-particle pseudorapidity density, $\mathrm{d}N_{\mathrm{ch}}/\mathrm{d}\eta$, and transverse-momentum spectrum, $\mathrm{d}N_{\mathrm{ch}}/\mathrm{d}p_{\mathrm{T}}$, can also be used to infer properties of the medium. The pseudorapidity density constrains the produced-particle multiplicity and, in turn, the medium's entropy production and initial density. The low-$p_{\mathrm{T}}$ shape of the spectrum and the mean transverse momentum, $\langle p_{\mathrm{T}} \rangle$, are sensitive to the medium's expansion and radial flow. These observables are interpreted together with models of the initial conditions and particle production.

### 2.5.4 Jet quenching and hard probes of the QGP

A signature of QGP formation is the interaction of the medium with hard (high-$Q^2$) partons, which modifies the momentum and direction of the experimentally observed jets. *Jet quenching* describes how jets interact with the QGP medium and lose energy through a redistribution of energy from inside a jet cone to outside it. Jets are produced very early in a heavy-ion collision and, as such, "see" the full lifecycle of the QGP. The pre-equilibrium phase has been shown to contribute to energy loss, but the dominant effect comes from the QGP medium [59].

Energetic partons lose energy to the medium via both elastic and inelastic collisions with the medium. Elastic $2 \to 2$ collisions with the medium are referred to as collisional energy loss. Parton splittings and gluon radiation, both present in the vacuum, become more prevalent inside the QGP due to multiple scatterings with the medium constituents. This mechanism is called radiative energy loss, and it also includes inelastic medium collisions [60]. The two types of quenching processes are pictured in Figure 22.

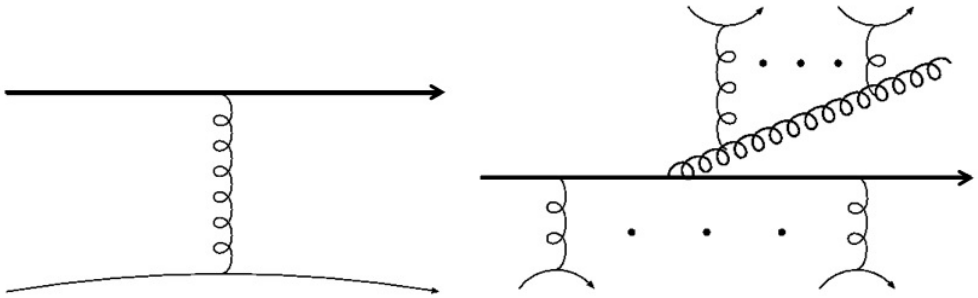


Figure 22: Schematic illustration of collisional (left) and radiative (right) energy loss of an energetic parton in the QGP. Taken from Ref. [60].

The strength of jet quenching can be described by $\hat{q}$, known as the *jet transport coefficient*.

$$\hat{q} = \frac{d\langle \Delta p_{\mathrm{T}}^2 \rangle}{dL} \tag{28}$$

The jet transport coefficient describes the average squared transverse momentum lost by the jet to the medium per unit length. Reference [61] shows that $\hat{q}/T^3$ can be related to $\eta/s$ in both a strongly and weakly coupled QGP and argues that $\hat{q}/T^3$ is a better measure of the in-medium QCD coupling strength. The left panel of Figure 23 shows that $\hat{q}/T^3$ increases as $T$ decreases, according to a Bayesian analysis by JETSCAPE [62].

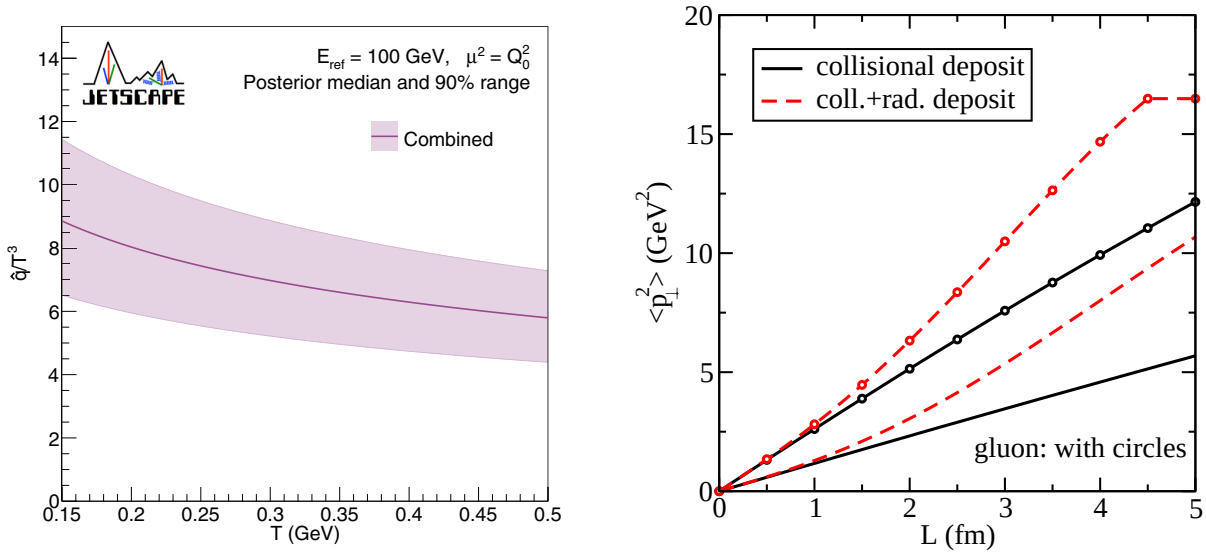



Figure 23: Left: Temperature dependence of $\hat{q}/T^3$ extracted through Bayesian analysis using jet and hadron measurements. Taken from Ref. [62]. Right: Path-length dependence of the mean transverse momentum squared lost by quark and gluon jets through collisional energy loss alone and through collisional plus radiative energy loss. Taken from Ref. [63].

The right panel shows the dependence of average jet quenching effects on the path length traversed through the medium [63]. Collisional energy loss grows linearly with path length, while radiative effects grow quadratically. Additionally, gluon jets are quenched more than quark jets are because they have a larger color charge and therefore interact more strongly with the medium. Additionally, a simple path-length dependence of jet quenching is complicated by jet-by-jet fluctuations in energy loss [64].

The jet-medium effects described above can be written as a factorized formula, with the "hot" jet-quenching effects expressed in Equation 29.

$$\begin{aligned} \mathrm{d}\sigma_{AB\to h+X} \approx \sum_{ijkk'l} \int \mathrm{d}x_i \mathrm{d}x_j \mathrm{d}z_k \, f_i^A\left(x_i, \mu_F^2\right) f_j^B\left(x_j, \mu_F^2\right) \\ \otimes \, \mathrm{d}\hat{\sigma}_{ij\to kl}\left(x_i, x_j, \alpha_s\left(\mu_R\right), Q^2\right) \\ \otimes \, P_{k\to k'}\left(z_k, \mu_F^2\right) \\ \otimes \, D_{k'\to h}\left(z_{k'}, \mu_F^2\right) \end{aligned} \tag{29}$$

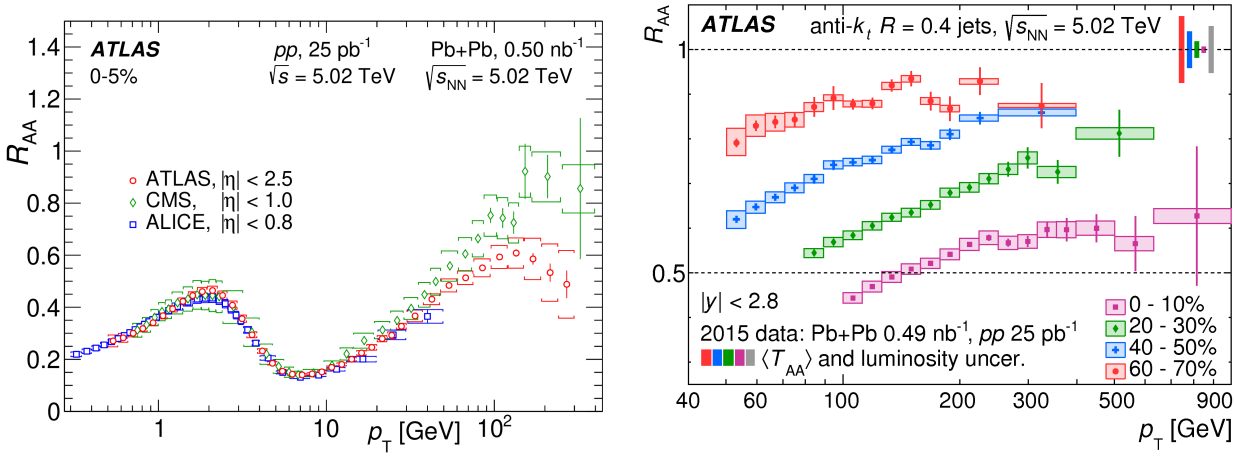
The interactions between the hard parton $k$ and the QGP are parameterized by $P_{k\to k'}(z_k)$. Note also that the PDFs in Equation 8 have been replaced by nPDFs for the relevant nuclear species. The "so-called" medium-modified FF is given by $\Sigma_{k'} P_{k\to k'}(z_k) \otimes D_{k'\to h}(z_{k'})$. Although this formula is widely used in phenomenological studies of the QGP, no formal proof of this type of factorization is currently available [60].

**Jet $R_{\mathrm{AA}}$** Jet quenching is studied experimentally by measuring hadron or jet observables in A+A collisions. One such observable is the nuclear modification factor, $R_{\mathrm{AA}}$, which describes how the production of hadrons or jets is modified relative to an expectation of $\langle N_{\mathrm{coll}} \rangle$ scaling. Note that calculating $R_{\mathrm{AA}}$ requires the reference $pp$ data to be taken under similar conditions and at a similar $\sqrt{s}$ to the ion data.

$$R_{AA} = \frac{\frac{1}{A^2}\frac{\mathrm{d}^2\sigma_{AA}^{h/j}}{\mathrm{d}p_{\mathrm{T}}\,\mathrm{d}y}}{\frac{\mathrm{d}^2\sigma_{pp}^{h/j}}{\mathrm{d}p_{\mathrm{T}}\,\mathrm{d}y}} \qquad R_{AA}|_{\mathrm{cent}} = \frac{\frac{1}{N_{\mathrm{evt}}\langle T_{AA}\rangle}\frac{\mathrm{d}^2 N_{AA}^{h/j}}{\mathrm{d}p_{\mathrm{T}}\,\mathrm{d}y}\Big|_{\mathrm{cent}}}{\frac{\mathrm{d}^2\sigma_{pp}^{h/j}}{\mathrm{d}p_{\mathrm{T}}\,\mathrm{d}y}} \tag{30}$$

$R_{\mathrm{AA}}$ is typically reported as a function of $p_{\mathrm{T}}$ and/or rapidity. By varying $p_{\mathrm{T}}$, one is sensitive to the dependence of quenching on $Q^2$. By varying rapidity, one is sensitive to the flavor of the initiating parton, since gluon jets lose more energy to the medium. The left-hand side of Equation 30 represents a centrality-integrated $R_{\mathrm{AA}}$. The prefactor of $A^2$ accounts for the number of nucleons in the colliding ion species. Note that this formula for $R_{\mathrm{AA}}$ is not sensitive to any Glauber-model uncertainties. The right-hand side of Equation 30 shows a centrality-differential $R_{\mathrm{AA}}$. Studying jet-quenching observables differentially in centrality is important because the energy loss is expected to be proportional to the path length in the medium [60], and centrality provides a systematic way to vary the medium size. Here, $N_{\mathrm{evt}}^{C}$ is the number of minimum-bias A+A events in a given centrality interval, and $\langle T_{AA}\rangle_C$ is the average nuclear thickness function in that interval, obtained from a Glauber calculation [43]. The nuclear thickness function

satisfies $\langle T_{AA} \rangle_{\text{cent}} = \langle N_{\text{coll}} \rangle_{\text{cent}} / \sigma_{pp}^{\text{inel}}$. Sometimes, the numerator on the right-hand side of Equation 30 is called the parton-flux-normalized yield because the Glauber factors normalize the overlapping flux of the two ions to that of a $pp$ collision.




(a) Charged hadrons, 0–5% centrality. ATLAS Fig. 20 [65].

(b) Inclusive jets. ATLAS Fig. 4(a) [66].

Figure 24: ATLAS measurements of the nuclear modification factor for charged hadrons and inclusive jets in Pb+Pb collisions.

Figure 24a displays the charged-hadron $R_{\text{AA}}$ in Pb+Pb measured by the LHC experiments, which show good agreement across different rapidities. The measurement shows strong suppression of the charged-hadron $R_{\text{AA}}$ over a broad range of $p_{\text{T}}$, indicative of strong partonic energy loss in the 0–5% centrality interval considered. At low-to-intermediate $p_{\text{T}}$, the charged-hadron $R_{\text{AA}}$ exhibits a broad local maximum around 2–3 GeV, before decreasing to stronger suppression at higher $p_{\text{T}}$. Figure 24b reports the jet $R_{\text{AA}}$ differential in $p_{\text{T}}$ and for various centrality selections. The magnitude of $R_{\text{AA}}$ increases with $p_{\text{T}}$. Additionally, suppression depends systematically on centrality: jet yields in the most central events are the most suppressed. This centrality scaling shows the connection between a larger QGP medium and stronger jet quenching.

**Dijet balance** Dijet observables are directly sensitive to asymmetric quenching experienced by the two jets. This asymmetry can result from different path lengths

through the medium or from jet-by-jet fluctuations in energy loss. The left panel of Figure 25 illustrates how the location of the hard scattering can cause the two jets to travel different path lengths through the QGP. An observable used to characterize the quenching of the two jets in a dijet system is the dijet momentum balance, shown in Equation 31.

$$x_{\mathrm{J}} = \frac{p_{\mathrm{T}_2}}{p_{\mathrm{T}_1}}, \tag{31}$$

Here, $p_{\mathrm{T}_1}$ and $p_{\mathrm{T}_2}$ are the transverse momenta of the leading and subleading jets, respectively, which are required to be nearly back-to-back in azimuth.

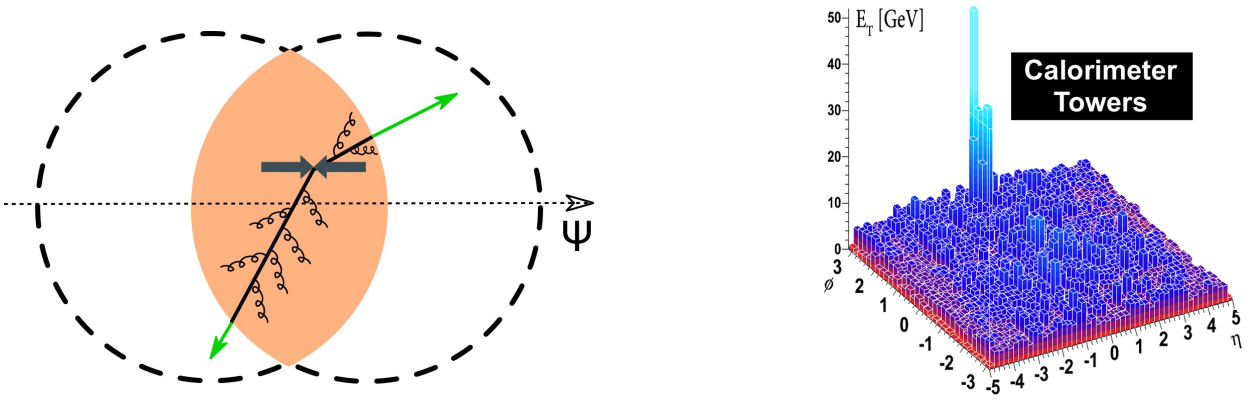



Figure 25: (left) Schematic illustration of asymmetric dijet energy loss due to different path lengths through the QGP, adapted from Reference [67] by V. Bailey. (right) ATLAS event display of a highly asymmetric dijet event [68].

Historically, the first measurement of fully reconstructed jets in heavy-ion collisions was the dijet balance[8] measured by ATLAS [68]. The right panel of Figure 25 shows a map of calorimeter energy deposits in $\eta - \phi$ space used in the analysis. The event displayed is characteristic of a highly asymmetric dijet event, with one clear large-jet signature at $(\eta, \phi) = 0, \pi$ but no recoiling jet at the expected location, $(\eta, \phi) = 0, 0$. This measurement revealed that the dijet balance was modified to favor more asymmetric momentum configurations in Pb+Pb collisions than in $pp$ collisions, with a stronger

[8]Technically, the observable measured was the dijet asymmetry, $A_J = \frac{|E_{\mathrm{T},1} - E_{\mathrm{T},2}|}{E_{\mathrm{T},1} + E_{\mathrm{T},2}}$, which probes the same physics as $x_{\mathrm{J}}$

effect in more central collisions. This discovery prompted theorists to incorporate the highly modified dijet balance into models [69, 70], as shown in the left panel of Figure 26. The ATLAS data agree much better with predictions from PYTHIA+medium than with plain PYTHIA, providing strong evidence of medium-induced jet quenching in Pb+Pb collisions at the LHC.

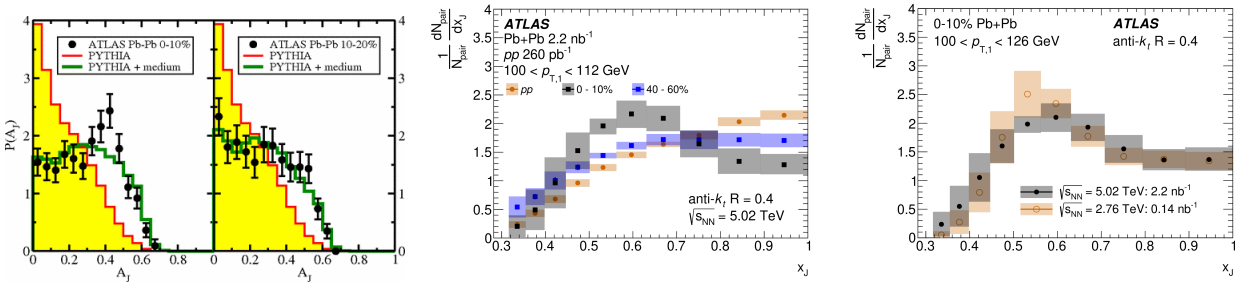



Figure 26: (left) ATLAS dijet asymmetry distributions in 0–10% and 10–20% Pb+Pb collisions, compared with PYTHIA and PYTHIA plus medium calculations [68, 70]. (middle) The $x_{\mathrm{J}}$ distributions in pp and in 0–10% and 40–60% Pb+Pb collisions at $\sqrt{s_{NN}} = 5.02$ TeV [71]. (right) Comparison of the $x_{\mathrm{J}}$ distributions in 0–10% Pb+Pb collisions at $\sqrt{s_{NN}} = 2.76$ and 5.02 TeV [71, 72].

Later ATLAS measurements [71–73] at 2.76 and 5.02 TeV included corrections for detector effects and jet energy resolution. The middle panel of Figure 26 shows the fully corrected $x_{\mathrm{J}}$ distribution for central (0–10%) and peripheral (40–60%) collisions, as well as a *pp* selection. The *pp* measurement agrees well with a PYTHIA8 prediction (not shown), while both Pb+Pb distributions are substantially modified relative to *pp*. As in Reference [68], the Pb+Pb distributions favor dijets with a lower average $x_{\mathrm{J}}$ (more imbalanced dijets), with the effect depending on centrality. The right panel of Figure 26 compares the measured dijet asymmetry in subsequent $x_{\mathrm{J}}$ measurements at 5.02 and 2.76 TeV and shows little dependence on $\sqrt{s_{\mathrm{NN}}}$.

All the $x_{\mathrm{J}}$ distributions discussed so far have been "area-normalized"; that is, they are divided by the total number of events so that the integral under each distribution is unity. Interpretations of these plots must account for the fact that shifts in the Pb+Pb distributions relative to *pp* represent relative enhancements or suppressions. ATLAS

addressed this lack of information about absolute dijet yields in Reference [71] by introducing absolutely normalized $x_{\mathrm{J}}$ yields, which introduce a Glauber-model dependence through $\langle T_{\mathrm{AA}} \rangle$[9]. The left panel of Figure 27 shows absolute $x_{\mathrm{J}}$ yields for several centrality intervals. A striking feature of this plot is that the suppression of balanced dijets relative to *pp* increases with centrality, with more central events showing greater suppression. The picture at low $x_{\mathrm{J}}$ is less clear and depends on the $p_{\mathrm{T_1}}$ selection considered. To study the differences between Pb+Pb and *pp* data directly, Reference [74] defines the $J_{\mathrm{AA}}$ observable as the ratio of the centrality-differential, absolutely normalized $x_{\mathrm{J}}$ distributions in Pb+Pb to those in *pp*. The $J_{\mathrm{AA}}$ distributions for different centrality selections are shown in the middle panel of Figure 27. The interpretation is similar to that of the previous plot, but the experimental precision is increased by canceling correlated sources of uncertainty between the Pb+Pb and *pp* measurements.

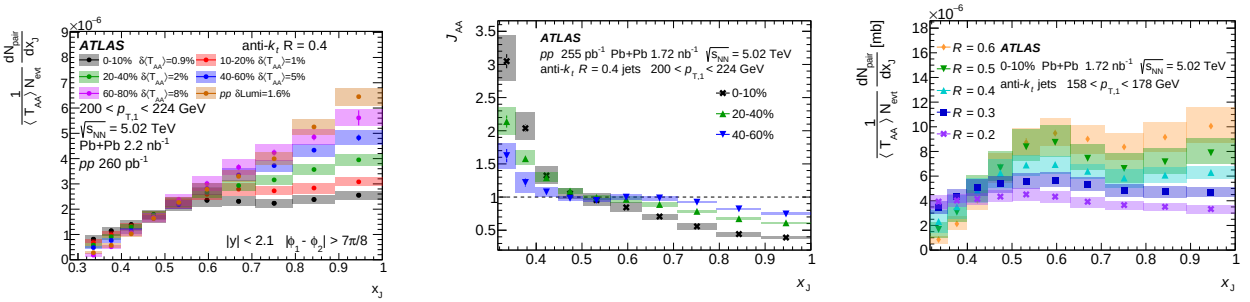



Figure 27: (left) Absolutely normalized $x_{\mathrm{J}}$ yields in Pb+Pb and *pp* collisions [71]. (middle) $J_{\mathrm{AA}}$, the ratio of the Pb+Pb absolute yields to the *pp* yields, for different centrality intervals [74]. (right) Radius dependence of the absolute $x_{\mathrm{J}}$ yields [74].

The right panel of Figure 27 shows the sensitivity of the absolutely normalized $x_{\mathrm{J}}$ distribution to the choice of jet radius, while still using the anti-$k_t$ clustering algorithm. Since jet quenching redistributes energy from inside the jet cone to outside it through collisional and radiative interactions with the medium, widening the jet cone captures more of this energy and thus increases the yield of balanced dijets.

[9]Technically, any centrality-differential measurement has a Glauber-model dependence. However, for wide centrality bins, such as those typically used in jet measurements, this uncertainty is usually negligible and is therefore neglected.

## 2.6 Signatures of the quark-gluon plasma in small systems

The umbrella term "small systems" covers many different collision scenarios, each with its own advantages and challenges. Generally, *pp*, *p*+A, $A^l$+$A^l$, and peripheral A+A collisions can all be considered small systems. Here, $A^l$+$A^l$ refers to light-ion collisions with mass number $A \lesssim 30$. Light-ion data have only recently become available at RHIC (2026) and the LHC (2025).

Evidence for QGP formation includes partonic energy loss and collective flow. As the system size decreases from that of central A+A collisions, these experimental signatures are expected to weaken as the fireball, if formed, becomes progressively smaller.

### 2.6.1 Collectivity in small systems

Flow-like signatures have been observed in *pp* [75], *p*+Pb [76], *p*+Au, *d*+Au, and $^3He$+Au [77], O+O, Ne+Ne [78], and peripheral Pb+Pb [58]. Discuss and add plots of flow observables. Collectivity has also been studied in systems even "smaller" than those listed above, such as electron–positron annihilation measured by the ALEPH detector at LEP [79]. The collectivity signal in $e^+e^-$ collisions was not significant and could be explained by PYTHIA6.

### 2.6.2 *pp* collisions as a no-quenching baseline

Jet energy loss is not observed in *pp* collisions, as demonstrated by the success of pQCD calculations in describing *pp* data. For example, measurements of the balance between an electroweak boson ($\gamma$ or $Z$) and a recoiling jet show that, on average, the boson and jet momenta agree with NLO predictions [80]. Since the electroweak boson does not interact strongly with the medium, any partonic energy loss would affect only the jet. Therefore, jet quenching in *pp* collisions would produce a net momentum imbalance that would disagree with pQCD predictions. Figure 28(a) shows that the photon–jet momentum balance in *pp* collisions is consistent with pQCD predictions, while the corresponding

Pb+Pb data exhibit an imbalance [81].

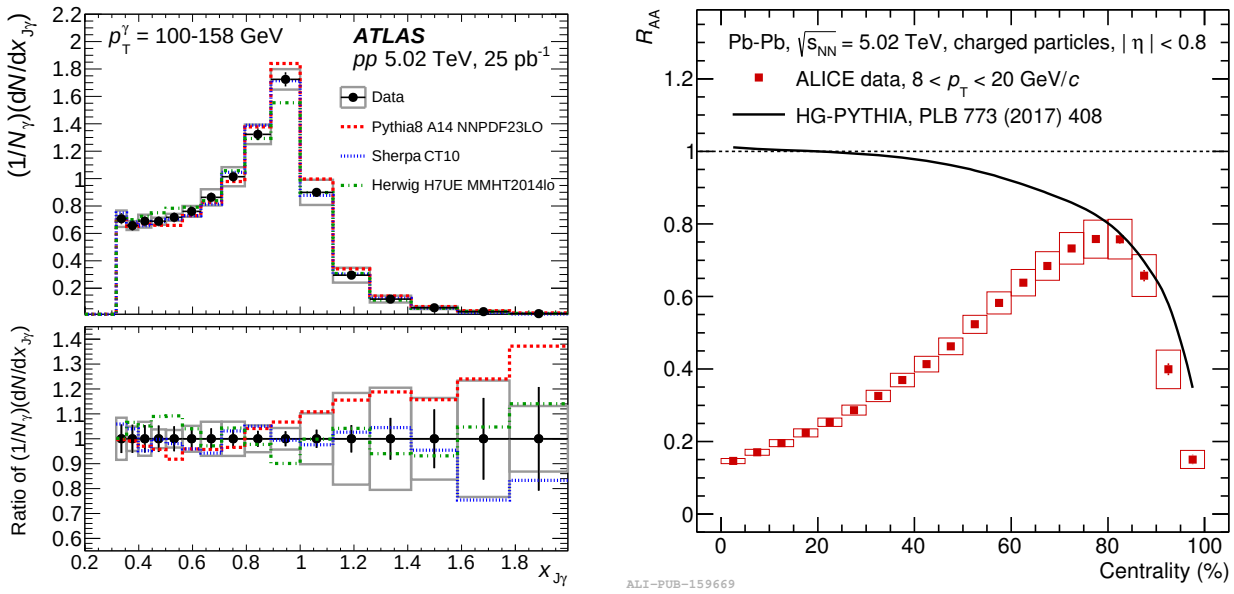



Figure 28: Measurements relevant to jet quenching in small and peripheral collision systems. In *pp* collisions, photon–jet momentum balance agrees with pQCD predictions, whereas the very peripheral charged-hadron suppression can be described by a model without jet quenching [81, 82].

### 2.6.3 Ambiguity of quenching signals in peripheral A+A collisions

Jet-quenching observables in peripheral A+A collisions show suppression relative to *pp* collisions; however, interpreting this suppression is challenging. Glauber modeling of peripheral collisions is associated with larger uncertainties because, at large impact parameters, the $N_{\text{elem}}$ dependence is dominated by multiplicity fluctuations rather than geometry. Additionally, events containing a hard process are correlated with higher soft underlying-event activity [83]. This effect is small at large $\langle N_{\text{part}} \rangle$, where it is largely washed out, but its relative impact increases as $\langle N_{\text{part}} \rangle$ decreases. In the right panel of Figure 24, the jet $R_{\text{AA}}$ is approximately 0.8 at a $p_{\text{T}}$ of 50 GeV, but substantial uncertainties in $\langle T_{\text{AA}} \rangle$ (about 10%) complicate its interpretation. For very peripheral collisions (greater than 80% centrality), the right panel of Figure 28 shows that the apparent suppression of charged-hadron $R_{\text{AA}}$ in Pb+Pb collisions can be described by a

PYTHIA8 toy model with a HIJING Glauber initial state and no jet quenching [82, 83]. Beyond $R_{\mathrm{AA}}$, correlation measurements (such as dijets and $\gamma$+jet pairs) have been shown to be more robust against centrality biases [84]. Figure 27 shows that the yield of balanced dijets is slightly suppressed relative to $pp$ in the 60–80% centrality interval, although the significance of this deviation has not been evaluated [71]. The most peripheral $J_{\mathrm{AA}}$ measurement reported is for the 40–60% centrality interval, where the ratio is clearly different from unity [74]. The effects of nPDFs and centrality biases on these measurements have not been rigorously evaluated, so definitive conclusions about the presence of jet quenching remain elusive.

### 2.6.4 Complications to jet quenching measurements in $p$+A collisions

Measurements of inclusive-hadron and jet $R_{\mathrm{AA}}$ in $p$+Pb collisions at the LHC and $d$+Au collisions at RHIC show no clear jet-quenching signal, as shown in Figure 29. The charged-hadron $R_{\mathrm{AA}}$ is approximately unity for hadrons with $p_{\mathrm{T}} > 5$ GeVand is consistent with the two pQCD predictions shown. The suppression at very low $p_{\mathrm{T}}$ can be explained by nPDF effects. The jet $R_{\mathrm{AA}}$ is at or above unity in measurements by ATLAS [85], ALICE [86], CMS [87], and PHENIX [88].

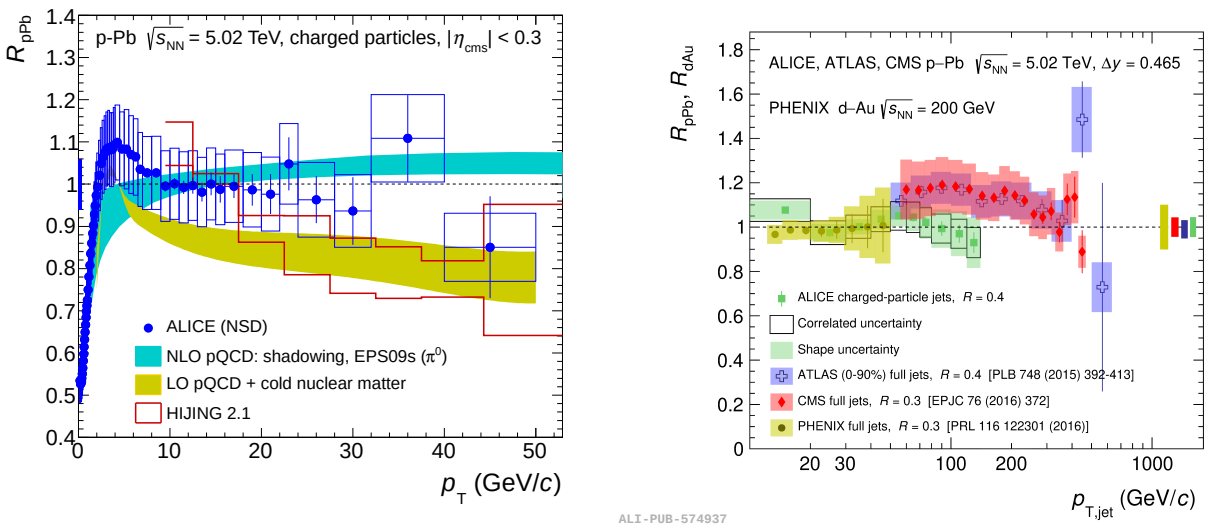



Figure 29: ALICE measurements of the nuclear modification factor in $p$+Pb collisions. Left: charged-hadron $R_{\mathrm{pPb}}$ compared with theoretical predictions [89]. Right: charged-particle jet $R_{\mathrm{pPb}}$ compared with reference data and theoretical predictions [86]. Other data from References [85, 87, 88].

Centrality-differential jet production has also been studied by PHENIX in $d$+Au collisions and by ATLAS in $p$+Pb collisions [85, 88]. The centrality-differential $R_{\mathrm{AA}}$ in $d$+Au collisions is presented in the left panel of Figure 30. Focusing initially on the most central 0–20% $d$+Au class, the suppressed $R_{\mathrm{AA}}$ is compatible with a small jet-quenching signal and is compared with a model that includes medium-induced radiation [90]. However, this model failed to describe the observed enhancement of the $R_{\mathrm{AA}}$ in peripheral collisions well above what could be expected from nPDF effects. These $R_{\mathrm{AA}}$ findings were qualitatively similar to the ATLAS measurement [85]. ATLAS defined the central-to-peripheral ratio of jet yields, $R_{\mathrm{CP}}$, as follows:

$$R_{\mathrm{CP}} = \frac{\left.\frac{1}{N_{\mathrm{evt}}\langle T_{p\mathrm{Pb}}\rangle}\frac{\mathrm{d}^2N}{\mathrm{d}p_{\mathrm{T}}\,\mathrm{d}y}\right|_{\mathrm{central}}}{\left.\frac{1}{N_{\mathrm{evt}}\langle T_{p+\mathrm{Pb}}\rangle}\frac{\mathrm{d}^2N}{\mathrm{d}p_{\mathrm{T}}\,\mathrm{d}y}\right|_{\mathrm{peripheral}}} \tag{32}$$

This represents the ratio of central to peripheral jet yields, corrected for the difference in $\langle N_{\mathrm{coll}}\rangle$ between the two classes. The right panel of Figure 30 shows that the $R_{\mathrm{CP}}$

decreases as a function of the *total jet energy*, defined as $p_{\mathrm{T}} \times \cosh(\langle y^* \rangle)$, where $y^*$ is the average rapidity of all jets in the event.

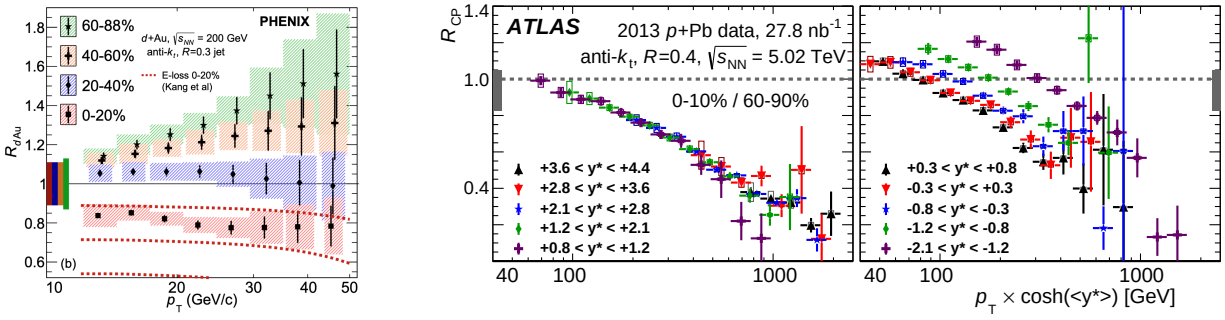



Figure 30: Centrality-dependent jet measurements in small collision systems. Left: PHENIX jet nuclear modification factor in d+Au collisions for several centrality intervals [88]. Right: ATLAS jet central-to-peripheral ratio, $R_{\mathrm{CP}}$, as a function of total jet energy in $p$+Pb collisions at 5.02 TeV [85].

This suppression of $R_{\mathrm{CP}}$ as a function of jet energy was interpreted as a color-transparency effect (see Section 2.2.3) rather than as evidence of jet energy loss. Events with higher jet energy are more likely to have been initiated by a high-$x$ parton in the proton and therefore to involve a spatially compact transverse configuration, leading to fewer nucleon–nucleon collisions. This centrality bias, discussed in Section 2.4.3, directly causes events to migrate to more peripheral centrality intervals, decreasing the $R_{\mathrm{CP}}$. ATLAS studied this effect in more detail by measuring the triple-differential dijet yield in 8.16 TeV $p$+Pb collisions [91]. The dijet system can be related to the Bjorken-$x$ of the proton through the average transverse momentum ($p_{\mathrm{T,Avg}}$), the average rapidity ($y_{\mathrm{b}}$), and the half-angle separation in rapidity ($y^*$). Figure 31 shows that the $R_{\mathrm{CP}}$ scales directly with the proton's $x$, further supporting the color-transparency hypothesis.

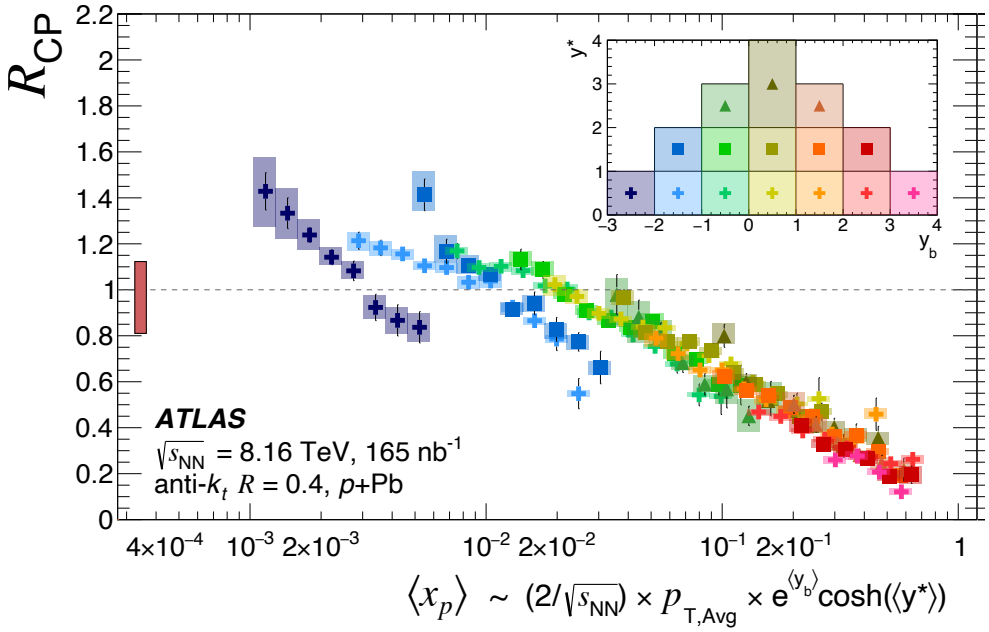



Figure 31: ATLAS jet central-to-peripheral ratio, $R_{\mathrm{CP}}$, as a function of proton momentum fraction $x_p$ in 8.16 TeV $p$+Pb collisions [91].

The observed suppression of the central $R_{\mathrm{AA}}$ in $d$+Au and $p$+Pb can be explained by an event-activity bias arising from color-transparency effects. Since any jet-quenching signal in $p$+A is expected to be small, and inclusive $p$+A measurements show no such evidence, it is of great interest to perform less-biased, centrality-differentiated measurements. One proposed way to reduce the CT bias is to use centrality estimators that are less sensitive to CT than event activity [92][10]. One such CE selects centrality classes in $p$+A events based on the number of spectator neutrons measured at $|\eta| > 8$ by a Zero-Degree Calorimeter (ZDC) located in the nucleus-going direction [93, 94]. This dissertation includes a measurement that directly probes the impact of CT effects on two CEs, event activity and forward neutrons, as a function of the Bjorken-$x$ of the parton extracted from the proton in the hard scattering. This inverts the previous approach [85, 91] by directly measuring the quantities used to estimate centrality as a function of the jet kinematics, providing the first quantitative investigation of global observables in the

[10]Event activity at forward rapidities has been used as the centrality estimator in the measurements mentioned above.

presence of CF effects. The details of this analysis are presented in Chapter 4 of this dissertation.

### 2.6.5 Jet quenching in $A^l$+$A^l$ collisions

Light-ion collisions provide a fertile new system for searching for the onset of partonic energy loss. As seen in Figure 32, minimum-bias O+O collisions have a similar $\langle N_{\text{part}} \rangle$ to peripheral Pb+Pb collisions, where observing jet quenching is precluded by the aforementioned issues. Figure 32 also shows hadron and jet $R_{\text{AA}}$ measurements in several systems as a function of $\langle N_{\text{part}} \rangle$. Following the trends of the experimental points and taking experimental uncertainties into account, this indicates that the $R_{\text{AA}}$ for minimum-bias O+O collisions should be around 0.8.

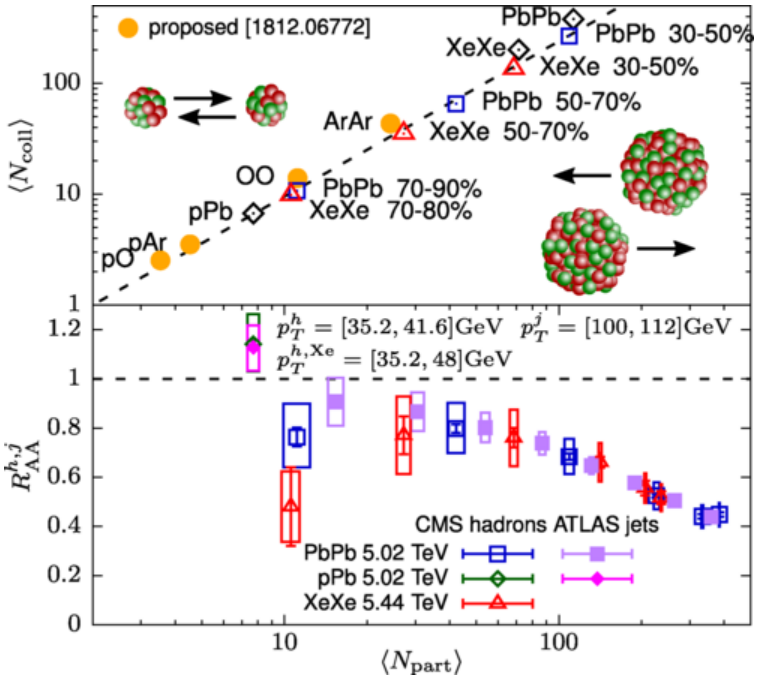



Figure 32: Figure 1 from Ref. [95].

However, this argument based simply on $\langle N_{\text{part}} \rangle$ is weak, as it ignores the geometry of the collision system. The $\hat{q}$ parameter is a function of path length in the medium [63], not $\langle N_{\text{coll}} \rangle$. For hard probes measured near midrapidity, the transverse extent of the overlap region controls the path length. For central collisions, the radius of the QGP

formed in an A+B nuclear collision is approximately limited by the radius of the smaller nucleus. Table 1 summarizes the radii of all collision species that have collided at the LHC. Light-ion collisions, such as O+O and Ne+Ne, provide a bridge between systems with small and large transverse sizes.

Table 1: Approximate radius scales for collision species studied at the LHC [5, 44, 45].

| Species | Proton | $^{16}$O | $^{20}$Ne | $^{129}$Xe | $^{208}$Pb |
|---|---|---|---|---|---|
| Radius (fm) | 0.84 | 2.61 | 2.81 | 5.36 | 6.62 |

As an aside, these transverse size arguments can shed light on why jet quenching is so elusive in $p$+Pb. A medium created in $p$+Pb collisions cannot be much larger than 1 fm since it is limited by the proton size. A jet could very easily escape from the overlap region before the hydrodynamic stage, given that a QGP formation time is $\approx$1 fm/c.

ALICE and CMS reported $R_{\mathrm{OO}}$ results in O+O collisions shortly after the data became available [96, 97]. The CMS results (left panel) show the evolution of the charged-hadron $R_{\mathrm{AA}}$ from $p$+Pb to Pb+Pb with the $R_{\mathrm{OO}}$ falling squarely in between the two. The $R_{\mathrm{OO}}$ is significantly below unity over a broad range of $p_{\mathrm{T}}$. Compared with models incorporating only nPDF effects and models incorporating both nPDF effects and jet quenching, the data favor the model that includes jet quenching. The ALICE measurement (right panel) shows that the $\pi^0$ $R_{\mathrm{OO}}$ is suppressed relative to nPDF-only baseline scenarios; however, this effect is less pronounced at high $p_{\mathrm{T}}$ when experimental uncertainties are taken into account. ALICE also reports an $R_{\mathrm{AA}}$ in $p$+O collisions, based on the expectation that if no jet quenching is observed in $p$+Pb then there should also be none in an even smaller system $p$+O. The $R_{p\mathrm{O}}$ is consistent with unity across a broad range of $p_{\mathrm{T}}$. All of these $R_{\mathrm{AA}}$ measurements are minimum-bias measurements (using the left expression in Equation 30), meaning they have no Glauber model dependence. This lack of centrality differentiation is motivated by expected centrality biases for $R_{\mathrm{AA}}$ measurements in light-ion collisions [84].

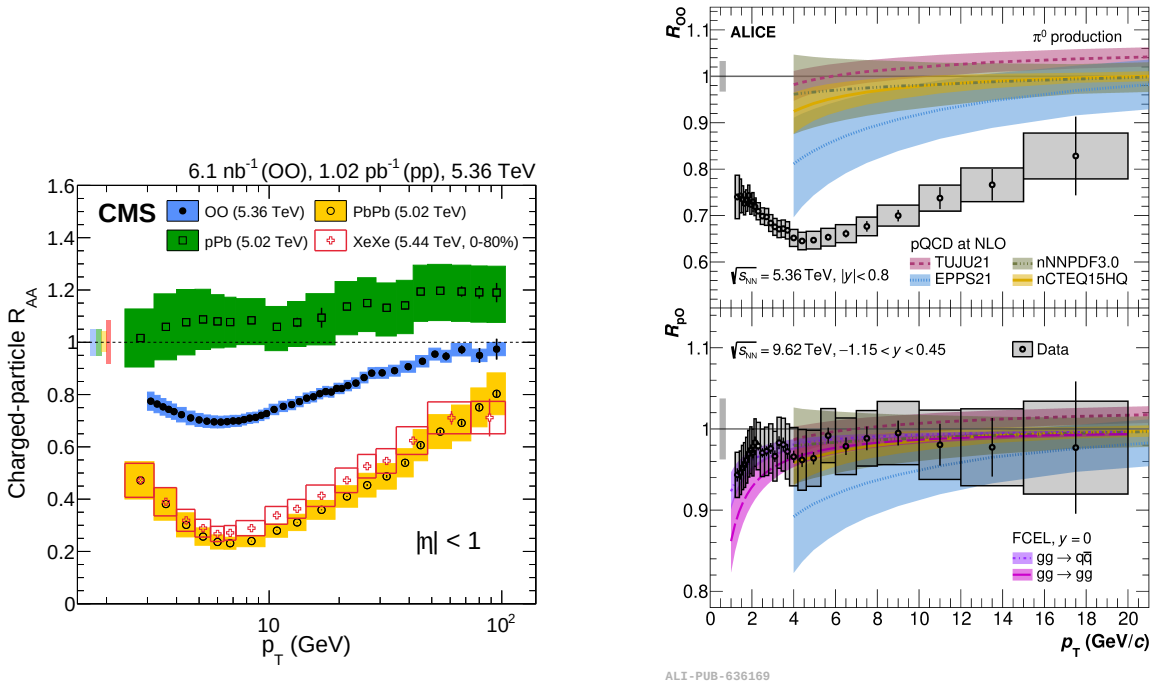



Figure 33: Nuclear modification measurements in light-ion collisions. Left: CMS charged-hadron $R_{\mathrm{OO}}$ in O+O collisions [96]. Right: ALICE neutral-pion $R_{\mathrm{OO}}$ and $R_{\mathrm{pO}}$ [97].

The CMS results report suppression relative to unity with a significance greater than $5\sigma$, while the ALICE results report a significance of $9.2\sigma$. To further constrain nPDF effects, ALICE measures the double ratio $R_{\mathrm{OO}}/\left(R_{p\mathrm{O}}\right)^2$ to factor out cold nuclear matter effects coming from the oxygen nuclei. The factor of two accounts for the two nuclei participating in the O+O collision[11]. Taking this ratio greatly reduces the impact of nPDFs on the interpretation, as shown in the left panel of Figure 34. The double ratio is suppressed beyond the nPDF-only predictions and is consistent with calculations from two separate jet-quenching models. Taking nPDF uncertainties into account, the double ratio is suppressed with a significance of $4.9\sigma$.

[11]Note that the O+O and $p$+O collisions were taken at different values of $\sqrt{s_{\mathrm{NN}}}$, so this ratio does not account for the $\sqrt{s_{\mathrm{NN}}}$ dependence of nPDFs

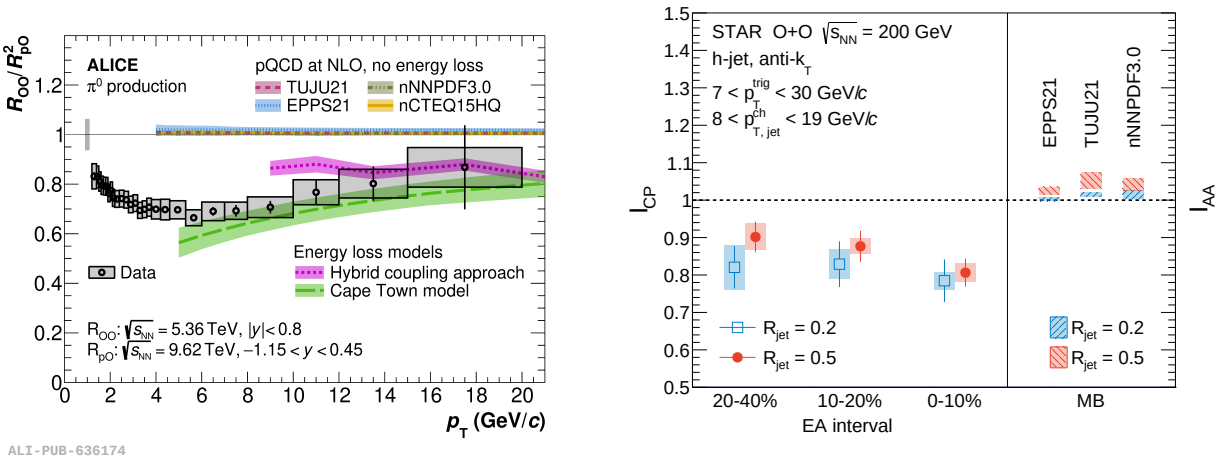



Figure 34: Additional probes of nuclear effects and jet quenching in oxygen collisions. Left: ALICE neutral-pion double ratio $R_{\mathrm{OO}}/(R_{\mathrm{pO}})^2$ [97]. Right: STAR central-to-peripheral ratio of semi-inclusive jets recoiling from a high-$p_{\mathrm{T}}$ hadron in O+O collisions [98].

At RHIC, STAR measured the semi-inclusive yield of hadrons and jets in O+O collisions at 200 GeV [98]. Figure 34 (right panel) shows the central-to-peripheral ratio of semi-inclusive jets ($I_{\mathrm{CP}}$). The jets used in the STAR analysis are charged-particle jets, which lack sensitivity to neutral hadrons. The numerator selection in the ratio is varied, while the peripheral selection is always 40–60%. For the most central selection (0–10%), the suppression of $I_{\mathrm{CP}}$ relative to unity has a significance of $4.3\sigma$, increasing to $5\sigma$ when the most extreme nPDF effects are considered. Using a correlation observable (hadron-triggered jets) instead of a spectrum ($R_{\mathrm{AA}}$) is expected to mitigate centrality biases in light-ion collisions [84]. However, using peripheral data as a reference can be susceptible to projectile-configuration biases affecting the nucleon from which the jet is extracted; these biases can appear as a reduction in the central-to-peripheral ratio [91].

Concurrently with the light-ion jet-quenching searches presented above, the author of this dissertation was the main analyzer for an ATLAS measurement of dijet balance in O+O and Ne+Ne collisions [99]. Since this analysis is of the correlations of dijets, the quenching observable can be studied differentially in centrality with minimal centrality-bias effects [84]. By measuring the dijet balance across centralities, the analysis

correlates the onset of jet quenching with increasing nuclear overlap. The measurement includes a reference *pp* sample taken at the same energy and with similar detector conditions; therefore, it avoids potential biases from projectile configurations while reducing experimental uncertainties. Furthermore, this is the first analysis of O+O data utilizing fully reconstructed calorimeter jets. ATLAS has a great deal of institutional knowledge from past measurements of dijet balance [68, 71–74], and focused that expertise on a fast analysis of O+O collisions, culminating in a preliminary result two months post data-taking. That preliminary result, presented by the author at the Initial Stages 2025 conference in Taipei, was quickly followed by a final publication in June 2026. The details of this analysis are presented in Chapter 5 of this dissertation.

# Chapter 3 Experimental Apparatus

## 3.1 the LHC

In addition to its flagship high luminosity *pp* program the LHC also provides special runs aimed specifically for the field of heavy ion physics. A list of these runs are compiled in Table 2. These include heavy ion runs (A+A) to study QGP physics, small systems runs ($p$+Pb and $A_{\text{light}} + A_{\text{light}}$) to study cold nuclear matter effects and search for the turn on of QGP physics, and *pp* reference runs as a comparision point for the other systems at the same energy. The author of this dissertation was on site assiting with operations and data taking for all Run 3 run sans the 2026 Pb+Pb run.

| **Year(s)** | **Run** | **Energy ($\sqrt{s_{\text{NN}}}$)** | **Species** | **Luminosity** |
|---|---|---|---|---|
| 2010 | Run 1 | 2.76 TeV | Pb+Pb | $\sim 10\ \mu\text{b}^{-1}$ |
| 2011 | Run 1 | 2.76 TeV | Pb+Pb | $\sim 0.15\ \text{nb}^{-1}$ |
| 2013 | Run 1 | 2.76 TeV | *pp* (ref.) | $\sim 5\ \text{pb}^{-1}$ |
| 2013 | Run 1 | 5.02 TeV | $p$+Pb | $\sim 30\ \text{nb}^{-1}$ |
| 2015 | Run 2 | 5.02 TeV | Pb+Pb | $\sim 0.5\ \text{nb}^{-1}$ |
| 2015 | Run 2 | 5.02 TeV | *pp* (ref.) | $\sim 28\ \text{pb}^{-1}$ |
| 2016 | Run 2 | 5.02 TeV | $p$+Pb | $\sim 0.5\ \text{nb}^{-1}$ |
| 2016 | Run 2 | 8.16 TeV | $p$+Pb | $160\ \text{nb}^{-1}$ |
| 2017 | Run 2 | 5.44 TeV | Xe+Xe | $\sim 0.3\ \mu\text{b}^{-1}$ |
| 2017 | Run 2 | 5.02 TeV | *pp* (ref.) | $\sim 270\ \text{pb}^{-1}$ |
| 2018 | Run 2 | 5.02 TeV | Pb+Pb | $\sim 1.8\ \text{nb}^{-1}$ |
| 2023 | Run 3 | 5.36 TeV | Pb+Pb | $\sim 1.8\ \text{nb}^{-1}$ |
| 2023 | Run 3 | 5.36 TeV | *pp* (ref.) | $\sim 400\ \text{pb}^{-1}$ |
| 2024 | Run 3 | 5.36 TeV | Pb+Pb | $\sim 1.7\ \text{nb}^{-1}$ |
| 2024 | Run 3 | 5.36 TeV | O+O | $\sim 8\ \text{nb}^{-1}$ |
| 2024 | Run 3 | 5.36 TeV | Ne+Ne | $\sim 1\ \text{nb}^{-1}$ |
| 2025 | Run 3 | 5.36 TeV | Pb+Pb | $\sim 2.8\ \text{nb}^{-1}$ |
| 2026 | Run 3 | 5.36 TeV | Pb+Pb | $\sim 2.8\ \text{nb}^{-1}$ |

Table 2: Summary of nuclear physics data-taking runs at the Large Hadron Collider (LHC), detailing runs for nuclear physics.

### 3.1.1 2016 $p$+Pb Data

In 2016, the LHC collided the asymmetric proton–Lead ($p$+Pb) system. Two different center of mass energies were used, $\sqrt{s_{\text{NN}}}$= 5.02 TeV and $\sqrt{s_{\text{NN}}}$= 8.16 TeV. The luminosity sampled by the higher energy sample far exceeded that of the lower energy sample. Part of the reason for that discrepency comes from the difference in the average number of collisions per bunch crossing ($\langle\mu\rangle$) in each sample is much higher in the $\sqrt{s_{\text{NN}}}$= 8.16 TeV sample. Higher $\langle\mu\rangle$ also means higher level of contamination from in-time and out-of-time pileup events.

Another particular feature of the 2016 $p$+Pb dataset it fact that the direction of circulation for the proton and lead beams switched in the middle of the run. In the first hi-$\langle\mu\rangle$ period, **Period B**, protons circulated in the accelerator in a clockwise direction[12] while lead ions circulated in the counterclockwise direction. In the second hi-$\langle\mu\rangle$ period, **Period C**, this directionality was reveresed.

### 3.1.2 2025 Light Ion Data

### 3.1.3 Luminosity

## 3.2 ATLAS detector

Placeholder text for this section.

## 3.3 Zero Degree Calorimeter

The ATLAS Zero Degree Calorimeter (ZDC) is designed to detect neutral particles that very close to the beampipe. It consists of two detectors located in absorbers along the beamline $\pm$140 m from the ATLAS interaction point. Each side (or arm) consists of four modules each with a material budet of 1.14 hadronic interaction lengths. The sampling calorimeter design consists of insterspered absorber layers, which cause the incident

[12] direction defined from viewing the LHC from above

particles to shower, and active layers, which output a signal proportinal to the energy deposited by charged particle shower crossing that layer. The absorber layers consist of tungstan plates, and the active layers consist of cherenkov radiators. These Cherenkov radiators transport Cherenkov radiation[13] induced by the particle shower to the lens of a Photomultiplier tupe (PMT). Each of the four modules are instamented with a single PMT. Cherenkov radiators were chosen since they offer substantial radiation hardness, quite necessary for a location that recives a very high dose. In Run 2, the active layers consisted of Quartz rods as the active layer. The radiation hardness of the deterctor was improved considerably in Run 3 when the active layer was replaced with doped fused silica rods.

As a calorimeter, the ZDC destructively samples the energy of the particles that enter it. Since dipole magnets between the IP and the ZDC swipe away forward charged particles, only neutral particles enter the ZDC. In ion running, the fragmentation of the ion's nucleus results in a population of neutrons interacting in the ZDC. Since these so called "spectator" neutrons do not interact in the collision at the IP they will impact the ZDC with roughly the same per nucleon energy as the beam. The number of spectator neutrons in non-trivially dependent on the event geometry, but they range from zero to roughly $(A - Z)/2$.

In order to capture signals from a zero to many neutrons, the ZDC operates with two sets of readout instrumentation, denoted as high and low gain readouts. The high gain readout is configured to have a dynamic range that saturates with a couple of neutrons. The low gain readout has a range of a roughly 10x that of the high gain readout.

The ZDC reconstruction software transforms the raw data read out from the above ADCs into module specific amplitures. Firstly, each waveform must have its pedestal subtracted in order to make sure that the output amplitude represents only strength of the module signal and not any offset caused by the electronics configuration. Then, the

[13]Cherenkov radiation is emmited in a mach cone when paerticles travel faster than the local speed of light in the material medium

subtracted samples can be fit to a functional waveform. The amplitude that minimize the $\chi^2$ of this fit is then considered to be the aplitude in that given event. Prob need to include more details about the ZDC reconstruction

Finally, the ZDC calibration transforms the above amplitudes, which have units of ADC counts, into units of energy.

**Missing ZDC Module in 2016 Data** During the 2016 $p$+Pb run, the ZDC side C module 0 (EM) was removed and replaced with the Arm2 calorimeter of the LHCf experiment [100]. Figure 35 displays the transverse acceptance of both the ATLAS ZDC and LHCf. The ATLAS ZDC EM module is the first of four sequential calorimeter modules, counting from the perspective of looking at the ZDCs from the ATLAS interaction point; therefore, the LHCf calorimeter would see forward neutral products before they reach the side C ZDC. The LHCf detector was put in place to measure the forward neutral products of the proton-going side in the **Period B** orientation; however, in the **Period C** orientation the LHCf detector was retracted into the "garage" position to avoid radiation damage [100]. The "garage" position indicates an upward vertical shift of 112 mm from the nominal position shown in Figure 35. This shift places the full LHCf Arm2 calorimeter outside of the full acceptance of the ATLAS ZDC. The side C ZDC has roughly 75% of the material budget of side A. This effectively reduces the interaction length of the side C ZDC from 4.6 to 3.4, which in turn reduces the neutrons expected to shower in the ZDC from 99% to 96.6%.

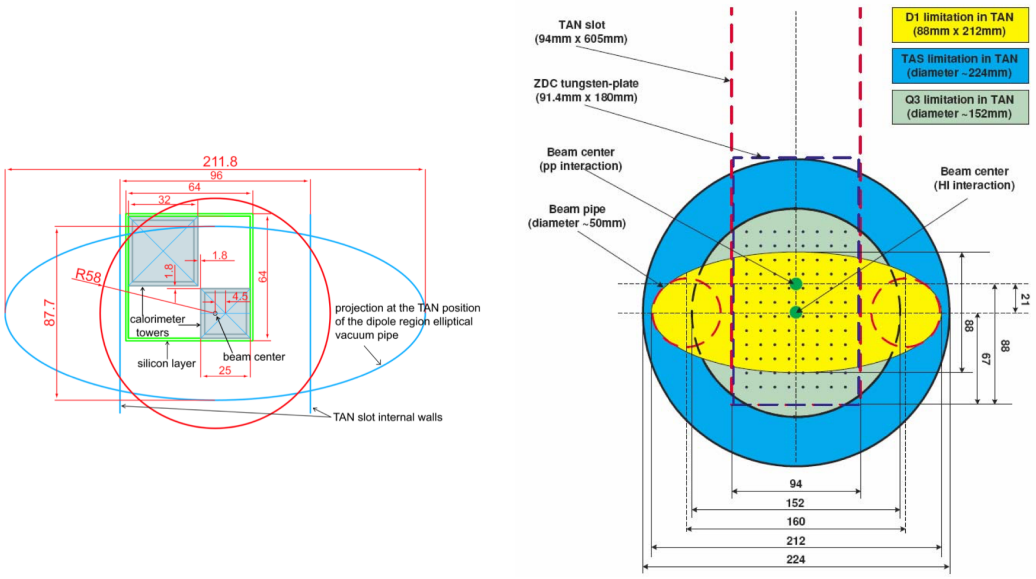



Figure 35: Detector geometry for LHCf Arm2 [101] (left) and the ATLAS ZDC [102] (right), viewed from the ATLAS interaction point. Distances are given in mm. The blue oval in the left image corresponds to the boundary of the yellow oval in the right image. The active area of LHCf is represented by the filled blue squares, and the active area of the ZDC is bounded by the purple dashed line.

### 3.3.1 2016 $p$+Pb ZDC Calibration

Due to the asymmetric nature of the $p$+Pb collisions system in each period only one arm of the ZDC detector will recive ion fragments.

In 2019, J. Nagle performed ref needed the calibration of the 5.02 TeV data, but concluded that the contamination from pileup in the high-$\langle\mu\rangle$ $p$+Pb data made a calibration unfeasible.

The author of this dissertation undertook the calibratrion of the high high-$\langle\mu\rangle$ $p$+Pb data in 2023 after substatial improvemnts had been made to the ZDC reconstruction in order to mitigate the previously untennable issues.

The first improvement made to the ZDC reconstruction comes from [103]. This note documents updates to the signal processing created mainly for the 2018 Pb+Pb data, but that are useful for the 2016 $p$+Pb data as well. These updates includes significant improvements to the reconstructions ability to fit pulses in the presence of out of time (OOT) pile up, both preceding and after the event in question. Large OOT pileup pulses would saturate the high gain ADC, leading to the pulse being analyzed with the low gain ADC which has a roughly 10x worse resolution. By excluding a few samples from the beginning or (end) of these pulses and looking at the high gain ADC values, the reconstruction will yield a reasonable fit for the current event. This is especially important given that the relatively high $\langle\mu\rangle$ in the 2016 8.16 TeV $p$+Pb has more events affected by out of time pileup.

The second significant improvement made to the ZDC reconstruction was the 'two-pass' analysis method. This method first uses a relatively high second derivative threshold to search for pulses in a ZDC event. This high threshold helps to suppress noise from small background and pileup pulses. However, if at least one module has a pulse that passes this threshold, all the modules on said ZDC arm are reanalyzed with a lower second derivative threshold. This 'repass' recovers finer pulses that were too small to be 'seen' on the first pass.

The third significant improvemnt[14], involves the studies on the pulse timing.

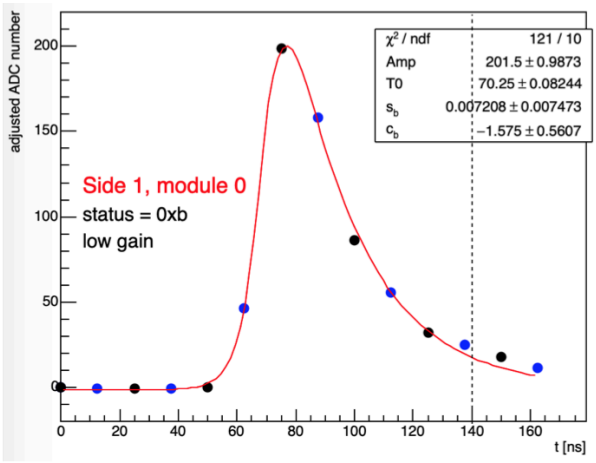



Figure 36: Example of a fitted ZDC pulse, taken from [103]

Base ZDC pulses (see Figure 36) are fit to a function of the form given in Equation 33, hereafter referred to as a FermiExp pulse.

$$f(t \mid A, \tau_1, \tau_2, t_0, C) = A\left(\frac{e^{-(t-t_0)/\tau_2}}{1+e^{-(t-t_0)/\tau_1}}\right) + C \tag{33}$$

Several of these parameters are fixed in order for the fitting function to have fewer degrees of freedom and thus converge more efficiently. For example, $\tau_1$ & $\tau_2$ are fixed in the 2016 ZDC reconstruction.

From a data quality standpoint, it would be useful to be able to reject pulses that were relatively early or late when compared to other pulses. One way to do this would be by defining a nominal $t_0$ range that is acceptable. However, there are two main challenges to defining a nominal $t_0$.

1. When the high voltage powering the a ZDC PMT changes, it affects the transit time in the tube which leads to a slightly different nominal $t_0$.

   **Solution:** Implement a time dependent nominal $t_0$ correction for the entire dataset.

[14]This improvment was done by the author of the dissertation, while the first two were done by Pengqi Yin of Columbia University

2. There is a nonlinear dependence of the $t_0$ on the pulse amplitude that must be corrected.

   **Solution:** Fit the curvature induced by the nonlinear dependence to a 5th order polynomial. Use that fit function to flatten the convex distribution.

The data used for the calibration of the high-$\langle\mu\rangle$ $p$+Pb was read out into a dedicated ZDC calibration stream. This dedicated readout only stores information about the ZDC detector for each accepted event[15] and can therefore accumulate substantial statistics needed for a precision calibration. Events entereing the stream fired the **L1_ZDC_A** or **L1_ZDC_C** triggers, requiring that there was some substatial energy in at least one arm of the ZDC. To get a set of events unbiased by trigger turn on thresholds, the set of events used to analyze Side A were triggered on a signal in Side C and vice versa.

The ZDC is designed to sample neutral ion fragments with the same per-nucelon energy as the beam. The lightest neutral fragment possible is a single beam energy neutron incident on the detector. A single neutron that fragmented from a lead ion will, on average, be as energetic as the per-nucleon beam energy, which here is 2.5625 TeV. Therefore, the energy of single neutrons gives a well defined energy to calibrate against. Of course single neutrons will not have the exact per-nucleon beam energy every time, since the exact energy of bonud nucleons is stochastically modified by the so-called 'fermi-momentum' citation needed. The variance of single neutrons given by fermi-momentum coupled with the experimental resolution of the ZDC should give the width of the detector response to a single neutron.

Therefore the single neutron peak [16] intrinsically contains information about the ZDC Energy Scale ($\mu_{1n}$) and ZDC Energy Resolution ($\sigma_{1n}$). However, we are intereted in not just any calibration, but the *optimal* one. The optimal calibration is defined to be the

[15] The per-event data footprint of the ZDC detector is roughly an order of magnitude smaller than the whole ATLAS detector including all detector subsystems

[16] observed detector response to single neutrons

one that puts the mean of the single neutron peak at 2.5625 TeV and minimizes the variance (therefore maximizing experimental resolution). The machinery of the calibraiton works as follows.

The goal is to find a set of gains $\mathbf{g} = \{g_0, g_1, g_2, g_3\}$ such that given the ZDC module amplitudes for a given event $\mathbf{A} = \{A_0, A_1, A_2, A_3\}$ the dot product $\mathbf{g}^T\mathbf{A}$ is equal to the energy of the ZDC in units of GeV(2562.5 GeV). The below calibration procedure is performed on a subset of ZDC events in the first neutron peak. Equations 34 and 35 are the mean an variance of the first neutron peak sample. The variable $N$ indicates the size of the sample used to perform the analysis.

$$E_{1\mathrm{n}} = \left\langle \mathbf{g}^T\mathbf{A} \right\rangle_N \tag{34}$$

$$\sigma^2 = \left\langle \left(\mathbf{g}^T\mathbf{A}\right)^2 \right\rangle_N - \left( \left\langle \mathbf{g}^T\mathbf{A} \right\rangle_N \right)^2 = \left\langle \left(\mathbf{g}^T\mathbf{A}\right)^2 \right\rangle_N - \left(\mathrm{E}_{1\mathrm{n}}\right)^2 \tag{35}$$

Using the method of Lagrange multipliers, one can minimize the variance under the constraint that mean of the one neutron peak is equal to the per nucleon beam energy.

$$\mathcal{L} = \left\langle \left(\mathbf{g}^T\mathbf{A}\right)^2 \right\rangle_{\mathrm{N}} - \left(\mathrm{E}_{1\mathrm{n}}\right)^2 + \lambda \left( \left\langle \mathbf{g}^T\mathbf{A} \right\rangle_{\mathrm{N}} - \mathrm{E}_{1\mathrm{n}} \right) \tag{36}$$

Solving 36 with respect to $\mathbf{g}$ and $\lambda$ yields the following system of equations.

$$\begin{bmatrix} 2\langle A_0A_0\rangle & 2\langle A_0A_1\rangle & 2\langle A_0A_2\rangle & 2\langle A_0A_3\rangle & \langle A_0\rangle \\ 2\langle A_1A_0\rangle & 2\langle A_1A_1\rangle & 2\langle A_1A_2\rangle & 2\langle A_1A_3\rangle & \langle A_1\rangle \\ 2\langle A_2A_0\rangle & 2\langle A_2A_1\rangle & 2\langle A_2A_2\rangle & 2\langle A_2A_3\rangle & \langle A_2\rangle \\ 2\langle A_3A_0\rangle & 2\langle A_3A_1\rangle & 2\langle A_3A_2\rangle & 2\langle A_3A_3\rangle & \langle A_3\rangle \\ \langle A_0\rangle & \langle A_1\rangle & \langle A_2\rangle & \langle A_3\rangle & 0 \end{bmatrix} \begin{bmatrix} g_0 \\ g_1 \\ g_2 \\ g_3 \\ \lambda \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ E_{1n} \end{bmatrix} \tag{37}$$

The ZDC energy response changes significantly over the period of a run. Therefore, it is imperative to create a time dependent calibration.

The software to implement this calibration was created by the author of this dissertation. Some important practical details are described below. Firstly, the data is broken a series of seperate consecutive time intervals with the condition that each time interval has about a million events, which was found to be sufficient for a stable calibration. Next, within each interval, a histogram is created out of the sum of ZDC module amplitudes for a given event $\Sigma_i^4 A_i$. The single neutron peak is apperent in the spectrum of $\Sigma_i^4 A_i$, and a window $\pm 1.5\sigma$ around the peak is defined by a simple Gaussian fit. Then, for each time interval, fourteen histograms are defined corresponding to the unaveraged matrix elements of Equation 37 and are filled with events falling inside the single neutron window defined in the previous step. Next the histograms are averaged over and Equation 37 can be solved for the optimal $\mathbf{g}$ for each time interval. Once the optimal $\mathbf{g}$ for every time interval is decided, a cubic spline is used to interpolate smoothly between the sucsessive time intervals. Then, on an event by event basis the calibraited ZDC energy is equal to $\mathbf{g}(\mathbf{t})^T\mathbf{A}$.

The optimized gains as a function of time ($\mathbf{g}(t)$) for Period B of the $p$+Pb run is displayed in Figure 37.Citation needed Each module is denoted by a different color, but the overall trend is common to all modules. Clusters of points on the time-axis represent individial LHC running periods. Throughout individual running periods, the gains increase. This increase is caused by accumulated radiation damage to the detector, which lowers the average module amplitudes ($\mathbf{A}$) requiering higher gain factors ($\mathbf{g}$) to calibrate to the same energy. Running periods with higher levels of detector degriation (higher $\frac{dg}{dt}$) corruspond to periods of data taking with higher-$\langle\mu\rangle$ and thus more incident radiation and integrated luminosity. The gain increses in Figure 37 are not monotonic as expected from continous radiation damage. This is because at certain points throught the run, the operating power supply for the module PMTs were manually adjusted[17] to elicit a stronger detector response. Each of these changes was correlted with 2016 logbook

[17]The high voltage of the ZDC PMTs was increased

entrires from Peter Steinberg. Additially, the data quality recommendations had to be adjusted to make sure that no data was used when the detector configuration was activly being changed.

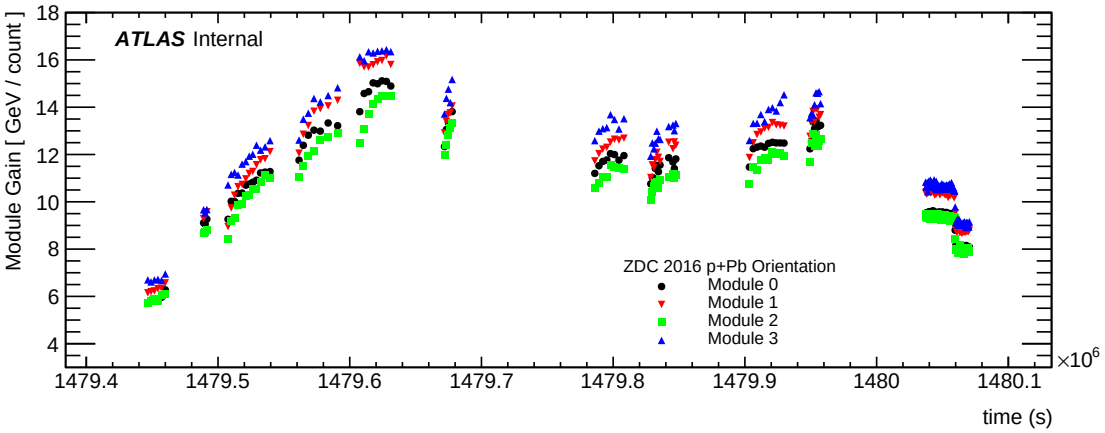



Figure 37: Citation and description needed

An example of the ZDC energy spectrum and single neutron peak is displayed in Figure 38. Following calibration, a simultaneous fit is performed to the first three neutron peaks, as well as a background term. Background sources include neutral pions, photons, and beam–-beam backgrounds. The functional form of the background term is chosen to facilitate fit convergence in the 1–3 neutron range [104].

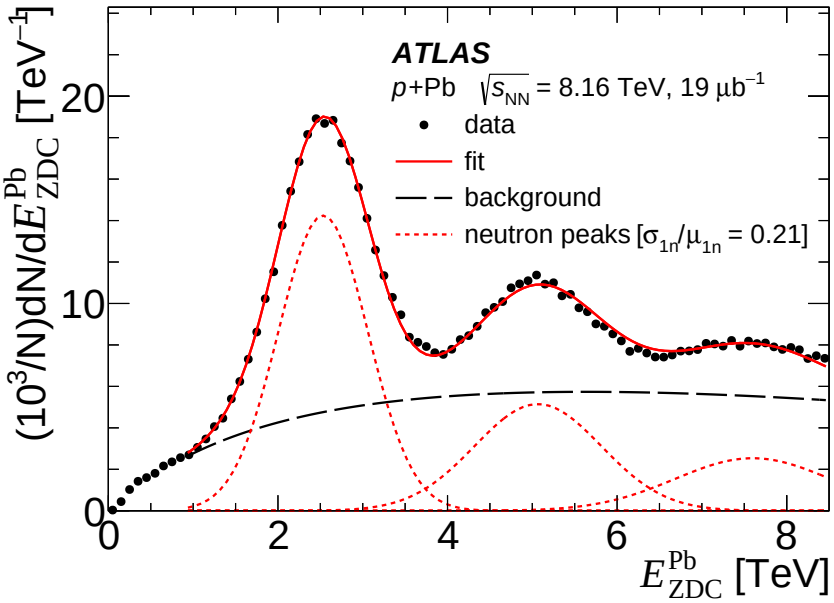



Figure 38: ZDC background estimation for run 313100, taken from [104].

### 3.4 ATLAS Jet Reconstruction

When studying HI collisions with the ATLAS detector, jets are reconstructed using the anti-$k_t$ jet clustering algorithm [105] with calorimeter towers as constituents. This algorithm uses collections of four-vectors corresponding to a set of tracks of charged particles in the inner tracking system (aka "track jets"), massless calorimeter towers with size $\Delta\eta \times \Delta\phi = 0.1 \times 0.1$ (aka "heavy ion jets"), or sets of simulated stable particles at the generator level (aka "truth jets"). The measurements presented in this dissertation make use of heavy ion jets reconstructed using the anti-$k_t$ algorithm with R = 0.4, called `AntiKt4HIJets`.

A detailed review of the performance of jet reconstruction in heavy ion collisions can be found in [106]. This note will only discuss the main features of the heavy ion jet reconstruction and their implications on the dijet balance measurement.

### 3.4.1 Underlying Event Subtraction

To reconstruct jets in HI collisions, a background arising from the underlying event (UE) must be subtracted from each tower to avoid significant rates of fake jets [107] and to avoid distortion of the jet energy scale. The energy registered in each cell can be decomposed into contributions from a jet and from the UE as

$$\frac{d^2E_{\mathrm{T}}^{\mathrm{total}}}{d\eta d\phi} = \frac{d^2E_{\mathrm{T}}^{\mathrm{jet}}}{d\eta d\phi} + \frac{d^2E_{\mathrm{T}}^{\mathrm{UE}}}{d\eta d\phi} \tag{38}$$

Only the first contribution should be used to reconstruct jets. Therefore, $\frac{d^2E_{\mathrm{T}}^{\mathrm{UE}}}{d\eta d\phi}$ must be estimated and subtracted. The UE subtraction procedure makes use of the UE average transverse energy density, $\rho(\eta, \phi)$. The $\phi$ dependence in $\rho$ arises from global azimuthal correlations in the particle production, usually referred to as flow, that arises from the hydrodynamic response of the medium to the geometry of the initial collision [58]. One can describe the UE contribution to the transverse energy of the calorimeter towers by using the amplitudes ($\nu_n$) and the phases ($\Psi_n$) of the Fourier components of the azimuthal angle distributions:

$$\rho(\eta, \phi) = \frac{1}{2\pi}\frac{\langle dE_T\rangle}{d\eta}\Big(1 + 2\sum_{n=2}^{N}\nu_n \cos(n(\phi - \Psi_n))\Big) \tag{39}$$

where $\phi$ and $n$ represent the azimuthal angle of the tower and the order of the flow harmonic, respectively. Although $\nu_2$ dominates, the third and the fourth harmonics are used to further improve the $\rho$ evaluation. N = 4 was used to reconstruct Run 2 Pb+Pb data, while for $p$+Pb and $pp$ data used for the analysis presented in this note, no anisotropic component was allowed in the fit, *i.e.* N = 0. This choice was made due to the relatively low multiplicity characterizing $p$+Pb and $pp$ collisions compared to Pb+Pb.

The average transverse energy density is estimated using an iterative procedure that excludes regions populated by jets. Please note that, for both $pp$ reference, $p$+Pb and

light ion data, the HIP reconstruction mode [108], where the HI jet reconstruction is deployed accounting for low underlying event conditions, is applied. HIP mode does not include any correction for azimuthal anisotropy of the UE.

A first estimate of the UE average transverse energy density, $\rho(\eta)$, is evaluated in 0.1 intervals of $\eta$ excluding $(\eta, \phi)$ regions of the calorimeter populated by "seed" jets. During this first step, the seeds are defined as a set of R = 0.2 HI jets and R = 0.4 track jets. To be considered as a seed, an R = 0.2 jet has to pass the discriminant cut on the value of maximal-over-mean tower energy, e.g. max/mean $> 4$. This cut enhances purity of the selected jet candidates against UE correlations. The background is then subtracted from the jet calorimeter towers to compute the subtracted jet kinematics. At this stage, $\rho$ and $\nu_n$ are recalculated by excluding $(\eta, \phi)$ region of the calorimeter containing a jet seed.

### 3.4.2 Jet Calibration

This section describes the general features of the HI jet calibration strategy. Since the two measurments reported in this dissertation were taken nine years apart, they use slightly different calibration strategies[18].

Individual calorimeter cells are themselves calibrated to the EM scale, but there is a variety of factors (*e.g.* passive material in the detector, gaps between cells, signal losses, etc. [109]) that allow only for a partial measurement of the energy deposited by a single jet in the detector. To compensate for these experimental effects, the energy scale of reconstructed jets needs to be calibrated to the truth energy scale. The calibration sequence for HI jets is presented in [110]. Here we briefly review the main steps of this procedure. After the UE subtraction, three different types of calibrations are applied, namely:

1. origin correction to account for the position of the primary vertex in each event.

[18] Differences in the calibration strategies between the two measurments will be reported on a per measurment level

2. EtaJES MC-based calibration, to recover the truth energy scale in MC.

3. In-situ $\gamma$+jet, Z+jet and $\eta$-intercalibration, to account for differences between MC and data.

A small pseudorapidity correction is also applied to the jet reconstructed $\eta$, to counteract the implicit assumption that the $z$-vertex[19] $= 0$ made during the reconstruction of the data. For this reason, this procedure is called "origin correction".

The EtaJES calibration is the larger between the two and is derived from PYTHIA8 simulated dijet events. The simulations are carried out using the MC conditions and the energy characteristic of the data taking. The scope of this calibration, common to both data and MC, is to correct from the EM scale to the truth hadronic scale. The per-jet energy correction factor is computed using a numerical inversion procedure [111, 112], see [113] for a review of the procedure's math. A typical jet energy scale correction, $E_{\mathrm{T}}^{\mathrm{true}}/E_{\mathrm{T}}^{\mathrm{reco}}$, ranges from $\sim$ 1.2 to 2.0.

The second calibration component is only applied to data to account for discrepancies between MC and data. The in-situ and $\eta$-intercalibrations implement residual corrections to the jet energy scale based on the $p_{\mathrm{T}}$ balance in vector boson ($\gamma$, $\mathrm{Z}^0$) - jet events (the *in-situ* correction) and dijet events where mid-rapity jets are used to calibrate the forward rapidity jets (the $\eta$-intercalibration). Calibration factors are derived in bins of $p_{\mathrm{T}}$ and $\eta$ and are applied as a correction to the jet $p_{\mathrm{T}}$.

[19] Position of the vertex associated to the jet along the beam axis.

# Chapter 4 Dependence of geometry estimators on hard scattering kinematics in p+Pb collisions

Just like A+A collisions, hard-process rates in $p$+A collisions are studied as a function of centrality, defined using global event-level observables sensitive to collision geometry. In the presence of a hard scattering, however, the proton configuration can modify the relation between those observables and the impact parameter, contaiminatign the above correlation. Measuring the estimators directly as a function of the reconstructed partonic kinematics inverts the usual centrality-selected approach and quantifies this bias.

This chapter presents the first measurment of event geometry estimators on hard-scattering kinematics in $p$+Pb collisions. Dijets are used as a proxy for the initial-state partonic scattering configuration [91] and their kinematics are correlated with the Pb-going flow forward transverse energy and zero-degree netral energy.

## 4.1 Measured Observables

The measured observables are the neutral energy deposited in the Pb-going Zero Degree Calorimeter (ZDC), $\mathrm{E}_{\mathrm{ZDC}}^{\text{Pb-side}}$, and the transverse energy measured in the Pb-going Forward Calorimeter (FCal), $\mathrm{E}_{\mathrm{T}}^{\mathrm{Pb}}$. The ZDC is sensitive primarily to spectator neutrons emitted during nuclear breakup, whereas the FCal transverse energy is an event-activity estimator sensitive to particle production by the collision participants. Both quantities have been used to characterize the geometry of proton–nucleus collisions.

The hard-scattering kinematics are accessed using the two highest-transverse momentum jets in each event. The momentum fraction carried by the parton in the proton is estimated at leading order as

$$x_p = \frac{p_{\mathrm{T},1} e^{y_1^{\mathrm{CM}}} + p_{\mathrm{T},2} e^{y_2^{\mathrm{CM}}}}{\sqrt{s_{\mathrm{NN}}}}, \tag{40}$$

where $p_{\mathrm{T},1}$ and $p_{\mathrm{T},2}$ are the transverse momenta of the leading and subleading jets and

$y_1^{\mathrm{CM}}$ and $y_2^{\mathrm{CM}}$ are their rapidities in the nucleon–nucleon center-of-mass frame. This estimator reconstructs the Bjorken-$x$ of the scattered proton parton from final-state jet kinematics. Simulation studies show that the reconstructed quantity is, on average, 6–8% lower than the parton-level value in the measurement phase space, with a relative resolution that improves from about 10% at low $x_p$ to 4% at high $x_p$.

Normalized $\mathrm{E}_{\mathrm{ZDC}}^{\text{Pb-side}}$ and $\mathrm{E}_{\mathrm{T}}^{\mathrm{Pb}}$ distributions are measured in intervals of $x_p$, along with their means, $\langle \mathrm{E}_{\mathrm{ZDC}}^{\text{Pb-side}} \rangle$ and $\langle \mathrm{FCal} \sum \mathrm{E}_{\mathrm{T}}^{\text{Pb-side}} \rangle$.The distributions expose changes in shape, while the means provide compact measures of the $x_p$ dependence. The correlation between $\mathrm{E}_{\mathrm{ZDC}}^{\text{Pb-side}}$ and $\mathrm{E}_{\mathrm{T}}^{\mathrm{Pb}}$ is also studied in each $x_p$ interval to connect the spectator-neutron and participant-activity observables.

## 4.2 Data Sets

### 4.2.1 2016 $p$+Pb Data

The $p$+Pb data at $\sqrt{s_{\mathrm{NN}}} = 8.16$ TeV used for this analysis were delivered by the LHC to ATLAS in November and December of 2016. In $p$+Pb collisions at 8.16 TeV the colliding beams have different energies per nucleon (6.5 TeV protons colliding with $(Z/A) \times 6.5\ TeV \simeq 2.6\ TeV$ per nucleon in a lead nucleus), resulting in a collision system with $\sqrt{s_{\mathrm{NN}}} = 8.16$ TeV per nucleon pair. Under these conditions, the center-of-mass frame is shifted in rapidity, $y$, by $\Delta$y= 0.465 in the direction of the proton beam. Data were collected in two different periods, characterized by opposite beam orientations. Due to technical issues with the ZDC instrumentation (see Section 3.3), only **Period B** was used for this measurement.

The runs in this period were characterized by a total integrated luminosity of 55.9 $\mathrm{nb}^{-1}$. Beam 1 consisted of protons and Beam 2 of Pb ions. The protons traveled toward negative rapidities in the "A→C" direction, corresponding to clockwise motion in the accelerator according to ATLAS conventions. All data sets used in this analysis, listed in Table 3, were processed using ATHENA Release 21.

| 2016 $\sqrt{s_{\text{NN}}}$ = 8.16 TeV Data Samples | Luminosity [nb$^{-1}$] |
|---|---|
| data16_hip8TeV.00313063.physics_Main.merge.AOD.r11416_p3868 | 0.021 |
| data16_hip8TeV.00313067.physics_Main.merge.AOD.r11416_p3868 | 0.486 |
| data16_hip8TeV.00313100.physics_Main.merge.AOD.r11416_p3868 | 8.787 |
| data16_hip8TeV.00313107.physics_Main.merge.AOD.r11416_p3868 | 11.053 |
| data16_hip8TeV.00313136.physics_Main.merge.AOD.r11416_p3868 | 9.602 |
| data16_hip8TeV.00313187.physics_Main.merge.AOD.r11416_p3868 | 3.030 |
| data16_hip8TeV.00313259.physics_Main.merge.AOD.r11416_p3868 | 4.659 |
| data16_hip8TeV.00313285.physics_Main.merge.AOD.r11416_p3868 | 4.272 |
| data16_hip8TeV.00313295.physics_Main.merge.AOD.r11416_p3868 | 9.923 |
| data16_hip8TeV.00313333.physics_Main.merge.AOD.r11416_p3868 | 3.791 |
| data16_hip8TeV.00313435.physics_Main.merge.AOD.r11416_p3868 | 0.301 |

Table 3: Data samples from $\sqrt{s_{\text{NN}}}$ = 8.16 TeV $p$+Pb collisions collected during the 2016 Heavy Ion Run. The luminosity collected for each run is reported. All the data were processed using ATHENA Release 21.

### 4.2.2 Trigger Configuration

Jet triggers used for Period B of the 2016 $p$+Pb run are listed in Table 6 All data comes from the physics_Main stream. The working points of the triggers are described further in Section 4.3.

### 4.2.3 Monte Carlo Samples

In order to evaluate the performance of jet reconstruction and the jet response within ATLAS, dijet Monte Carlo samples generated using Pythia8 [114] with the A14 tune and the NNPDF23LO PDFs [115] were produced. Dijet events are filtered using the JZ∗R04 filters, which select events containing an anti-$k_t$ $R = 0.4$ jet in specific truth $p_{\text{T}}$ ranges, referred to using the prefix "JZ" and an integer ranging from 1–5 in this analysis.

The detector response was simulated using the full ATLAS detector simulation based on the Geant4 toolkit [116]. For $p$+Pb collisions, Pythia8 truth events are overlaid with real minimum-bias $p$+Pb data. For each event, the Pythia generation and subsequent simulations were run with conditions matching the collected data, allowing the simulation to account for underlying-event effects in the jet reconstruction. In this MC

sample, the event activity originates mainly from the overlay contribution and therefore has no associated truth information.

Tables 4 and 5 list the $p_{\mathrm{T}}$ range, Pythia8 cross section and filter efficiency, and the number of events for the inclusive and forward-filtered $p$+Pb MC samples. For forward-filtered samples, a specific filter characterized by its own efficiency is applied to preferentially select events in which a jet exists at forward $\eta$, where the cross section is lower than for central jets. This allows smaller MC productions to provide increased statistics for events characterized by dijets in the forward region, namely low $x_{\mathrm{Pb}}$ and high $x_p$, providing satisfactory MC population over the full phase space considered. For the forward-filtered samples generated for this analysis, only jets in the proton-going direction are considered forward.

| JZ | $p_{\mathrm{T}}^{\mathrm{truth}}$ [GeV] | $\sigma$ [nb] | $\epsilon$ | $N_{\mathrm{evt}}$ |
|---|---|---|---|---|
| 1 | 20–60 | $7.3070 \times 10^{7}$ | $5.8782 \times 10^{-3}$ | 4M |
| 2 | 60–160 | $1.2916 \times 10^{6}$ | $5.6911 \times 10^{-3}$ | 4M |
| 3 | 160–400 | $1.1775 \times 10^{4}$ | $7.4572 \times 10^{-3}$ | 4M |
| 4 | 400–800 | $8.9941 \times 10^{1}$ | $8.1320 \times 10^{-3}$ | 4M |
| 5 | >800 | 1.1951 | $7.1577 \times 10^{-3}$ | 4M |

Table 4: Summary of inclusive $\sqrt{s_{\mathrm{NN}}} = 8.16$ TeV $p$+Pb Pythia8 MC samples.

| JZ | $p_{\mathrm{T}}^{\mathrm{truth}}$ [GeV] | $\sigma$ [nb] | $\epsilon$ | $N_{\mathrm{evt}}$ |
|---|---|---|---|---|
| 1 | 20–60 | $7.3070 \times 10^{7}$ | $9.0671 \times 10^{-4}$ | 4M |
| 2 | 60–160 | $1.2916 \times 10^{6}$ | $5.3628 \times 10^{-4}$ | 2M |

Table 5: Summary of forward-filtered $\sqrt{s_{\mathrm{NN}}} = 8.16$ TeV $p$+Pb Pythia8 MC samples.

The ATLAS data-set names for the MC samples used in this analysis are listed below.

$\sqrt{s_{\mathrm{NN}}} = 8.16$ TeV $p$+Pb Inclusive

---

mc16_pPb8TeV.420011.Pythia8EvtGen_A14NNPDF23LO_jetjet_JZ1R04.merge.AOD.e6518_d1603*

mc16_pPb8TeV.420012.Pythia8EvtGen_A14NNPDF23LO_jetjet_JZ2R04.merge.AOD.e6394_d1603*

mc16_pPb8TeV.420013.Pythia8EvtGen_A14NNPDF23LO_jetjet_JZ3R04.merge.AOD.e6518_d1603*

mc16_pPb8TeV.420014.Pythia8EvtGen_A14NNPDF23LO_jetjet_JZ4R04.merge.AOD.e6518_d1603*

mc16_pPb8TeV.420015.Pythia8EvtGen_A14NNPDF23LO_jetjet_JZ5R04.merge.AOD.e6518_d1603*

$\sqrt{s_{\mathrm{NN}}}$ = 8.16 TeV $p$+Pb Forward-filtered

---

mc16_pPb8TeV.420018.Pythia8EvtGen_A14NNPDF23LO_jetjet_JZ1R04_MaxEta_m3p0.merge.AOD.e8288_d1603*

mc16_pPb8TeV.420019.Pythia8EvtGen_A14NNPDF23LO_jetjet_JZ2R04_MaxEta_m3p0.merge.AOD.e8288_d1603*

## 4.3 Event Selection and Jet Reconstruction

### 4.3.1 Event Selection

To be included in the analysis, events must be pass the relevant good run list (GRL) selection. A new GRL for the 2016 8.16 TeV $p$+Pb data was created for this analysis to remove periods where the ZDC had an intolerable defect. This GRL can be found at:

GoodRunsLists/data16_hip/20240611/data16_hip8TeV.periodAllYear_HEAD_Unknown_PHYS_HeavyIonP_All_Good_WITH_IGNORES.xml.

### 4.3.2 Pileup Rejection

To have a meaningful determination of per-event centrality, it is essential to analyze events where only one hadronic $p$+Pb collision occurred. To meet this requirement, an effective pile-up rejection strategy is needed, in particular for data with high $\mu$ (by Heavy Ion standards) like those analyzed in this note ($\mu$ ranging from 0.15 to 0.3 in most of the runs). The same pile-up rejection strategy described in [117], specifically established for this data-set, is chosen for this analysis. Events characterized by more than one vertex with 7 or more associated tracks are rejected at the level of event selection. By applying this pile-up rejection, residual pile-up contributions can be still noticed in the ZDC, see Figure 39 where the correlation between $\mathrm{E}_{\mathrm{ZDC}}^{\mathrm{Pb\text{-}side}}$ and $\mathrm{E}_{\mathrm{T}}^{\mathrm{Pb}}$ is demonstrated.

This residual pile-up is not seen in the $\mathrm{E}_{\mathrm{T}}^{\mathrm{Pb}}$ spectrum, indicating that these backgrounds come from interactions that either only produce particles at very far forward rapidities or happen away from the IP. This residual was found to be present even when removing all the secondary vertices, and is suspected to be related to UPC $p$+Pb events.

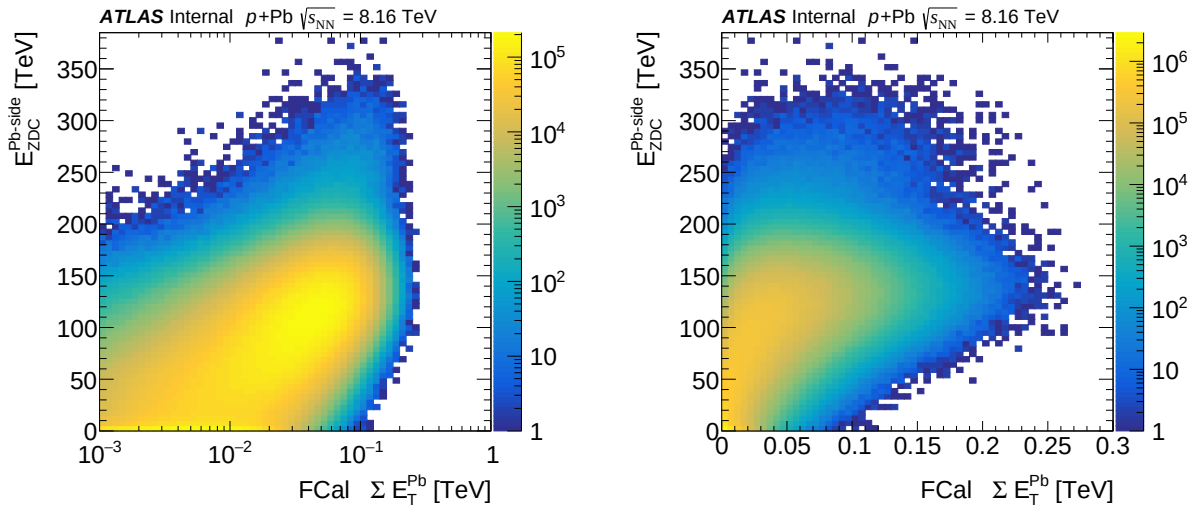



Figure 39: Correlation between the Pb going neutral forward energy ($E_{ZDC}^{Pb\text{-side}}$) and the transverse energy accumulated in the Pb-going FCal arm ($E_T^{Pb}$) for the $p$+Pb orientation. $E_T^{Pb}$ is binned a in logarithmic (left) and linear (right) fashion. $E_{ZDC}^{Pb\text{-side}}$ is normalized to the single neutron peak in order to display energy in terms of number of neutrons.

A procedure to characterize and remove this additional pile-up from the ZDC distributions is discussed in Section

ATLAS Liquid Argon calorimeter [118] cells receive ionization pulses resultant from the passage of charged particle tracks. These analog pulses are delivered to a Front-End Board (FEB) where they are amplified and shaped. The characteristic shape of these FEB pulses is a large positive amplitude pulse lasting around 100 ns proceeded immediately by a smaller negative amplitude pulse lasting around 400 ns. In the 2016 8.16 TeV $p$+Pb dataset, bunches of colliding particles circulate with 100 ns spacing between them. The measurement of $E_T^{Pb}$ is then biased negatively in events with preceding bunches having large event activity. Due to the physical constraints of transmission lines and the ZDCs location far from the interaction point, pulses seen in the ZDC have long characteristic decay time. Therefore, by sampling the ZDCs ADC value just before a pulse arrives and comparing that to a common baseline it is possible to identify these events with preceding large event activity.

Due to the difference in Z between the Pb ion (82) and the proton (1), most of the

events from UPC $p$+Pb collisions are expected to involve the emission of a virtual photon from the Pb ion to the $p$, with subsequent low activity on the Pb-going side of the FCal. By applying a 0–90% selection for the per-event dijet yield analysis, one can expect a substantial fraction of these events to be effectively rejected. Diffractive events are associated with significant rapidity gaps in the particle production. To further enforce the UPC events rejection, a cut based on rapidity gaps, equal to one utilized in the centrality analysis [117], where events with $\Delta\eta^{\mathrm{Pb}}_{\mathrm{gap}} > 1.8$ were rejected. In this analysis, such a cut rejects only a handful of events in the 0-90% centrality class, as demonstrated by Figure ADD Gaps figure.

### 4.3.3 Jet Reconstruction

The jet reconstruction is common to both measurments reported in this dissertation and is detailed in Section 3.4.

### 4.3.4 Jet Calibration

For heavy ion jets in the 2016 dataset[20], the in-situ portion of the jet calibration was taken from `EMTopoJets`[21]. A cross-calibration, derived using the direct balance method in events containing a Z boson or photon opposite to a jet, is used to bridge between `AntiKt4HIJets` and fully calibrated `EMTopoJets` jets by accounting for additional differences in response between them [112]. Thanks to the cross-calibration, it is possible to characterize the performance of the `AntiKt4HIJets` without deriving a whole new in-situ calibration. This procedure also has an associated systematic uncertainty.

[20] For an overview of the heavy ion jet claibration, see section 3.4.2. This section will only describe peculiarities of the calibration specific to the 2016 dataset.

[21] `EMTopoJets` are calorimeter jets reconstructed somewhat similarly to `AntiKt4HIJets` but without any UE subtraction. The `EMTopoJets` are calibrated by the ATLAS JetEtMiss group which has much more personpower than the ATLAS heavy ion jet group.

### 4.3.5 Jet Selection

Jets are reconstructed using the anti-$k_t$ jet algorithm with a radius parameter of R = 0.4. Jets are required to pass the 'LooseBad' jet selection working point.

### 4.3.6 Jet Trigger Strategy

Events are considered only if the leading jet in the trigger has fired the corresponding trigger chosen for the ($\eta$,$p_{\mathrm{T}}$) region where the jet is reconstructed. The trigger selection for $|\eta|<3.0$ and for $|\eta|>3.5$ above a $p_{\mathrm{T}}$ of 40 GeV is straightforward. In these regions, the fully efficient efficient trigger associated with the lowest prescale value is chosen. Efficiency values used to determine what trigger to choose in each ($\eta$,$p_{\mathrm{T}}$) region are reported in Table 6. For triggers covering the same $\eta$ region, prescale values decrease as the $p_{\mathrm{T}}$ threshold increases.

A different treatment is requested in the $\eta$ region corresponding to the transition between central and forward triggers, i.e. around $\eta = \pm 3.2$[22]. To handle the trigger analysis in this region, an inclusive approach (see [119] and references therein), hereafter referred to as the Fully Efficient Trigger Combination (FETC) method, was chosen. In the FETC approach, a combined weight based on all the considered trigger items for the given kinematic region is determined for the entire event sample. For each event, at least one actual trigger item bit from the set of considered items is required to be set. The weight ($w_{j,k}$) calculation for an event $j$ in a given run $k$ is based on the probability to accept the event after the application of the prescale to all the $i$-th trigger items considered. If $r_{i,j}$ is the bit for the trigger item $i$ in the event $j$, and the same trigger item is prescaled by a factor $p_{i,k}$ during the run $k$, the probability that the event $j$ in the

[22] Sign here is determined by the beam orientation, since this analysis covers only central and forward regions, discarding the backward where the energy accumulated in the FCal is used for $\langle \mathrm{FCal} \sum \mathrm{E}_{\mathrm{T}}^{\mathrm{Pb\text{-}side}} \rangle$.

run $k$ is accepted by the trigger item $i$ after the prescale can be expressed as

$$q_{i,j,k} = \frac{r_{i,j}}{p_{i,k}} \tag{41}$$

Assuming the prescale decisions for each trigger item are independent from each other, the probability that at least one of the $N_{\text{items}}$ trigger items accepts the event is given by

$$q_{j,k} = 1 - \prod_{i=1}^{N_{\text{items}}} \left(1 - \frac{r_{i,j}}{p_{i,k}}\right) \tag{42}$$

Starting from this, one can define the run dependent weight for event $j$ as

$$w_{j,k} = \frac{1}{q_{j,k}} \tag{43}$$

When combining all the considerations outlined above, one obtains a ($\eta$,$p_{\text{T}}$) map that provides, per each jet with $p_{\text{T}}>$ 40 GeV and -4.5 $< \eta <$ 3.1 (-3.1 $< \eta <$ 4.5), the jet trigger(s) considered for the analysis of the $p$+Pb (Pb+$p$) data. The three maps corresponding to the trigger configurations used in the $p$+Pb run are shown in Figure add trigger map.

The analysis reported in this note will use dijet events with a leading jet $p_{\text{T}}>$ 40 GeV and a sub-leading jet $p_{\text{T}}>$ 30 GeV. Only the dataset using $p$+Pb-ion trigger scheme is considered for the analysis presented in the CONF note, while the other two periods are used just for studies included in the Appendix of this Note and reported here for completeness.

### 4.4 Analysis Procedure

The Pb-going FCal transverse energy is constructed by summing the transverse projection of the electromagnetic-scale tower energies in the hemisphere facing the Pb beam. This definition is consistent with previous ATLAS $p$+Pb centrality measurements.

| Trigger Chain | $p_{\mathrm{T}}^{99}$ [GeV] | $\eta$ | $\mathscr{L}$ [ub$^{-1}$] | $p_{\mathrm{T}}^{99}$ Monitor |
|---|---|---|---|---|
| HLT_j30_ion_0eta490_L1TE10 | 36 | [−4.9,4.9] | 113.76 | HLT_mb_sptrk |
| HLT_j40_ion_L1J5 | 44 | [−3.2,3.2] | 468.1 | HLT_mb_sptrk |
| HLT_j50_ion_L1J10 | 57 | [−3.2,3.2] | 1120 | HLT_mb_sptrk |
| HLT_j60_ion_L1J20 | 70 | [−3.2,3.2] | 1562 | HLT_mb_sptrk |
| HLT_j75_ion_L1J20 | 80 | [−3.2,3.2] | 3468 | HLT_mb_sptrk |
| HLT_j100_ion_L1J20 | 111 | [−3.2,3.2] | 56671 | HLT_mb_sptrk |
| HLT_j15_ion_n320eta490_L1MBTS_1_1 | 44 | [−3.2,−4.9] | 77.5 | HLT_mb_sptrk |
| HLT_j25_ion_n320eta490_L1TE5 | 40 | [−3.2,−4.9] | 247.7 | HLT_mb_sptrk |
| HLT_j35_ion_n320eta490_L1TE10 | 57 | [−3.2,−4.9] | 179.9 | HLT_j25_ion_n320eta490_L1TE5 |
| HLT_j45_ion_n320eta490 | 65 | [−3.2,−4.9] | 5288 | HLT_j25_ion_n320eta490_L1TE5 |
| HLT_j55_ion_n320eta490 | 80 | [−3.2,−4.9] | 16635 | HLT_j25_ion_n320eta490_L1TE5 |
| HLT_j65_ion_n320eta490 | 85 | [−3.2,−4.9] | 53571 | HLT_j25_ion_n320eta490_L1TE5 |

Table 6: Overview of $p$+Pb ion triggers used in the analysis. The sampled luminosity, $\mathscr{L}$, recorded by each trigger is reported. The L1 for triggers that do not specify this detail in the chain name are: L1_J15.31ETA49 for HLT_j45_ion_n320eta490 and HLT_j55_ion_n320eta490, and L1_J20.31ETA49 for HLT_j65_ion_n320eta490. $p_{\mathrm{T}}^{99}$ represents the 99% efficiency point for a given chain, calculated taking into account both L1 and HLT efficiencies, and is reported in GeV.

In typical events containing a hard scattering, the energy observed in the Pb-going ZDC is dominated by spectator neutrons evaporating from the struck nucleus.

The ZDC energy scale is determined run by run from the response to a single spectator neutron. The low-energy spectrum contains distinguishable neutron peaks, and a simultaneous fit to the first three peaks and a background term validates the calibration. The fitted single-neutron peak is constrained to the nominal per-nucleon beam energy. The detector response below 1 TeV is limited by detector noise and low-energy backgrounds; this region is treated as a dedicated zero-energy bin. A residual in-time pileup contribution to the $\mathrm{E}_{\mathrm{ZDC}}^{\text{Pb-side}}$distribution is removed with a data-driven template method.

Finite jet-resolution effects cause migration in $x_p$. These effects are corrected with a two-dimensional iterative Bayesian unfolding procedure using RooUnfold. The response matrix contains $x_p$ and, in turn, either $\mathrm{E}_{\mathrm{ZDC}}^{\text{Pb-side}}$or $\mathrm{E}_{\mathrm{T}}^{\mathrm{Pb}}$. Since the available simulation does not model nuclear breakup in the ZDC or the full FCal response for the overlaid

minimum-bias event, the calorimeter observable is propagated through the unfolding with diagonal migration matrices. This retains the measured correlation of calorimeter energy with $x_p$ while correcting the jet-related kinematic migration.

The response matrix is reweighted as a function of reconstructed $x_p$ to match the data spectrum. For the ZDC measurement, pileup-subtracted data distributions are sampled to assign a ZDC energy to each simulated dijet event. The procedure is repeated with independent random seeds, and closure tests verify that the unfolding reproduces the particle-level simulation. Two iterations are used, balancing the convergence of the unfolded result against the growth of statistical uncertainty. Statistical uncertainties are evaluated by bootstrap resampling of both the data and response matrix.

### 4.5 Systematic Uncertainties

Systematic uncertainties arise from the jet energy scale (JES), jet energy resolution (JER), unfolding prior, disabled HEC sector, and the ZDC energy calibration. Except for the HEC and ZDC-scale variations, the full analysis is repeated with a response matrix modified according to each systematic variation. The difference relative to the nominal result defines the corresponding uncertainty.

The JES and JER variations quantify the mean and width of the $p_{\mathrm{T}}^{\mathrm{reco}}/p_{\mathrm{T}}^{\mathrm{gen}}$ response for matched jets. The HEC uncertainty is evaluated by enlarging the jet-exclusion region by 0.1 in both pseudorapidity and azimuth. To test the sensitivity to the unfolding prior, the response matrices are constructed without the nominal event-level reweighting and the full unfolding is repeated.

The ZDC-specific uncertainty includes the stability of the single-neutron peak, the residual energy-scale variation, and the non-linear energy correction. The FCal stability across the data-taking period is negligible relative to the other sources. The independent variations are added in quadrature. For ratios between $x_p$ intervals, JES, JER, HEC, and ZDC uncertainties are treated as fully correlated, while the unfolding-prior contribution

is treated as uncorrelated.

For the normalized distributions, the $E_{ZDC}^{Pb\text{-}side}$uncertainty is typically dominated by the ZDC energy-stability term and is below 7% over most of the measured range. The $E_T^{Pb}$distribution uncertainty is primarily driven by JES, JER, and the unfolding prior and is generally below 10%. The systematic uncertainties on $\langle E_{ZDC}^{Pb\text{-}side}\rangle$and $\langle \mathrm{FCal} \sum E_T^{Pb\text{-}side}\rangle$are below 2% in all but the lowest $x_p$ interval.

### 4.6 Results

The normalized $E_T^{Pb}$and $E_{ZDC}^{Pb\text{-}side}$distributions change systematically with $x_p$. Relative to a low-$x_p$ selection, high-$x_p$ dijet events exhibit an $E_T^{Pb}$distribution shifted toward lower energies. This is a direct confirmation of the event-activity bias previously observed in centrality-dependent jet measurements. The $E_{ZDC}^{Pb\text{-}side}$distribution shows a qualitatively similar trend: at high $x_p$, a larger fraction of events contains fewer forward neutrons. The effect in the ZDC is, however, substantially weaker than that in the FCal.

The mean values $\langle E_{ZDC}^{Pb\text{-}side}\rangle$and $\langle \mathrm{FCal} \sum E_T^{Pb\text{-}side}\rangle$both decrease with increasing $x_p$, most visibly for $x_p \gtrsim 0.02$, where earlier ATLAS measurements found evidence for small proton-size configurations. Across this range, $\langle E_{ZDC}^{Pb\text{-}side}\rangle$decreases by up to about 5%, corresponding on average to two fewer beam-energy neutrons, whereas $\langle \mathrm{FCal} \sum E_T^{Pb\text{-}side}\rangle$decreases by up to 40%. When the two means are normalized to their maximum values, the FCal estimator is found to be approximately six times more sensitive to hard-process kinematics than the ZDC estimator.

The relation between $E_T^{Pb}$and $E_{ZDC}^{Pb\text{-}side}$is also $x_p$ dependent. For ZDC energies below approximately 150 TeV, the average FCal transverse energy is consistent with a linear dependence on the ZDC energy in low-, middle-, and high-$x_p$ selections. The slope decreases progressively from low to high $x_p$, and the Pearson correlation coefficient changes by approximately 3% across the measured range. At higher ZDC energies, deviations from the linear trend are consistent with larger fluctuations in nuclear breakup

for events with the greatest event activity.

These results show that both common geometry estimators retain sensitivity to the proton configuration associated with a hard scattering. The ZDC observable is more robust than Pb-going FCal transverse energy, but is not independent of the hard-scattering kinematics. This observation provides experimental input for simultaneous descriptions of event activity, nuclear breakup, and color fluctuations in $p$+Pb collisions.

# Chapter 5 Observation of centrality-dependent dijet transverse momentum imbalance in O+O and Ne+Ne collisions

## 5.1 observable

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## 5.2 Data Sets

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## 5.3 Event Selection and Jet Reconstruction

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### 5.3.1 Event Selection

### 5.3.2 Pileup Rejection

### 5.3.3 Jet Reconstruction

The jet reconstruction is common to both measurments reported in this dissertation and is detailed in Section 3.4.

### 5.3.4 Jet Calibration

In Run 3, the in-situ and $\eta$-intercalibration[23] calibrations were investigated in detail for `AntiKt4HIJets` for the first time using high-$\mu$ $pp$ data at 13.6 TeV, leveraging the independence of the calibration on $\sqrt{s_{\mathrm{NN}}}$. The $\eta$-intercalibration is performed using the matrix-minimization method described in [120], with the central reference method used as a cross check. Calibration factors are derived in bins of $p_{\mathrm{T}}$ and $\eta$ and are applied as a correction to the jet $p_{\mathrm{T}}$. Cross-checks of the calibration have been established using

[23] In Run 3 the ATLAS JetEtMiss group stopped supporting the `EMTopoJets` so the cross calibration stretegy used in Run 2 was not an option.

dedicated MC samples at $\sqrt{s_{\text{NN}}}$= 5.36 TeV and residual non-closure resulted in additional systematic uncertainty. The work also included studies of the systematic uncertainty associated with each component of the calibration, as well as the jet energy resolution (JER).

### 5.3.5 Jet Selection

### 5.3.6 Jet Trigger Strategy

## 5.4 analysis

Placeholder text for this section.

## 5.5 uncertainties

Placeholder text for this section.

## 5.6 results

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# Chapter 6 Summary and Conclusions

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## References


[1] Murray Gell-Mann, *A Schematic Model of Baryons and Mesons*, Phys. Lett. **8** (1964) 214–215 (cit. on p. 4).

[2] Michael E. Peskin and Daniel V. Schroeder, *An Introduction to quantum field theory*, Addison-Wesley, 1995 (cit. on pp. 4, 6).

[3] Mark Thomson, *Modern Particle Physics*, Cambridge University Press, 2013, ISBN: 9781107034266 (cit. on pp. 4, 7, 8).

[4] M.E. Peskin, *Concepts of Elementary Particle Physics*, Oxford Master Series in Condensed Matter Physics Series, Oxford University Press, 2019, ISBN: 9780198812180, [books.google.com/books?id=BpGqDwAAQBAJ] (cit. on p. 4).

[5] F. Takahashi et al., *Review of Particle Physics*, Int. J. Mod. Phys. A **41** (2026) 2630011 (cit. on pp. 6, 9, 11, 16–18, 20, 26, 62).

[6] Richard P. Feynman, *Photon-Hadron Interactions*, W. A. Benjamin, 1972 (cit. on pp. 8, 9).

[7] J. D. Bjorken, *Asymptotic Sum Rules at Infinite Momentum*, Physical Review **179** (1969) 1547–1553 (cit. on p. 10).

[8] C. G. Callan and David J. Gross, *High-Energy Electroproduction and the Constitution of the Electric Current*, Phys. Rev. Lett. **22** (4 1969) 156–159, [link.aps.org/doi/10.1103/PhysRevLett.22.156] (cit. on p. 10).

[9] A. Bodek et al., *Experimental studies of the neutron and proton electromagnetic structure functions*, Phys. Rev. D **20** (7 1979) 1471–1552, [link.aps.org/doi/10.1103/PhysRevD.20.1471] (cit. on p. 10).

[10] H1 and ZEUS Collaborations, *Combination of Measurements of Inclusive Deep Inelastic $e^{\pm}p$ Scattering Cross Sections and QCD Analysis of HERA Data*, 2015, arXiv: 1506.06042v3 [hep-ex], [arxiv.org/abs/1506.06042v3] (cit. on pp. 11, 16, 17).

[11] John C. Collins, Davison E. Soper, and George Sterman, *Factorization of Hard Processes in QCD*, 2004, arXiv: hep-ph/0409313 [hep-ph], [arxiv.org/abs/hep-ph/0409313] (cit. on pp. 13, 15, 31).

[12] R. K. Ellis, W. J. Stirling, and B. R. Webber, "Deep inelastic scattering", *QCD and Collider Physics*, Cambridge Monographs on Particle Physics, Nuclear Physics and Cosmology, Cambridge University Press, 1996 86–156 (cit. on p. 15).

[13] G. Altarelli and G. Parisi, *Asymptotic freedom in parton language*, Nuclear Physics B **126** (1977) 298–318, ISSN: 0550-3213, [www.sciencedirect.com/science/article/pii/0550321377903844] (cit. on p. 16).

[14] I. I. Balitsky and L. N. Lipatov,
*The Pomeranchuk Singularity in Quantum Chromodynamics*,
Sov. J. Nucl. Phys. **28** (1978) 822–829 (cit. on p. 17).

[15] E. A. Kuraev, L. N. Lipatov, and Victor S. Fadin,
*The Pomeranchuk singularity in nonabelian gauge theories*,
Sov. Phys. JETP **45** (1977) 199–204 (cit. on p. 17).

[16] Krzysztof Kutak and Sebastian Sapeta, *Gluon saturation in dijet production in<mml:math xmlns:mml="http://www.w3.org/1998/Math/MathML" display="inline"><mml:mi>p</mml:mi></mml:math>-Pb collisions at the Large Hadron Collider*, Physical Review D **86** (2012), ISSN: 1550-2368,
[dx.doi.org/10.1103/PhysRevD.86.094043] (cit. on p. 18).

[17] Kari J. Eskola, Petja Paakkinen, Hannu Paukkunen, and Carlos A. Salgado,
*EPPS16: nuclear parton distributions with LHC data*,
The European Physical Journal C **77** (2017), ISSN: 1434-6052,
[dx.doi.org/10.1140/epjc/s10052-017-4725-9] (cit. on p. 19).

[18] Néstor Armesto, *Nuclear shadowing*,
Journal of Physics G: Nuclear and Particle Physics **32** (2006) R367–R393,
ISSN: 1361-6471, [dx.doi.org/10.1088/0954-3899/32/11/R01] (cit. on p. 19).

[19] J.J. Aubert et al.,
*The ratio of the nucleon structure functions F2N for iron and deuterium*,
Physics Letters B **123** (1983) 275–278, ISSN: 0370-2693,
[www.sciencedirect.com/science/article/pii/0370269383904379]
(cit. on p. 19).

[20] Or Hen, Gerald A. Miller, Eli Piasetzky, and Lawrence B. Weinstein,
*Nucleon-nucleon correlations, short-lived excitations, and the quarks within*,
Rev. Mod. Phys. **89** (4 2017) 045002,
[link.aps.org/doi/10.1103/RevModPhys.89.045002] (cit. on p. 19).

[21] A. Bodek and J. L. Ritchie,
*Fermi-motion effects in deep-inelastic lepton scattering from nuclear targets*,
Phys. Rev. D **23** (5 1981) 1070–1091,
[link.aps.org/doi/10.1103/PhysRevD.23.1070] (cit. on p. 20).

[22] C.F. Perdrisat, V. Punjabi, and M. Vanderhaeghen,
*Nucleon electromagnetic form factors*,
Progress in Particle and Nuclear Physics **59** (2007) 694–764, ISSN: 0146-6410,
[dx.doi.org/10.1016/j.ppnp.2007.05.001] (cit. on p. 20).

[23] Stanley J. Brodsky and Guy F. de Teramond,
*Onset of Color Transparency in Holographic Light-Front QCD*,
MDPI Physics **4** (2022) 633–646, arXiv: 2202.13283 [hep-ph]
(cit. on pp. 21–24).

[24] M. Benali et al., *Deeply virtual Compton scattering off the neutron*,
Nature Physics **16** (2020) 191–198, [doi.org/10.1038/s41567-019-0774-3]
(cit. on pp. 21, 22).

[25] Michel Guidal, Hervé Moutarde, and Marc Vanderhaeghen, *Generalized parton distributions in the valence region from deeply virtual compton scattering*, Reports on Progress in Physics **76** (2013) 066202, ISSN: 1361-6633, [dx.doi.org/10.1088/0034-4885/76/6/066202] (cit. on p. 22).

[26] Holly Szumila-Vance, Dipangkar Dutta, Gerald A. Miller, and Misak Sargsian, *Seeing Through the Nucleus: A Review of Color Transparency Phenomena*, 2025, arXiv: 2508.13311 [nucl-ex], [arxiv.org/abs/2508.13311] (cit. on p. 22).

[27] G. H. Burkhardt, *The suppression effect in ionization by fast electron pairs*, Il Nuovo Cimento (1955-1965) **9** (1958) 375–399, [doi.org/10.1007/BF02725095] (cit. on p. 22).

[28] M. Alvioli, L. Frankfurt, V. Guzey, and M. Strikman, *Revealing "flickering" of the interaction strength in <mml:math xmlns:mml="http://www.w3.org/1998/Math/MathML"> <mml:mrow> <mml:mi>p</mml:mi> <mml:mi>A</mml:mi> </mml:mrow> </mml:math> collisions at the CERN LHC*, Physical Review C **90** (2014), ISSN: 1089-490X, [dx.doi.org/10.1103/PhysRevC.90.034914] (cit. on p. 24).

[29] M. Alvioli, B. A. Cole, L. Frankfurt, D. V. Perepelitsa, and M. Strikman, *Evidence for<mml:math xmlns:mml="http://www.w3.org/1998/Math/MathML"><mml:mi>x</mml:mi></mml:math> dependent proton color fluctuations in<mml:math xmlns:mml="http://www.w3.org/1998/Math/MathML"><mml:mrow><mml:mi>p</mml:mi>‹ lisions at the CERN Large Hadron Collider*, Physical Review C **93** (2016), ISSN: 2469-9993, [dx.doi.org/10.1103/PhysRevC.93.011902] (cit. on p. 24).

[30] M. Alvioli, L. Frankfurt, D. V. Perepelitsa, and M. Strikman, *Global analysis of color fluctuation effects in proton– and deuteron–nucleus collisions at RHIC and the LHC*, Physical Review D **98** (2018), ISSN: 2470-0029, [dx.doi.org/10.1103/PhysRevD.98.071502] (cit. on pp. 24, 25).

[31] G. Hanson et al., *Evidence for Jet Structure in Hadron Production by e+ e- Annihilation*, Phys. Rev. Lett. **35** (1975), ed. by Martha C. Zipf 1609–1612 (cit. on p. 25).

[32] George Sterman and Steven Weinberg, *Jets from Quantum Chromodynamics*, Phys. Rev. Lett. **39** (23 1977) 1436–1439, [link.aps.org/doi/10.1103/PhysRevLett.39.1436] (cit. on p. 25).

[33] John E. Huth et al., "Toward a Standardization of Jet Definitions", *1990 DPF Summer Study on High-energy Physics: Research Directions for the Decade (Snowmass 90)*, 1990 0134–136 (cit. on p. 26).

[34] ATLAS Collaboration, *High-mass dijet events summer 2015*, ATLAS public event display record UNSG-2021-48, 2015, [atlaspo.cern.ch/public/event_display/] (cit. on p. 27).

[35] Gavin P. Salam, *Towards jetography*, The European Physical Journal C **67** (2010) 637–686, ISSN: 1434-6052, [dx.doi.org/10.1140/epjc/s10052-010-1314-6] (cit. on pp. 28, 30).

[36] Gavin P Salam and Grégory Soyez,
*A practical seedless infrared-safe cone jet algorithm*,
Journal of High Energy Physics **2007** (2007) 086–086, ISSN: 1029-8479,
[dx.doi.org/10.1088/1126-6708/2007/05/086] (cit. on p. 28).

[37] Stephen D. Ellis and Davison E. Soper,
*Successive combination jet algorithm for hadron collisions*,
Physical Review D **48** (1993) 3160–3166, ISSN: 0556-2821,
[dx.doi.org/10.1103/PhysRevD.48.3160] (cit. on p. 29).

[38] M. Wobisch and T. Wengler,
*Hadronization Corrections to Jet Cross Sections in Deep-Inelastic Scattering*,
1999, arXiv: hep-ph/9907280 [hep-ph], [arxiv.org/abs/hep-ph/9907280]
(cit. on p. 29).

[39] Matteo Cacciari, Gavin P Salam, and Gregory Soyez,
*The anti- k <sub> t </sub> jet clustering algorithm*,
Journal of High Energy Physics **2008** (2008) 063–063, ISSN: 1029-8479,
[dx.doi.org/10.1088/1126-6708/2008/04/063] (cit. on p. 29).

[40] Matteo Cacciari, Gavin P. Salam, and Gregory Soyez,
*FastJet user manual: (for version 3.0.2)*,
The European Physical Journal C **72** (2012), ISSN: 1434-6052,
[dx.doi.org/10.1140/epjc/s10052-012-1896-2] (cit. on p. 29).

[41] ATLAS Collaboration, *A precise measurement of the jet energy scale derived from single-particle measurements and in situ techniques in proton–proton collisions at* $\sqrt{s} = 13$ *TeV with the ATLAS detector*, Eur. Phys. J. C **85** (2025) 927,
arXiv: 2407.15627 [hep-ex] (cit. on p. 31).

[42] Zhong-Bo Kang, Felix Ringer, and Ivan Vitev, *The semi-inclusive jet function in SCET and small radius resummation for inclusive jet production*,
Journal of High Energy Physics **2016** (2016), ISSN: 1029-8479,
[dx.doi.org/10.1007/JHEP10(2016)125] (cit. on p. 32).

[43] Michael L. Miller, Klaus Reygers, Stephen J. Sanders, and Peter Steinberg,
*Glauber Modeling in High-Energy Nuclear Collisions*,
Annual Review of Nuclear and Particle Science **57** (2007) 205–243,
ISSN: 1545-4134, [dx.doi.org/10.1146/annurev.nucl.57.090506.123020]
(cit. on pp. 33, 50).

[44] H. De Vries, C.W. De Jager, and C. De Vries,
*Nuclear charge-density-distribution parameters from elastic electron scattering*,
Atomic Data and Nuclear Data Tables **36** (1987) 495–536, ISSN: 0092-640X,
[www.sciencedirect.com/science/article/pii/0092640X87900131]
(cit. on pp. 34, 62).

[45] Constantin Loizides, *Glauber predictions for oxygen and neon collisions at energies available at the CERN Large Hadron Collider*,
Physical Review C **113** (2026), ISSN: 2469-9993,
[dx.doi.org/10.1103/mkp8-zgxh] (cit. on pp. 34, 62).

[46] David J. Gross, Robert D. Pisarski, and Laurence G. Yaffe,
*QCD and instantons at finite temperature*, Rev. Mod. Phys. **53** (1 1981) 43–80,
[link.aps.org/doi/10.1103/RevModPhys.53.43] (cit. on p. 39).

[47] R. Hagedorn, *Statistical thermodynamics of strong interactions at high-energies*,
Nuovo Cim. Suppl. **3** (1965) 147–186 (cit. on p. 39).

[48] Roman Pasechnik and Michal Šumbera,
*Phenomenological Review on Quark–Gluon Plasma: Concepts vs. Observations*,
Universe **3** (2017) 7, ISSN: 2218-1997, [dx.doi.org/10.3390/universe3010007]
(cit. on p. 40).

[49] John C. Collins and M. J. Perry,
*Superdense Matter: Neutrons Or Asymptotically Free Quarks?*,
Phys. Rev. Lett. **34** (1975) 1353 (cit. on p. 40).

[50] N. Cabibbo and G. Parisi, *Exponential Hadronic Spectrum and Quark Liberation*,
Phys. Lett. B **59** (1975) 67–69 (cit. on p. 40).

[51] Andreas S. Kronfeld, *Twenty-First Century Lattice Gauge Theory: Results from the Quantum Chromodynamics Lagrangian*,
Annual Review of Nuclear and Particle Science **62** (2012) 265–284,
ISSN: 1545-4134, [dx.doi.org/10.1146/annurev-nucl-102711-094942]
(cit. on p. 41).

[52] KRISHNA RAJAGOPAL and FRANK WILCZEK,
"THE CONDENSED MATTER PHYSICS OF QCD",
*At The Frontier of Particle Physics*, WORLD SCIENTIFIC, 2001 2061–2151,
ISBN: 9789812810458, [dx.doi.org/10.1142/9789812810458_0043]
(cit. on p. 41).

[53] Dariusz Miśkowiec, *Heavy-ion collision basics*, 2023,
arXiv: 2305.11521 [nucl-ex], [arxiv.org/abs/2305.11521] (cit. on pp. 42, 45).

[54] T. Lappi and L. McLerran, *Some features of the glasma*,
Nuclear Physics A **772** (2006) 200–212, ISSN: 0375-9474,
[dx.doi.org/10.1016/j.nuclphysa.2006.04.001] (cit. on p. 43).

[55] Ulrich Heinz and Raimond Snellings,
*Collective Flow and Viscosity in Relativistic Heavy-Ion Collisions*,
Annual Review of Nuclear and Particle Science **63** (2013) 123–151,
ISSN: 1545-4134, [dx.doi.org/10.1146/annurev-nucl-102212-170540]
(cit. on pp. 43–45).

[56] Paul Romatschke and Ulrike Romatschke, *Viscosity Information from Relativistic Nuclear Collisions: How Perfect is the Fluid Observed at RHIC?*,
Physical Review Letters **99** (2007), ISSN: 1079-7114,
[dx.doi.org/10.1103/PhysRevLett.99.172301] (cit. on p. 44).

[57] ATLAS Collaboration, *Evidence for the Collective Nature of Radial Flow in Pb+Pb Collisions with the ATLAS Detector*, Phys. Rev. Lett. **136** (2026) 032301,
arXiv: 2503.24125 [nucl-ex] (cit. on p. 45).

[58] ATLAS Collaboration, *Measurement of the azimuthal anisotropy of charged particles produced in $\sqrt{s_{NN}} = 5.02$ TeV Pb+Pb collisions with the ATLAS detector*, Eur. Phys. J. C **78** (2018) 997, arXiv: 1808.03951 [nucl-ex] (cit. on pp. 46, 55, 78).

[59] Daniel Pablos and Adam Takacs, *Bayesian constraints on pre-equilibrium jet quenching and predictions for oxygen collisions*, Physical Review C **113** (2026), ISSN: 2469-9993, [dx.doi.org/10.1103/nw2z-pr7j] (cit. on p. 47).

[60] Guang-You Qin and Xin-Nian Wang, *Jet quenching in high-energy heavy-ion collisions*, International Journal of Modern Physics E **24** (2015) 1530014, ISSN: 1793-6608, [dx.doi.org/10.1142/S0218301315300143] (cit. on pp. 47, 48, 50).

[61] Abhijit Majumder, Berndt Müller, and Xin-Nian Wang, *Small Shear Viscosity of a Quark-Gluon Plasma Implies Strong Jet Quenching*, Physical Review Letters **99** (2007), ISSN: 1079-7114, [dx.doi.org/10.1103/PhysRevLett.99.192301] (cit. on p. 48).

[62] JETSCAPE Collaboration, *Bayesian inference analysis of jet quenching using inclusive jet and hadron suppression measurements*, Phys. Rev. C **111** (5 2025) 054913, [link.aps.org/doi/10.1103/PhysRevC.111.054913] (cit. on pp. 48, 49).

[63] G.-Y. Qin, A. Majumder, H. Song, and U. Heinz, *Energy and Momentum Deposited into a QCD Medium by a Jet Shower*, Physical Review Letters **103** (2009), ISSN: 1079-7114, [dx.doi.org/10.1103/PhysRevLett.103.152303] (cit. on pp. 49, 61).

[64] José Guilherme Milhano and Korinna Christine Zapp, *Origins of the di-jet asymmetry in heavy ion collisions*, 2016, arXiv: 1512.08107 [hep-ph], [arxiv.org/abs/1512.08107] (cit. on p. 49).

[65] ATLAS Collaboration, *Charged-hadron production in pp, p+Pb, Pb+Pb, and Xe+Xe collisions at $\sqrt{s_{NN}} = 5$ TeV with the ATLAS detector at the LHC*, JHEP **07** (2023) 074, arXiv: 2211.15257 [hep-ex] (cit. on p. 51).

[66] ATLAS Collaboration, *Measurement of the nuclear modification factor for inclusive jets in Pb+Pb collisions at $\sqrt{s_{NN}} = 5.02$ TeV with the ATLAS detector*, Phys. Lett. B **790** (2019) 108, arXiv: 1805.05635 [nucl-ex] (cit. on p. 51).

[67] ATLAS Collaboration, *ATLAS Heavy Ion Jet Cartoons for Presentations*, ATLAS Heavy Ion Jet Working Group, 2019, [twiki.cern.ch/twiki/bin/viewauth/AtlasProtected/CartoonsForPresentations] (cit. on p. 52).

[68] ATLAS Collaboration, *Observation of a Centrality-Dependent Dijet Asymmetry in Lead–Lead Collisions at $\sqrt{s_{NN}} = 2.76$ TeV with the ATLAS Detector at the LHC*, Phys. Rev. Lett. **105** (2010) 252303, arXiv: 1011.6182 [hep-ex] (cit. on pp. 52, 53, 65).

[69] Jorge Casalderrey-Solana, José Guilherme Milhano, and Urs Achim Wiedemann, *Jet quenching via jet collimation*, Journal of Physics G: Nuclear and Particle Physics **38** (2011) 035006, ISSN: 1361-6471, [dx.doi.org/10.1088/0954-3899/38/3/035006] (cit. on p. 53).

[70] Guang-You Qin and Berndt Müller, *Explanation of Dijet Asymmetry in Pb-Pb Collisions at the Large Hadron Collider*, Physical Review Letters **106** (2011), ISSN: 1079-7114, [dx.doi.org/10.1103/PhysRevLett.106.162302] (cit. on p. 53).

[71] ATLAS Collaboration, *Measurements of the suppression and correlations of dijets in Xe+Xe collisions at* $\sqrt{s_{NN}} = 5.44$ *TeV*, Phys. Rev. C **108** (2023) 024906, arXiv: 2302.03967 [nucl-ex] (cit. on pp. 53, 54, 57, 65).

[72] ATLAS Collaboration, *Measurement of jet* $p_T$ *correlations in Pb+Pb and pp collisions at* $\sqrt{s_{NN}} = 2.76$ *TeV with the ATLAS detector*, Phys. Lett. B **774** (2017) 379, arXiv: 1706.09363 [hep-ex] (cit. on pp. 53, 65).

[73] ATLAS Collaboration, *Measurements of the suppression and correlations of dijets in Pb+Pb collisions at* $\sqrt{s_{NN}} = 5.02$ *TeV*, Phys. Rev. C **107** (2023) 054908, arXiv: 2205.00682 [nucl-ex] (cit. on pp. 53, 65), Erratum: Phys. Rev. C **109** (2024) 029901.

[74] ATLAS Collaboration, *Jet radius dependence of dijet momentum balance and suppression in Pb+Pb collisions at* 5.02 *TeV with the ATLAS detector*, Phys. Rev. C **110** (2024) 054912, arXiv: 2407.18796 [nucl-ex] (cit. on pp. 54, 57, 65).

[75] ATLAS Collaboration, *Observation of Long-Range Elliptic Azimuthal Anisotropies in* $\sqrt{s} = 13$ *and* 2.76 *TeV pp Collisions with the ATLAS Detector*, Phys. Rev. Lett. **116** (2016) 172301, arXiv: 1509.04776 [hep-ex] (cit. on p. 55).

[76] ATLAS Collaboration, *Observation of Associated Near-Side and Away-Side Long-Range Correlations in* $\sqrt{s_{NN}} = 5.02$ *TeV Proton–Lead Collisions with the ATLAS Detector*, Phys. Rev. Lett. **110** (2013) 182302, arXiv: 1212.5198 [hep-ex] (cit. on p. 55).

[77] Nature Physics **15** (2018) 214–220, ISSN: 1745-2481, [dx.doi.org/10.1038/s41567-018-0360-0] (cit. on p. 55).

[78] ATLAS Collaboration, *Measurement of the azimuthal anisotropy of charged particles in* $\sqrt{s_{NN}} = 5.36$ *TeV* $^{16}O$+$^{16}O$ *and* $^{20}Ne$+$^{20}Ne$ *collisions with the ATLAS detector*, Phys. Rev. C **113** (2026) 045205, arXiv: 2509.05171 [nucl-ex] (cit. on p. 55).

[79] Anthony Badea et al., *Measurements of Two-Particle Correlations in <mml:math xmlns:mml="http://www.w3.org/1998/Math/MathML" display="inline"><mml:msup><mml:mi>e</mml:mi><mml:mo>+</mml:mo></mml:msup><mml:msu Collisions at 91 GeV with ALEPH Archived Data*, Physical Review Letters **123** (2019), ISSN: 1079-7114, [dx.doi.org/10.1103/PhysRevLett.123.212002] (cit. on p. 55).

[80] ATLAS Collaboration, *Measurement of the production cross section for $Z/\gamma^*$ in association with jets in pp collisions at $\sqrt{s} = 7$ TeV with the ATLAS detector*, Phys. Rev. D **85** (2012) 032009, arXiv: 1111.2690 [hep-ex] (cit. on p. 55).

[81] ATLAS Collaboration, *Measurement of photon-jet transverse momentum correlations in 5.02 TeV Pb+Pb and pp collisions with ATLAS*, Phys. Lett. B **789** (2019) 167, arXiv: 1809.07280 [nucl-ex] (cit. on p. 56).

[82] ALICE Collaboration, *Analysis of the apparent nuclear modification in peripheral Pb–Pb collisions at 5.02 TeV*, Physics Letters B **793** (2019) 420–432, ISSN: 0370-2693, [dx.doi.org/10.1016/j.physletb.2019.04.047] (cit. on pp. 56, 57).

[83] Constantin Loizides and Andreas Morsch, *Absence of jet quenching in peripheral nucleus–nucleus collisions*, Physics Letters B **773** (2017) 408–411, ISSN: 0370-2693, [dx.doi.org/10.1016/j.physletb.2017.09.002] (cit. on pp. 56, 57).

[84] JaeBeom Park, J. L. Nagle, Dennis V. Perepelitsa, Sanghoon Lim, and Constantin Loizides, *Selection-bias effects on high- <mml:math xmlns:mml="http://www.w3.org/1998/Math/MathML"> <mml:msub> <mml:mi>p</mml:mi> <mml:mi>T</mml:mi> </mml:msub> </mml:math> yield and correlation measurements in oxygen <mml:math xmlns:mml="http://www.w3.org/1998/Math/MathML"> <mml:mo>+</mml:mo> </mml:math> oxygen collisions*, Physical Review C **112** (2025), ISSN: 2469-9993, [dx.doi.org/10.1103/yr6k-97mn] (cit. on pp. 57, 62, 64).

[85] ATLAS Collaboration, *Centrality and rapidity dependence of inclusive jet production in $\sqrt{s_{NN}} = 5.02$ TeV proton–lead collisions with the ATLAS detector*, Phys. Lett. B **748** (2015) 392, arXiv: 1412.4092 [hep-ex] (cit. on pp. 57–60).

[86] ALICE Collaboration, *Measurement of inclusive charged-particle jet production in pp and p-Pb collisions at*

$$\sqrt{s_{NN}}$$

*= 5.02 TeV*, Journal of High Energy Physics **2024** (2024), ISSN: 1029-8479, [dx.doi.org/10.1007/JHEP05(2024)041] (cit. on pp. 57, 58).

[87] CMS Collaboration, *Measurement of inclusive jet production and nuclear modifications in pPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV*, Eur. Phys. J. C **76** (2016) 372, arXiv: 1601.02001 [hep-ex] (cit. on pp. 57, 58).

[88] PHENIX Collaboration, *Centrality-Dependent Modification of Jet-Production Rates in Deuteron-Gold Collisions at<mml:math xmlns:mml="http://www.w3.org/1998/Math/MathML" display="inline"><mml:mrow><mml:msqrt><mml:mrow><mml:msub><mml:mrow><mml:mi>s</mml:*
Physical Review Letters **116** (2016), ISSN: 1079-7114, [dx.doi.org/10.1103/PhysRevLett.116.122301] (cit. on pp. 57–59).

[89] ALICE Collaboration, *Transverse momentum dependence of inclusive primary charged-particle production in p–Pb collisions at*

$$\sqrt{s_{\mathrm{NN}}} = 5.02 \ TeV$$

*s NN = 5.02 TeV*, The European Physical Journal C **74** (2014), ISSN: 1434-6052, [dx.doi.org/10.1140/epjc/s10052-014-3054-5] (cit. on p. 58).

[90] Zhong-Bo Kang, Ivan Vitev, and Hongxi Xing, *Effects of cold nuclear matter energy loss on inclusive jet production in p+A collisions at energies available at the BNL Relativistic Heavy Ion Collider and the CERN Large Hadron Collider*, Phys. Rev. C **92** (2015) 054911, arXiv: 1507.05987 [hep-ph] (cit. on p. 58).

[91] ATLAS Collaboration, *Measurement of the Centrality Dependence of the Dijet Yield in p+Pb collisions at* $\sqrt{s_{NN}} = 8.16$ *TeV with the ATLAS Detector*, Phys. Rev. Lett. **132** (2024) 102301, arXiv: 2309.00033 [nucl-ex] (cit. on pp. 59, 60, 64, 81).

[92] Dennis V. Perepelitsa, *Contribution to differential <mml:math xmlns:mml="http://www.w3.org/1998/Math/MathML"> <mml:msup> <mml:mi>π</mml:mi> <mml:mn>0</mml:mn> </mml:msup> </mml:math> and <mml:math xmlns:mml="http://www.w3.org/1998/Math/MathML"> <mml:msub> <mml:mi>γ</mml:mi> <mml:mtext>dir</mml:mtext> </mml:msub> </mml:math> modification in small systems from color fluctuation effects*, Physical Review C **110** (2024), ISSN: 2469-9993, [dx.doi.org/10.1103/PhysRevC.110.L011901] (cit. on p. 60).

[93] ATLAS Collaboration, *Strong Constraints on Jet Quenching in Centrality-Dependent p + Pb Collisions at* 5.02 *TeV from ATLAS*, Phys. Rev. Lett. **131** (2023) 072301, arXiv: 2206.01138 [nucl-ex] (cit. on p. 60).

[94] ALICE Collaboration, *Study of very forward energy and its correlation with particle production at midrapidity in pp and p-Pb collisions at the LHC*, Journal of High Energy Physics **2022** (2022), ISSN: 1029-8479, [dx.doi.org/10.1007/JHEP08(2022)086] (cit. on p. 60).

[95] Alexander Huss et al., *Discovering Partonic Rescattering in Light Nucleus Collisions*, Phys. Rev. Lett. **126** (19 2021) 192301, [link.aps.org/doi/10.1103/PhysRevLett.126.192301] (cit. on p. 61).

[96] CMS Collaboration, *Discovery of Suppressed Charged-Particle Production in Ultrarelativistic Oxygen–Oxygen Collisions*, Phys. Rev. Lett. **136** (2025) 162301, arXiv: 2510.09864 [nucl-ex] (cit. on pp. 62, 63).

[97] ALICE Collaboration, *Evidence for parton energy loss in oxygen—oxygen collisions at* $\sqrt{\mathbf{s}_{\mathrm{NN}}} = \mathbf{5.36}$ *TeV*, 2026, arXiv: 2606.19967 [nucl-ex], [arxiv.org/abs/2606.19967] (cit. on pp. 62–64).

[98] STAR Collaboration, *Measurement of jet quenching in O+O collisions at $\sqrt{s_{\mathrm{NN}}} = 200$ GeV by the STAR experiment at RHIC*, 2026, arXiv: 2604.13935 [nucl-ex], [arxiv.org/abs/2604.13935] (cit. on p. 64).

[99] ATLAS Collaboration, *Observation of centrality-dependent dijet transverse momentum imbalance in O+O and Ne+Ne collisions at $\sqrt{s_{NN}} = 5.36$ TeV with the ATLAS detector* (2026), arXiv: 2606.20463 [nucl-ex] (cit. on p. 64).

[100] O Adriani et al., *Study of forward physics in sqrt($s_N N$) = 8.1 TeV proton-Lead ion collisions with the LHCf detector at the LHC*, tech. rep., CERN, 2016, [cds.cern.ch/record/2135330] (cit. on p. 69).

[101] Yusuf Can Cekmecelioglu, Clara Elisabeth Leitgeb, Lydia Audrey Beresford, Maciej Trzebinski, and Cigdem Issever, *Physics potential of a combined data taking of the LHCf and ATLAS Roman Pot detectors*, tech. rep., All figures including auxiliary figures are available at https://atlas.web.cern.ch/Atlas/GROUPS/PHYSICS/PUBNOTES/ATL-PHYS-PUB-2023-024, CERN, 2023, [cds.cern.ch/record/2867369] (cit. on p. 70).

[102] Peter Jenni, Marzio Nessi, and Markus Nordberg, *Zero Degree Calorimeters for ATLAS*, tech. rep., CERN, 2007, [cds.cern.ch/record/1009649] (cit. on p. 70).

[103] *Optimization of the New Signal Processing in Zero Degree Calorimeter*, 2019, [cds.cern.ch/record/2703904] (cit. on pp. 71, 72).

[104] ATLAS Collaboration, *Characterization of nuclear breakup as a function of hard-scattering kinematics using dijets measured by ATLAS in p+Pb collisions*, Phys. Lett. B **877** (2026) 140440, arXiv: 2504.02638 [nucl-ex] (cit. on pp. 76, 77).

[105] Matteo Cacciari, Gavin P. Salam, and Gregory Soyez, *The anti-$k_t$ jet clustering algorithm*, JHEP **04** (2008) 063, arXiv: 0802.1189 [hep-ph] (cit. on p. 77).

[106] ATLAS Collaboration, *Jet energy scale and its uncertainty for jets reconstructed using the ATLAS heavy ion jet algorithm*, ATLAS-CONF-2015-016, 2015, [cds.cern.ch/record/2008677] (cit. on p. 77).

[107] J. A. Hanks et al., *Method for separating jets and the underlying event in heavy ion collisions at the BNL Relativistic Heavy Ion Collider*, Phys. Rev. C **86** (2 2012) 024908, [link.aps.org/doi/10.1103/PhysRevC.86.024908] (cit. on p. 78).

[108] Tech. rep., [acode-browser1.usatlas.bnl.gov/lxr/source/athena/Control/AthenaConfiguration/python/Enums.py#0112] (cit. on p. 79).

[109] ATLAS Collaboration, *Jet energy measurement with the ATLAS detector in proton–proton collisions at $\sqrt{s} = 7$ TeV*, Eur. Phys. J. C **73** (2013) 2304, arXiv: 1112.6426 [hep-ex] (cit. on p. 79).

[110] Martin Krivos, Martin Rybar, and Martin Spousta, *Numerical inversion and cross-calibration for heavy-ion jet collections in 2017 pp and 2018 Pb+Pb*, tech. rep., CERN, 2020, [cds.cern.ch/record/2706256] (cit. on p. 79).

[111] T Kosek et al., *Numerical inversion for calibration of Heavy-ion jets*, tech. rep., CERN, 2014, [cds.cern.ch/record/1742072] (cit. on p. 80).

[112] Jeff Ouellette, *Calibration of Heavy Ion Jets in 2016 8.16 TeV p+Pb Collisions*, tech. rep., CERN, 2019, [cds.cern.ch/record/2655891] (cit. on pp. 80, 87).

[113] Aviv Cukierman and Benjamin Nachman, *Mathematical Properties of Numerical Inversion for Jet Calibrations*, Nucl. Instrum. Meth. A **858** (2017) 1–11, arXiv: 1609.05195 [physics.data-an] (cit. on p. 80).

[114] Christian Bierlich et al., *A comprehensive guide to the physics and usage of PYTHIA 8.3*, SciPost Phys. Codebases (2022) 8, arXiv: 2203.11601 [hep-ph] (cit. on p. 83).

[115] NNPDF Collaboration, Richard D. Ball, et al., *Parton distributions with LHC data*, Nucl. Phys. B **867** (2013) 244, arXiv: 1207.1303 [hep-ph] (cit. on p. 83).

[116] S. Agostinelli et al., *GEANT4 – a simulation toolkit*, Nucl. Instrum. Meth. A **506** (2003) 250 (cit. on p. 83).

[117] Kurt Hill and Dennis Perepelitsa, *Centrality Calibration in 8.16 TeV p+Pb Collisions*, tech. rep., CERN, 2018, [cds.cern.ch/record/2301540] (cit. on pp. 85, 87).

[118] ATLAS Collaboration, *ATLAS Liquid Argon Calorimeter: Technical Design Report*, ATLAS-TDR-2; CERN-LHCC-96-041, 1996, [cds.cern.ch/record/331061] (cit. on p. 86).

[119] Victor Lendermann et al., *Combining Triggers in HEP Data Analysis*, Nucl. Instrum. Meth. A **604** (2009) 707–718, arXiv: 0901.4118 [hep-ex] (cit. on p. 88).

[120] Debjani Banerjee, *Summary of Run 3 Eta-intercalibration factors for HI Jets*, tech. rep., 2025, [indico.cern.ch/event/1568043/contributions/6606137/attachments/3102743/5498083/07-11-2025-HI-InSituJet-DB.pdf] (cit. on p. 94).